Social Ties and Scientists' Productivity: Evidence from Correspondence Networks during the Quantum Revolution

Roman Dimakov

November 9, 2026

How do pre-existing social ties shape scientists' responses to a major intellectual breakthrough? Combining newly digitized correspondence records for 5,600 physicists with publication and citation data, I study the effect of direct access to the social networks of founders of quantum mechanics on productivity of scientists following the 1925 breakthrough. Because connected scientists were already on a steeper productivity trajectory, I employ a doubly robust inverse-propensity-weighted difference-in-differences estimator to account for pre-discovery observables. I find that prior correspondence access generated an approximately 60% increase in research impact on average, measured by citations, persisting but gradually declining over roughly twelve years after the 1925 breakthrough. Exploring heterogeneity, I show that these gains concentrate among early career "rookies" rather than established "veterans", and are surprisingly larger for weaker ties (fewer exchanged letters) than for strong ones. Furthermore, scientists embedded in the broader network benefit more than isolated peers. Taken together, the results reveal how pre-existing social structures facilitate knowledge diffusion and shape scientific productivity during a Kuhnian paradigm shift.

Roman Dimakov: Centre for Economic Research and Graduate Education – Economic Institute, a joint workplace of Charles University and the Economics Institute of the Czech Academy of Sciences, Politickych veznu 7, 111 21 Prague, Czech Republic. Correspondence e-mail: Roman.Dimakov@cerge-ei.cz. I want to give special thanks to the following people for their invaluable comments, guidance and infinite support: Paolo Zacchia, Sebastian Ottinger, Christian Ochsner, Stepan Jurajda and other CERGE-EI faculty members; faculty and students at SDU, Denmark; Cagin Keskin, Vladimir Shchukin, and Samir Orujov.

1. Introduction

What is the role of preexisting social ties for scientists' responses to a major intellectual breakthrough? New scientific knowledge is accessible through articles and conferences, yet researchers also exchange unfinished arguments, technical advice, and early results through personal relationships. Such interactions may be especially valuable when a new theory changes the questions, methods, and language of a field. Scientists with direct access to the researchers developing the theory may learn sooner, resolve uncertainty more quickly, or identify productive applications before otherwise comparable peers. In such situation of significant changes in a scientifc enterprise, it might be important who is at the frontier of it and therefore will benefit the most. And to some degree this might be a matter of who is at the right place about the right time.

This paper studies these questions during the quantum-mechanics revolution of 1925–1927. The setting combines a sharp intellectual break with rich evidence on communication among scientists. By the early 1920s, the old quantum theory could no longer reconcile a growing set of experimental findings with classical mechanics. Heisenberg's matrix mechanics in 1925, followed by the work of Born and Jordan, replaced classical trajectories with a noncommutative mathematical formalism. Schrödinger's wave mechanics, developed independently in 1926 from de Broglie's matter-wave hypothesis, offered a formally equivalent route. Contributions by Bohr, Pauli, and Dirac helped consolidate the new mechanics and its probabilistic interpretation. These advances emerged from closely connected circles centred on Copenhagen and Göttingen, where seminars, visits, and correspondence moved ideas across institutions with unusual speed (Kojevnikov 2020). Therefore, year 1925 – the year of the publication of the new mechanics in Heisenberg (1925) – is used as a common scientific shock that affected the physics profession as a whole. However, the historical setting provides an opportunity to track the heterogeneous prior access to founders of the new theory.

The analysis in this paper asks whether scientists who already corresponded with the founders experienced a larger change in research output after the breakthrough than scientists without a direct link. The estimand is therefore the relative post-1925 advantage associated with pre-existing direct access to the founders. It can capture faster transmission of ideas, tacit knowledge, feedback, influence and power of established networks or entry into productive research networks. It does not capture the aggregate effect of quantum mechanics, since unconnected scientists could learn the

breakthrough from journals, conferences, and indirect contacts.[1]

To perform the analysis, I construct the communication network of physicists from a newly digitized source, the *Inventory of Sources for History of Twentieth-Century Physics* (ISHTCP), an archival inventory covering approximately 5,600 physicists and nearly 750,000 letters written between 1896 and 1952 (Wheaton 1993). Letters were an important medium for discussing scientific problems and circulating work before publication, which makes the inventory an unusually direct record of interpersonal access. Athough I digitized the whole inventory data, I focus on a subset of data on connections to founders of quantum mechanics *before* its arrival in 1925.

I digitize the list of physicists in the inventory and their characteristics together with the recorded direction, volume, and period of their correspondence. I link these data to publications and citations in leading journals indexed by the Web of Science. The resulting estimation sample is a balanced panel of around 1,500 scientists observed annually from 1915 through 1945, for a total of ... scientist-year observations.[2]

The baseline treatment indicator equals one if a scientist had a recorded correspondence link, in either direction, to at least one of eight founders of quantum mechanics before the breakthrough: Niels Bohr, Max Born, Louis de Broglie, Paul Dirac, Werner Heisenberg, Pascual Jordan, Wolfgang Pauli, or Erwin Schrödinger. This definition treats the presence of a direct link as a measure of access, affinity to an intellectual circle responsible for the main components of the breakthrough.[3] It does *not* assume that every recorded letter transmitted a specific quantum-mechanical idea.

In my setting, the selection issue plays a huge role. I rely on correspondence networks of the founders, and social networks are highly endogenous. As a result, an identification concern is that connected scientists may have selected themselves to the founders' circle (smarter or talented people could attract equally smarter/ talented ones, they might sort by the common preferences towards methods, theories and views in their profession, and so on). This proves to be the case in my setting and data. Because connected scientists were already more productive before 1925, the preferred

[1]The discovery of the new mechanics affected the physics profession as a whole, so the analysis cannot identify the total effect of quantum mechanics by comparing physics before and after 1925.

[2]The reason why the resulting number of scientists is less than the total number of scientists covered in the correspondence inventory is, firstly because I focus only on scientists that were already active in profession by 1925, while the inventory covers correspondence up to 1950s; and secondly because I need to include scientists for whom there is at least one publication record during 1915–1945 in the subset of journals in the Web of Science database. Also, observations for the eight founders are excluded from the estimation sample.

[3]It has been often the case in the history of science that the most innovative and influential ideas had been produced by a few super-star scientists.

specifications use pre-treatment observables to create a comparison group similar to the connected group.

I rely on the recent advances in the difference-in-differences methodology literature for construction of a proper comparison group and estimation of the effect. The empirical analysis employes the augmented inverse-propenstiy weighted (dobuly-robust) difference-in-differences specification comparing changes after 1925 for connected and unconnected scientists. To be precise, I use the doubly robust difference-in-differences estimator in Sant'Anna and Zhao (2020) for identification, and I equip the machinery in Callaway and Sant'Anna (2021) for event study estimates. The estimator in Sant'Anna and Zhao (2020) allows constructing a more comparable control group by inclduing covariates in a more flexible, non-parametric way. The propensity score conditions on pre-1925 publication activity, fractional authorship, short-horizon citations, physics specialty (field), principal country of work, and first publication year. The event-study coefficients trace the differential response by year and provide evidence on pre-treatment dynamics, with wild-bootstrap confidence intervals clustered by scientist.

The analysis also uses a standard two-way fixed-effects model to estimate the effect and demonstrate that even under a less robust specification, the main results still hold. In the model, scientist fixed effects absorb permanent differences in ability and career-long productivity, while publication-year fixed effects absorb shocks common to the sampled profession. The model also includes a rich set of pre-treatment scientist covariates interacted with post dummy (equal to 1 from 1925 and after, and 0 otherwise) and with year dummies in a preferred specification.

I measure annual scientific output on both the extensive and intensive margins: whether the scientist published, the number of publications, and citations received by that year's publications over five- and twenty-five-year forward windows. Publications measure the production of research, while citation-weighted publications measure its subsequent reception and influence (Waldinger 2012).

The doubly-robust difference-in-differences benchmark shows a positive post-1925 differential for directly connected scientists. Their probability of publishing increased by 5.4 percentage points relative to the contemporaneous change among unconnected scientists. Against a pre-period mean of 0.428 for the connected group, this corresponds to a 12.6 percent increase. The estimate for log(1+publications) is positive but imprecise. The citation estimates are larger: connected scientists' post-1925 work received 0.123 log points more citations within five years and 0.131 log points more citations within twenty-five years. The latter coefficient corresponds to an increase of approximately 14

percent in one plus citations. The evidence therefore points more clearly to a change in the production and recognition of influential work than to a precisely estimated increase in the number of papers.

The weighted event studies reinforce this interpretation while showing the uncertainty around individual horizons. For twenty-five-year citations, the inverse-probability-weighted estimates are centred near zero before 1925 and positive in every displayed post-treatment year. The doubly robust specification produces a similar profile. Confidence intervals remain wide at several horizons, so the figures provide stronger evidence of a sustained post-1925 differential than of its exact year-by-year path. Moreover, citations to papers published before 1925 may themselves arrive after the breakthrough. Leads in a long-horizon citation outcome are therefore not clean placebo tests for anticipation; publication outcomes provide the more informative evidence on pre-treatment output trends.

Two field-specific exercises help assess whether the pattern is connected to the intellectual content of the shock. Restricting the sample to theoretical quantum studies yields a pooled estimate for twenty-five-year citations of approximately 0.13 log points, close to the full-sample benchmark, and the post-1925 event-study estimates remain positive. In contrast, a negative-control comparison using astrophysics and physical chemistry shows no sustained positive post-treatment path. This contrast is consistent with pre-existing access mattering most for scientists working near the new theory. It is nevertheless suggestive rather than a formal test of equality across fields and cannot by itself eliminate field-specific confounding.

The research design has important limitations. Correspondence links were not randomly assigned, and weighting balances only observed characteristics. Selection on changing, unobserved determinants of productivity may remain. The treatment measure records the existence of a link rather than its content or strength. The inventory may omit interactions through unrecorded letters, face-to-face meetings, or other channels. Finally, knowledge can diffuse through indirect network paths, so an ostensibly unconnected scientist may still be exposed through connected colleagues. Such interference can blur the difference between direct and indirect effects (Vazquez-Bare 2023). For these reasons, I interpret the results as conditional evidence of a differential change associated with direct founder access, rather than as a design-based causal estimate free of selection and spillovers.

The paper contributes to three related literatures. First, it contributes to the economics of science by studying how interpersonal access is associated with scientific out-

put during a period of rapid intellectual change. Existing work demonstrates that peers and mentors can shape researchers' productivity and creative direction (Waldinger 2012; Borowiecki 2022). The correspondence data add a measure of direct communication that does not require coauthorship, common employment, or geographic proximity.

Second, the paper contributes to research on social networks and knowledge diffusion. Network position can affect when agents receive and adopt information (Banerjee et al. 2013), while sustained interpersonal exchange can support the production and recombination of useful knowledge (Mokyr 2016). The historical correspondence network makes it possible to observe links among scientists before a major shock and then follow their research output over time. It also exposes a central empirical difficulty in network settings: information may reach the comparison group through indirect paths, making spillovers part of the substantive mechanism as well as a threat to conventional identification.

Third, the paper provides a micro-level quantitative account of a scientific revolution. Historical scholarship emphasizes that discoveries are collective processes whose meaning and consequences develop across researchers and institutions rather than at a single instant (Kuhn 1962). By combining this historiography with scientist-level communication and output data, the paper examines the role of informal institutions such as existing professional networks – in mediation of the responses to a major intellectual transformation. This approach connects quantitative economic history to the study of innovation by treating a scientific breakthrough not only as a new stock of codified knowledge, but also as a change whose productive use may depend on prior social access.

The remainder of the paper proceeds as follows. Section 2 describes the setting of the development of quantum mechanics and networks of physicists. Section 3 introduces data sources, treatment definition, and estimation sample construction. Section 4 documents the structure of the pre-1925 correspondence network. Section 5 presents the identification strategy. Section 6 reports the doubly-robust difference-in-differences event-study results. Section ... presents robustness checks, section ... presents results of heterogeneity and mechanisms investigation, and section 9 concludes and provides a discussion of the limitation of provided evidence in the paper.

2. Historical Setting

The emergence and rapid development of quantum mechanics between 1925 and 1927 was the outcome of a crisis in the old quantum theory and of intense exchange among several research centres, including Copenhagen, Göttingen, Munich and Berlin. Journals, seminars, visits, proof sheets, and private correspondence circulated unfinished arguments quickly. The resulting theory changed both the mathematical language of physics and its account of what could be predicted about nature. This combination of rupture and networked discovery makes the quantum revolution a useful setting in which to study social ties and the diffusion of knowledge (van der Waerden 1967; Kojevnikov 2022).

2.1. From classical physics to the crisis of the old quantum theory

At the beginning of the twentieth century, mechanics and electrodynamics were built around continuous motion, definite trajectories, and deterministic laws. Planck's quantum hypothesis of 1900 and Einstein's light quantum of 1905 challenged this framework but did not supply a general mechanics of atomic processes. Bohr's 1913 model explained the stability and spectrum of hydrogen by combining classical electron orbits with discontinuous transitions between stationary states. His correspondence principle connected the new rules to classical physics in the appropriate limit and guided much subsequent work (Heilbron and Kuhn 1969).

The Bohr–Sommerfeld theory produced striking successes, yet its mixture of classical dynamics and quantum conditions was not self-contained. By the early 1920s it could not satisfactorily explain helium, many-electron atoms, the anomalous Zeeman effect, or spectral-line intensities. Physicists accumulated increasingly elaborate rules without a consistent account of atomic motion. This crisis called for a reformulation rather than another correction of classical mechanics (Mehra and Rechenberg 1982; van der Waerden 1967).

2.2. The Copenhagen–Munich–Göttingen circle and new mechanics

The breakthrough grew out of a network rather than a single laboratory. Bohr's Institute for Theoretical Physics, opened in Copenhagen in 1921, became an international meeting point for young theorists. Bohr shaped the agenda through his atomic theory, seminars, hospitality, and extensive correspondence. Göttingen supplied a comple-

mentary centre under Max Born, linked to its mathematical community and James Franck's experimental institute. Researchers moved among Copenhagen, Göttingen, Munich, Hamburg, Cambridge, and other centres. The "Copenhagen–Göttingen school" is therefore best understood as a connected circle, not a group united on every question (Kojevnikov 2020, 2022).

Werner Heisenberg occupied a central position in this circle. Trained by Arnold Sommerfeld in Munich, he worked with Born in Göttingen and spent the winter of 1924–1925 with Bohr and Hendrik Kramers in Copenhagen. His 1925 paper abandoned hypothetical electron orbits and built a mechanics from observable transition frequencies and amplitudes. Born recognized Heisenberg's noncommutative multiplication rule as matrix multiplication; with Pascual Jordan he formulated the canonical commutation relation, after which Born, Heisenberg, and Jordan extended the theory in their *Dreimännerarbeit*. Matrix mechanics replaced visualized trajectories with algebraic relations among infinite arrays, changing what counted as a legitimate physical description (van der Waerden 1967; Mehra and Rechenberg 1982).

The achievement remained collective. Jordan supplied essential mathematical technique. Pauli, a sharp critic whose letters connected the main centres, showed that matrix mechanics reproduced the hydrogen spectrum. Paul A. M. Dirac, working in Cambridge outside the immediate Göttingen group, independently related quantum commutators to classical Poisson brackets. The transformation theories of Dirac and Jordan then moved quantum mechanics toward its standard abstract operator language. The 1925 breakthrough was thus a rapid chain of linked advances, not a finished theory appearing in one paper (van der Waerden 1967).

2.3. Wave mechanics, duality, and the probabilistic interpretation

A second route began with Louis de Broglie, who in 1923–1924 extended the wave–particle duality of radiation to matter. Erwin Schrödinger developed this insight into wave mechanics in early 1926. His differential equation reproduced the hydrogen spectrum using tools more familiar than Heisenberg's matrices. Schrödinger and others soon demonstrated the close formal equivalence of wave and matrix mechanics, although their physical pictures differed. Their joint success established wave–particle duality as central to the new physics: quantum objects could be described as neither classical particles nor classical waves alone (van der Waerden 1967; Bacciagaluppi and Valentini 2009).

The common formalism did not settle its meaning. Schrödinger sought a continu-

ous, realist account of the wave, while the Copenhagen–Göttingen circle increasingly interpreted the theory statistically. In 1926 Born argued that the wave function determines probabilities rather than a definite trajectory. In 1927 Heisenberg formulated the uncertainty relation: conjugate quantities such as position and momentum cannot simultaneously be arbitrarily precise. Bohr placed this limit within complementarity, according to which mutually exclusive experimental arrangements reveal different but jointly necessary aspects of a quantum phenomenon.

These ideas broke with classical determinism, but "acausality" requires care. Born, Heisenberg, and Jordan presented irreducible probability and the limitation of classical causality as fundamental lessons of quantum mechanics. Einstein, Schrödinger, and de Broglie resisted and pursued causal or realist alternatives. Historians therefore treat the reception of indeterminism not only as a consequence of equations, but also as a process shaped by intellectual commitments and scientific communities (Forman 1971; Kojevnikov 2022).

2.4. The Fifth Solvay Congress and Kuhn's quantum revolution

The Fifth Solvay Congress in Physics, held in Brussels in October 1927 under the title *Electrons and Photons*, assembled the leading participants when the formalism was largely in place but its interpretation remained open. It confronted Born and Heisenberg's probabilistic mechanics, Schrödinger's wave mechanics, and de Broglie's pilot-wave theory, while giving lasting visibility to the Bohr–Einstein disputes over measurement and completeness. Bohr had introduced complementarity at Como shortly beforehand. The later image of Solvay as a decisive victory for a unified "Copenhagen interpretation" is too simple: de Broglie, Einstein, and Schrödinger remained unconvinced. Yet the congress clarified the competing positions and became a focal point in the subsequent consolidation of the Copenhagen approach (Bacciagaluppi and Valentini 2009).

Thomas Kuhn treated quantum mechanics as a paradigmatic scientific revolution. Anomalies in black-body radiation, specific heats, and the photoelectric effect produced a crisis that the established framework could not absorb. For Kuhn, a revolution is not cumulative: it reconstructs a field's problems, methods, standards, and objects from new fundamentals. Quantum mechanics did this by replacing classical trajectories with states and operators, deterministic prediction with probabilities, and a single visual model with complementary descriptions. Kuhn also stressed that a new paradigm is assimilated collectively and cannot be reduced to one scientist or instant. Accordingly,

this paper treats 1925 as the sharp beginning of the new mechanics and 1925–1927 as the broader period in which its formalism and dominant interpretation were constructed, contested, and diffused (Kuhn 1970).

3. Data and Construction of A Panel Data Set of Physicists

This paper draws on three main data sources. The first is an inventory of correspondence among physicists over roughly 1900–1952, providing information on the number and direction of letters exchanged. The second source is the schematic biograhical data on physicists from a report accompanying the inventory. Finally, the third source is the publication and citation data from the Web of Science online database over 1900–1945.

3.1. Data on Historical Correspondence of Physicists

This paper uses a novel data set of historical mail correspondence among 5,600 physicists over the period from 1896 to the middle 1950s from the *Inventory of Sources for History of the Twentieth Century Physics (ISHTCP)*. The inventory was curated by historians of science with Bruce R. Wheaton in charge at the end of 1980s, resulting in a publication of a microfilm index to collected materials and an accompanying report in Wheaton (1993). Letters comprise the largest part of the index, with the main microfilm reels containing the correspondence entries for 5,600 physicists whose names are listed in the report.

Overall, the inventory presents information on nearly 750,000 letters written to or by physicists active in the period from 1896 to 1952. The main criterion for selection of a physicist into the inventory was that they actively published in one of the seven leading journals of pure physics in the period.[4]

For the purpose of studying how preexisting social ties to founders of quantum mechanics shape scientists' productivity following the 1925 breakthrough, I focus on correspondence networks formed before the breakthrough. I reconstruct historical correspondence networks of physicists for the period roughly from 1900 to 1925. I use the networks to track physicists' connections with the eight founders of quantum mechanics established before 1925. Physicists who exchanged any amount of letters with any of the eight founders are considered "treated" and comprise the treated group of connected scientists.

[4] *Annalen der Physik*; *Annales de Physique*; *Philosophical Magazine*; *Proceedings of the Royal Society of London A*; *Physical Review*; *Uspekhi Fizicheskikh Nauk*; and *Zeitschrift für Physik*.

By this treatment measure, I obtain 159 connected physicists.

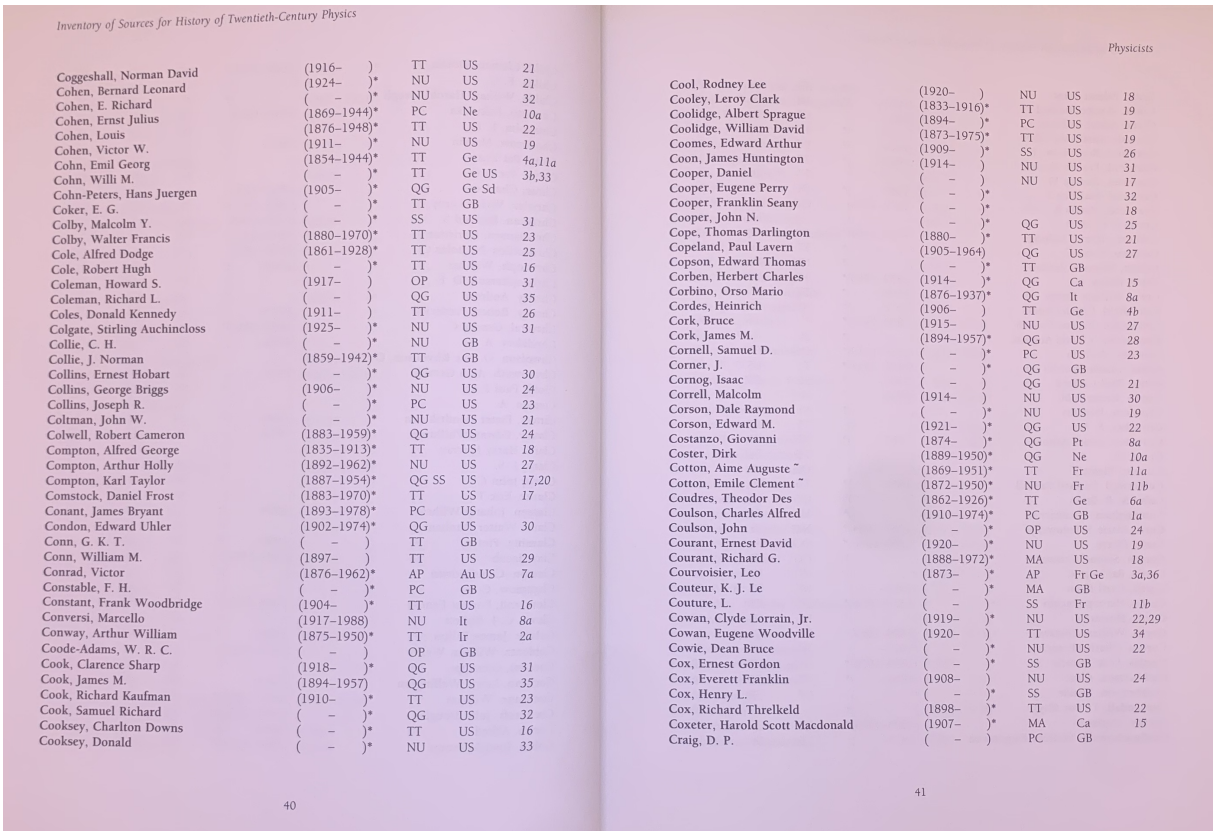
3.2. Schematic Biographical Data on Physicists

As was already mentioned, the correspondence inventory is supplemented by the book-report Wheaton (1993). It provides a user guide to the microfiche reel index (white and yellow tapes) and consists of several sections, including a section on descriptive statistics on the social correspondence networks. Importantly, it has a section with a list of physicists' full names that appear in the index. Figure 1 presents two typical pages from the *Physicists* section in the book with the physicists' names list. The inventory also contains a separate yellow-color card index which separately lists all non-physicists' correspondents for each physicist name in the main index, though the non-physicists correspondence are out of the scope of the analysis of this paper.

I digitize the full list of physicists' names from the "Physicists" section (120 pages, 5,600+ names), together with their biographical and career attributes: life years, primary and secondary country(ies) of work, primary and secondary specialty(ies) in physics. I construct a physicist data set from the digitized list which represents the benchmark data set of the universe of physicists that I study in the paper.

Table ... shows the full list of physics specialty codes, and Table ... in the Appendix shows the full list of country codes used in Wheaton (1993).

This is the main list of physicists which I use to match against the author names from the Web of Science articles. For matching algorithm details, see Appendix C.



Inventory of Sources for History of Twentieth-Century Physics

Coggeshall, Norman David	(1916–)	TT	US	21
Cohen, Bernard Leonard	(1924–)*	NU	US	21
Cohen, E. Richard	(–)*	NU	US	32
Cohen, Ernst Julius	(1869–1944)*	PC	Ne	10a
Cohen, Louis	(1876–1948)*	TT	US	22
Cohen, Victor W.	(1911–)*	NU	US	19
Cohn, Emil Georg	(1854–1944)*	TT	Ge	4a,11a
Cohn, Willi M.	(–)*	TT	Ge US	3b,33
Cohn-Peters, Hans Juergen	(1905–)*	QG	Ge Sd	
Coker, E. G.	(–)*	TT	GB	
Colby, Malcolm Y.	(–)*	SS	US	31
Colby, Walter Francis	(1880–1970)*	TT	US	23
Cole, Alfred Dodge	(1861–1928)*	TT	US	25
Cole, Robert Hugh	(–)*	TT	US	16
Coleman, Howard S.	(1917–)	OP	US	31
Coleman, Richard L.	(–)	QG	US	35
Coles, Donald Kennedy	(1911–)	TT	US	26
Colgate, Stirling Auchincloss	(1925–)*	NU	US	31
Collie, C. H.	(–)*	NU	GB	
Collie, J. Norman	(1859–1942)*	TT	GB	
Collins, Ernest Hobart	(–)*	QG	US	30
Collins, George Briggs	(1906–)*	NU	US	24
Collins, Joseph R.	(–)*	PC	US	23
Coltman, John W.	(–)*	NU	US	21
Colwell, Robert Cameron	(1883–1959)*	QG	US	24
Compton, Alfred George	(1835–1913)*	TT	US	18
Compton, Arthur Holly	(1892–1962)*	NU	US	27
Compton, Karl Taylor	(1887–1954)*	QG SS	US	17,20
Comstock, Daniel Frost	(1883–1970)*	TT	US	17
Conant, James Bryant	(1893–1978)*	PC	US	
Condon, Edward Uhler	(1902–1974)*	QG	US	30
Conn, G. K. T.	(–)	TT	GB	
Conn, William M.	(1897–)	TT	US	29
Conrad, Victor	(1876–1962)*	AP	Au US	7a
Constable, F. H.	(–)*	PC	GB	
Constant, Frank Woodbridge	(1904–)*	TT	US	16
Conversi, Marcello	(1917–1988)	NU	It	8a
Conway, Arthur William	(1875–1950)*	TT	Ir	2a
Coode-Adams, W. R. C.	(–)	OP	GB	
Cook, Clarence Sharp	(1918–)*	QG	US	31
Cook, James M.	(1894–1957)	QG	US	35
Cook, Richard Kaufman	(1910–)*	TT	US	23
Cook, Samuel Richard	(–)*	QG	US	32
Cooksey, Charlton Downs	(–)*	TT	US	16
Cooksey, Donald	(–)*	NU	US	33

40

Physicists

Cool, Rodney Lee	(1920–)	NU	US	18
Cooley, Leroy Clark	(1833–1916)*	TT	US	19
Coolidge, Albert Sprague	(1894–)*	PC	US	17
Coolidge, William David	(1873–1975)*	TT	US	19
Coomes, Edward Arthur	(1909–)*	SS	US	26
Coon, James Huntington	(1914–)	NU	US	31
Cooper, Daniel	(–)	NU	US	17
Cooper, Eugene Perry	(–)*		US	32
Cooper, Franklin Seany	(–)*		US	18
Cooper, John N.	(–)*	QG	US	25
Cope, Thomas Darlington	(1880–)*	TT	US	21
Copeland, Paul Lavern	(1905–1964)	QG	US	27
Copson, Edward Thomas	(–)*	TT	GB	
Corben, Herbert Charles	(1914–)*	QG	Ca	15
Corbino, Orso Mario	(1876–1937)*	QG	It	8a
Cordes, Heinrich	(1906–)	TT	Ge	4b
Cork, Bruce	(1915–)	NU	US	27
Cork, James M.	(1894–1957)*	QG	US	28
Cornell, Samuel D.	(–)*	PC	US	23
Corner, J.	(–)*	QG	GB	
Cornog, Isaac	(–)	QG	US	21
Correll, Malcolm	(1914–)	NU	US	30
Corson, Dale Raymond	(–)*	NU	US	19
Corson, Edward M.	(1921–)*	QG	US	22
Costanzo, Giovanni	(1874–)*	QG	Pt	8a
Coster, Dirk	(1889–1950)*	QG	Ne	10a
Cotton, Aime Auguste ~	(1869–1951)*	TT	Fr	11a
Cotton, Emile Clement ~	(1872–1950)*	NU	Fr	11b
Coudres, Theodor Des	(1862–1926)*	TT	Ge	6a
Coulson, Charles Alfred	(1910–1974)*	PC	GB	1a
Coulson, John	(–)*	OP	US	24
Courant, Ernest David	(1920–)*	NU	US	19
Courant, Richard G.	(1888–1972)*	MA	US	18
Courvoisier, Leo	(1873–)*	AP	Fr Ge	3a,36
Couteur, K. J. Le	(–)*	MA	GB	
Couture, L.	(–)	SS	Fr	11b
Cowan, Clyde Lorrain, Jr.	(1919–)*	NU	US	22,29
Cowan, Eugene Woodville	(1920–)	TT	US	34
Cowie, Dean Bruce	(–)*	NU	US	22
Cox, Ernest Gordon	(–)*	SS	GB	
Cox, Everett Franklin	(1908–)	NU	US	24
Cox, Henry L.	(–)*	SS	GB	
Cox, Richard Threlkeld	(1898–)*	TT	US	22
Coxeter, Harold Scott Macdonald	(1907–)*	MA	Ca	15
Craig, D. P.	(–)	PC	GB	

41

FIGURE 1. Two pages from the *Physicists* section in the ISHTCP report.

Note: The figure presents two pages, 40–41. The columns in the appearing order are: physicist full name, birth and death years, primary specialty(ies) of work in physics, primary country(ies) of work, and the microfiche reel number where the information on physicist's letters can be found. The asterisk after the lifetime years is an indicator of letters have been found for the physicist.

Source: Wheaton (1993)

3.3. Construction of the Sample of Scienitsts

3.4. Correspondence-Based Treatment Definition

In this paper, treatment is defined as preexisting correspondence connection with founders of quantum mechanics established *before* the breakthrough. In the next passages, I explain steps to define the treated group of scientists.

I start with reconstruction of the pre-1925 correspondence networks of physicists. It is important to highlite that I restrict analysis to the *pre-existing* correspondence networks of physicists, i.e. formed *before* the arrival of Heisenberg's matrix mechanics in 1925. I deliberately ignore the network ties formed after 1925 because they could be formed in a highgly endogenous way with respect to the future potnetial outcomes in quantum mechanics.[5] In this subsection, I describe how I prepare the subsample of the incumbent scientists – scientists observed in the correspondence networks prior to 1925.

3.5. Construction of the Estimation Panel Data Set

I start with creating of a list of the "incumbent" physicists and their correspondents.

First, I restrict the correspondence edges data set to keep only edges in the year intervals before 1925: 1900–06, 1907–13, 1914–21, and 1922–25. Keeping correspondence of 1922–25 somewhat poses a risk of including scientists who selected themselves already after the arrival of quantum mechanics. I perform a robustness exercise without the links formed in 1922-25 in the Appendix.

Before dealing with edges in the network, I restrict my set of physicists to be scientists who published their first paper between 1900 and 1924, and they stay alive in 1925, the year of discovery of matrix mechanics (Heisenberg 1925).[6] After applying these filters, I obtain the total number of incumbent physicists, up to the fifth neighbors, to be 1,626.

My incumbent physicists sample is built starting from the core – the "fathers of quantum mechanics": Niels Bohr, Werner Heisenberg, Max Born, Pascual Jordan, Wolfgang Pauli, Paul Dirac, Erwin Schrödinger and Louise de Broglie; and Albert Einstein. These scientists contributed outstandingly to quantum studies. Niels Bohr and Albert Einstein are always in the top-5 physicists by their total degree, betweenness and eigenvector

[5] Scientists could have strategically reach out to e.g. Bohr or Heisenberg after the breakthrough occurred because of their increased importance in the field.

[6] The first year of publication is extracted from articles only from a set of the seven leading journals in physics over 1900-1945, i.e. one can treat it as the year when one publishes in one of the elite physics journals.

centrality in the pre-1925 network. Starting from the core, I track all their neighbors up to the fifth order of the neighborhood: the first, the second, the third, the fourth and the fifth neighbors of the core. The first, second, third, fourth and fifth sets of neighbors constructed in a way that they are not overlapping sets of physicists. These core and the neighbors comprise *the giant connected component* of Niels Bohr, the other founding fathers of quantum mechanics, and Albert Einstein.

It is important to notice that all other important physicists, such as Arnold Sommerfeld, Max Planck, Max von Laue, etc. automatically fall into the giant component: they are mostly appearing as the second neighbor edges.

The Figure 2 presents the graph of the full pre-1925 network of physicists: the core, the first, the second and the third order neighbors. I also picture the graphs for periods 00-06, 07-13, and 14-21 separately in the Figures in the Appendix.

The Table 2 gives counts of unique physicists by correspondence-network set. One can see that the network is expanding very fast – the largest set is the second neighbors set, – but it is rather shallow too: in the fifth neighbor set, we have only single physicist. This means that on the fifth order of neighborhood, the giant component of Bohr and Einstein becomes practically exhaustive.

In network terms, these numbers suggest a rapidly expanding but relatively shallow correspondence network around the core physicists. The first-neighbor group consists of 344 scientists directly connected to the core, while the second-neighbor group is substantially larger, comprising 1,085 scientists. In contrast, the number of more distant scientists declines sharply to 165 third neighbors, 8 fourth neighbors, and a single fifth neighbor. Thus, most scientists in the connected component are within two steps of the core, indicating a relatively compact network in terms of path length. The sharp decline beyond the second-neighbor group further suggests network saturation, as most reachable scientists have already been identified within two degrees of separation. While these descriptive statistics alone are insufficient to establish that the correspondence network exhibits a small-world structure, they are consistent with a relatively shallow connected network.

Let D_i indicate whether scientist i exchanged at least one letter, in either direction, with at least one of the eight founders of quantum mechanics—Niels Bohr, Max Born, Louis de Broglie, Paul Dirac, Werner Heisenberg, Pascual Jordan, Wolfgang Pauli, and Erwin Schrödinger—before the breakthrough. Thus,

$$D_i = \mathbf{1}\{\text{scientist } i \text{ had a direct correspondence link to a founder before 1925}\}. \tag{1}$$

The baseline definition treats a single letter as sufficient and does not use direction or volume. It is consequently a measure of direct network access rather than of the intensity or scientific content of the relationship. Scientists with $D_i = 0$ form the comparison group. Treatment status is fixed over the analysis window, while the post indicator is $Post_t = \mathbf{1}\{t \geq 1925\}$.

The archival source groups part of the correspondence into a 1922–1925 interval. Unless exact dates from the Archive for History of Quantum Physics or another source establish that a link was formed by the end of 1924, this interval cannot cleanly separate pre-existing access from contact initiated during 1925. I therefore interpret the baseline estimates as effects conditional on the recorded timing of correspondence. A strict causal interpretation requires either an end-of-1924 treatment measure or a conservative treatment measure using links observed through 1921.

The target estimand is the average differential post-1925 change among scientists who were already active at the scientific frontier before the breakthrough. In potential-outcome notation, for $t \geq 1925$ it is

$$ATT(t) = \mathbb{E}\left[Y_{it}(1) - Y_{it}(0) \mid D_i = 1\right], \tag{2}$$

where $Y_{it}(1)$ is the outcome scientist i would produce in year t with pre-existing direct access to a founder and $Y_{it}(0)$ is the corresponding outcome without that direct access. Because unconnected scientists could learn through journals, conferences, colleagues, and indirect paths in the correspondence network, $ATT(t)$ is a relative advantage of direct access. It is not the total effect of the quantum-mechanics revolution on physics.

The analysis uses an annual scientist panel for 1919–1932. The intended risk set consists of physicists who had begun publishing by the end of 1924 and were alive in 1925; the founders themselves are excluded. The current results reported below contain 18,802 scientist-year observations for 1,343 scientists. The sample-selection rules, including the treatment flag and the handling of years before entry or after death, must be applied in a single reproducible analysis file because coding inactive years as zero can otherwise confound a productivity response with changes in sample composition.

3.6. Outcomes and pre-treatment covariates

The propensity-score model uses characteristics determined before 1925: mean annual fractional authorship, the probability of publishing, short-horizon citations,

field, country of work, birth year, and first publication year. Pre-treatment levels and slopes in the production outcomes are the most informative diagnostics of comparability. Long-horizon forward citations require a separate timing caveat: citations assigned to a pre-1925 publication can be received after 1925. Consequently, a lead coefficient for twenty-five-year citations is not a clean placebo for the absence of anticipation. I use publication outcomes to assess pre-trends and interpret the long-horizon citation event study as evidence on the later reception of papers from different publication cohorts.

I use the data on letters exchanged between physicists to track pre-existing connections with the scientists who laid the foundations of the future quantum mechanics by publishing the first key papers on it. I call these scientists founders. To identify the founders, I consulted various works of historians of science, including articles and most monumental studies of the development of quantum physics/mechanics, such as Mehra and Rechenberg (2000). The data on correspondence among physicists allows me to construct various measures of connectedness based on the intensity and frequency of scientific letters exchanged. What is crucial, is that the data allows me to construct the social networks of scientists based on their inter-personal communication, by different periods in the development of physics.

3.7. Publication and Citation Data

To measure productivity of scientists, I download data on scientific articles published in the period 1900–1945, and data on their citations up until 1952 from the Clarivate's Web of Science online database.[7] I collect publication data from academic journals in physics, chemistry, maths and general science that were considered as leading journals in that period. The journals cover German-, French- and English-speaking countries. They represent all major fields of pure physics, chemistry and mathematics. I collect records of document type "Article" only. Table 1 shows the list of selected journals and the earliest year of publication in WoS.

To answer the research question in this paper, I introduce three measures of scientist-level productivity outcomes. The first is simply the number of articles published by a scientist i in publication year t. I count the counts the number of publications per scientist in a given year adjusted by the weight calculated as the 1 over the number of authors per paper. For instance, if a published article has three authors, it contributes 1/3 to each author's outcome, irrelevant of who is the principal author in the article.

[7] I used the Charles University's subscription to access the database.

The number of publications is the simplest researcher's productivity measure which reflects only the quantity of the produced reseach.

To measure also the quality of produced research, I introduce the second measure of scientist producitivity – the number of citations to all articles publsihed by a scientist i during all years up to calendar year t. This measure counts citations to all existing works of a scientist by year t, with citations coming from all sources (journals) covered in the Web of Science. The measure is not restricted to the subset of journals selected for the analysis. Hence, it reflects the recognition of a scientist's work coming from a broad scientific community and from all domains of science. It is important to note that in the post-treatment period of my analysis (after 1925), the measure counts citations not only to works that start to be published in 1925, but also for the previous works (from 1900 to 1924). Hence, it mixes counts of citations to "novel" (after 1925) and "older" (before 1925) works. Later, I introduce an alternative construction of the measure that separates the two.

The third measure of scientist productiviy is citations per a publication-cohort received in the next 2, 3, 5 and 25 years since the next year following the publication year. It is defined as the sum of citations to the cohort of publications (published in the same year) over the time window. This measure sums up citations from all sources covered in the Web of Science, not being restricted to the subset of journals where the articles were published. This is a baseline measure of scientific productivity that reflects quality of the work (approximation of the recognition of contribution by broad scientific community). Waldinger (2012) calls this measure *citation weighted publications*.

These are the standard measures of scientific productivity used in the economics of science literature. For instance, Aghion et al. (2024), Pauskiewicz and ... (2026, JMP), and Waldinger (2012), Moser ("Red-ucators") use the same or similar scientific production outcomes. For robustness checks, I also use alternative measures of publications and citaitons such as the natural logarithm of $(1 + \text{publications}_{it})$ and of $(1 + \text{citations}_{it})$, and the inverse hyperbolic sinus transformation of citations_{it}.

TABLE 1. Earliest publication years available in the Web of Science database by journal

Journal	**Earliest year in WoS**	**Included in the analysis**
Annalen der Physik	1900	Yes
Annales de Physique	1946	No
Philosophical Magazine	1900	Yes
Proceedings of the Royal Society of London Series A	1905	Yes
Physical Review	1900	Yes
Uspekhi Fizicheskikh Nauk	1955	No
Zeitschrift für Physik	1920	Yes
Physikalische Zeitschrift	1900	Yes
Die Naturwissenschaften	1913	Yes
Nature	1900	Yes
Science	1900	Yes

Notes: The table reports the earliest publication year available in Clarivate's Web of Science (WoS) for each journal. Information on the earliest publication year as of October 2026. For *Uspekhi Fizicheskikh Nauk*, no records are available in WoS before 1955, and for *Annales de Physique* – before 1946. Therefore, I exclude the two journals from my sample.

Source: Web of Science

4. Correspondence Networks

This section provides a descriptive analysis of the pre-1925 correspondence networks.

Table 4 presents network coverage from Bohr and Einstein across correspondence periods. Table 6 reports the top-20 Bohr's strongest direct correspondence connections, 1900–1925. Similarly, Table 7 reports the same information for Einstein.

Figure A22 in the Appendix presents the same network as in Figure 3 but now visualized using the ForceAtlas2 layout algorithm, which is particularly suitable for larger networks and for displaying their community structure. The algorithm treats connected nodes as attracting one another, while all nodes exert a repelling force that prevents them from occupying the same position. Consequently, scientists with many connections among themselves tend to form visible clusters, whereas weakly connected groups are positioned farther apart. A gravitational force keeps the overall network together. The layout was run for 1,000 iterations, giving node positions more time to stabilize, and a fixed random seed was used to make the resulting visualization reproducible. The observed clusters should be interpreted as visual indications of network structure rather than formally estimated communities.

4.1. Star Networks of Niels Bohr and Albert Einstein

Essentially, the networks provided above are mostly the networks of two physicists – Bohr and Einstein. I therefore continue the further descriptive analysis around these two giant nodes.

Below, Figure 5 shows the social graph of first order connections of Niels Bohr and Albert Einstein which deliberately excludes connections among the neighbors themselves, so the overlap is visually clear.

The overlap between Bohr and Einstein correspondence circles can be shown as the Jaccard similarity. The Jaccard similarity between the two networks is: 0.166. Bohr had 180 direct correspondence neighbors, while Einstein had 213. They shared 56 common neighbors, representing 31.1% of Bohr's neighbors and 26.3% of Einstein's neighbors.

In the Appendix, Figures **??**, **??**, **??**, and **??** show how the correspondence network in the physics profession in the years approximately 1900 to 1925 was formed primarily around two major hubs: Niels Bohr and Albert Einsten. The overlap in the number of connections between the two hubs in the bunched years network is around 30%, which means that around 30% of active physicists were corresponding with both of the giants.

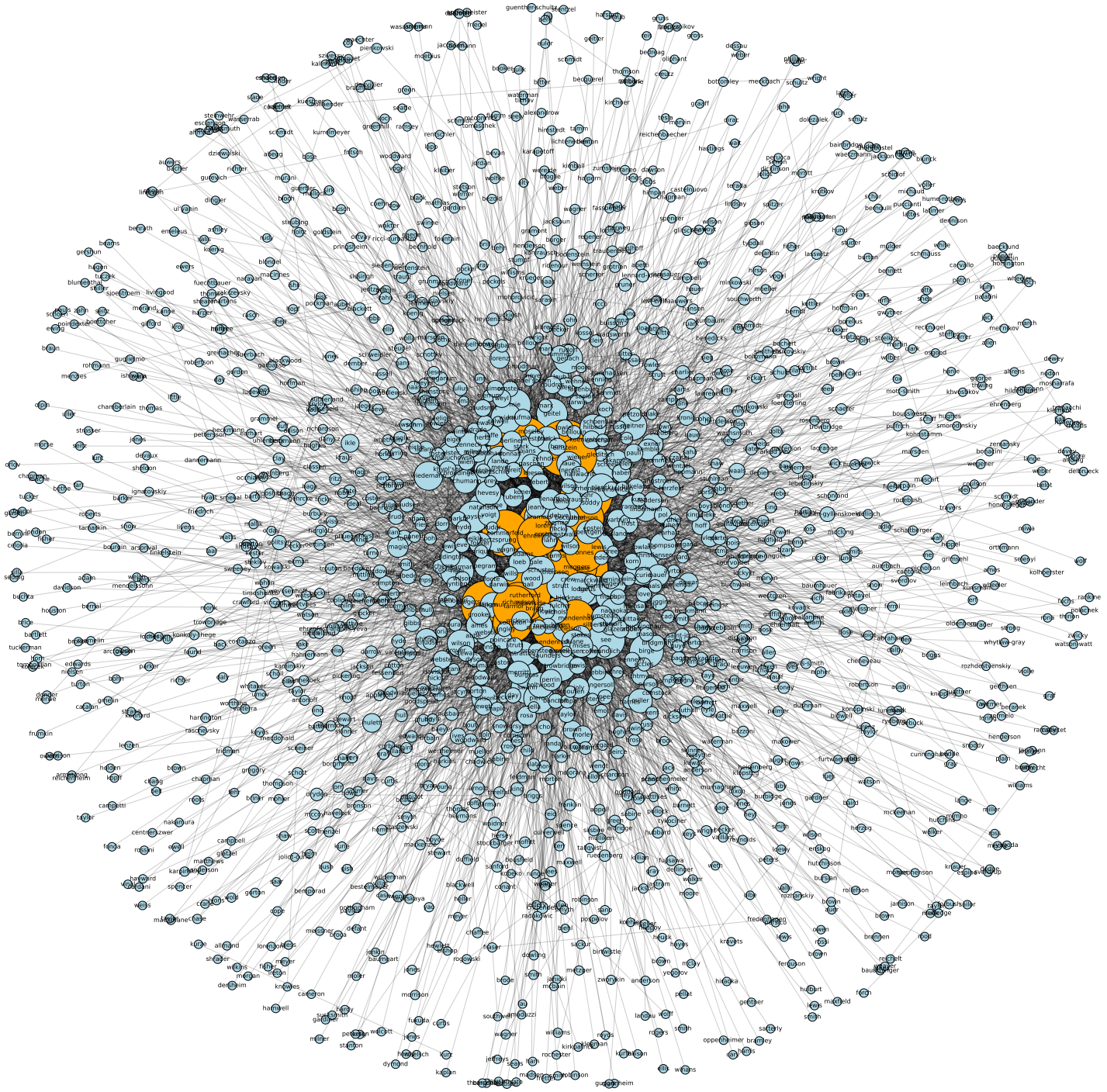


FIGURE 2. Social graph of correspondence, 1900-1925.

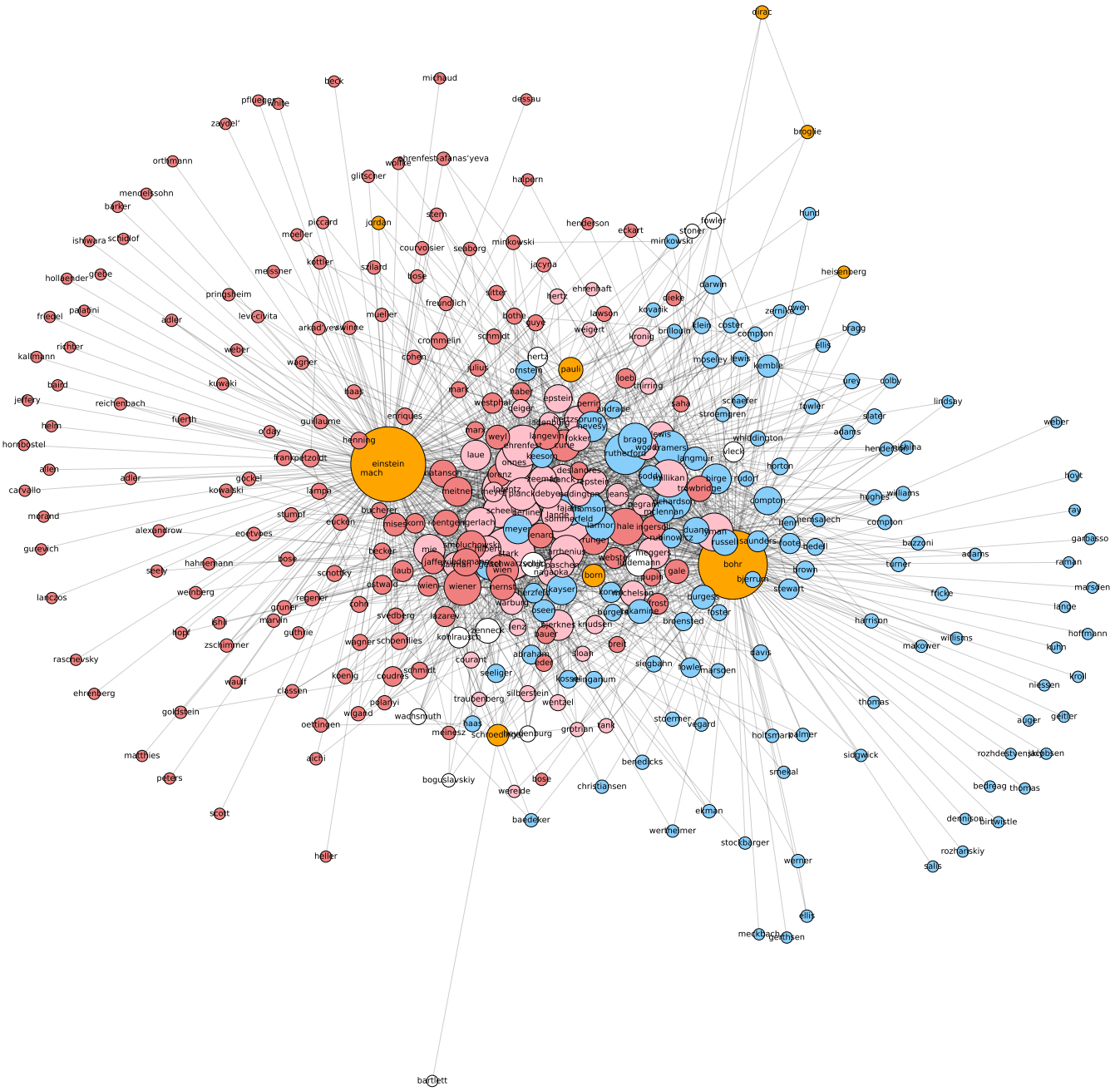


FIGURE 3. Social graph of the core and their first neighbors, 1900-1925.

Note: Undirected network. Years 1900-06, 1907-13, 1914-21, and 1922-25 are bunched together. The size of the nodes reflects the degree of a node (in or out) – the total number of connections. Colors: core (orange) vs. first neighbors. First neighbors: Bohr-only neighbors (lightskyblue), Einstein-only neighbors (coral), Bridges: Bohr- and Einstein-neighbors (pink). Spring layout algorithm, k=0.80, iterations=300. Graph built using Python networkx library.

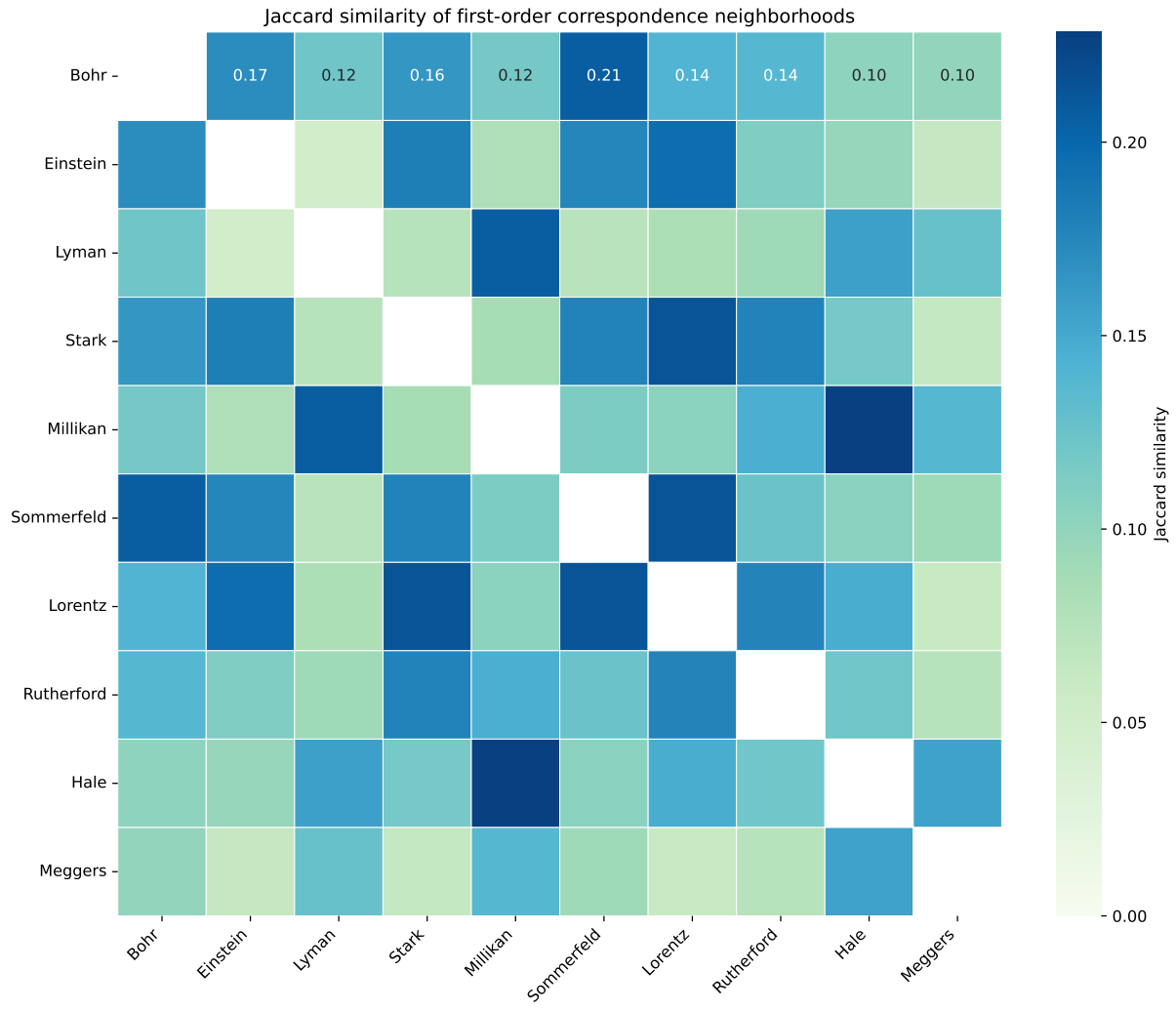


FIGURE 4. Heatmap of first neighbours overlap (Jaccard similairty).

Note: The figure reports pairwise Jaccard similarities between physicists' sets of first-order contacts in the undirected and unweighted correspondence network for 1900–1925. Jaccard similarity is defined as the number of contacts shared by two physicists divided by the total number of unique contacts belonging to either physicist's network. Darker blue indicates greater neighborhood overlap. The matrix is symmetric, and diagonal values are omitted because each physicist's contact set is perfectly similar to itself.

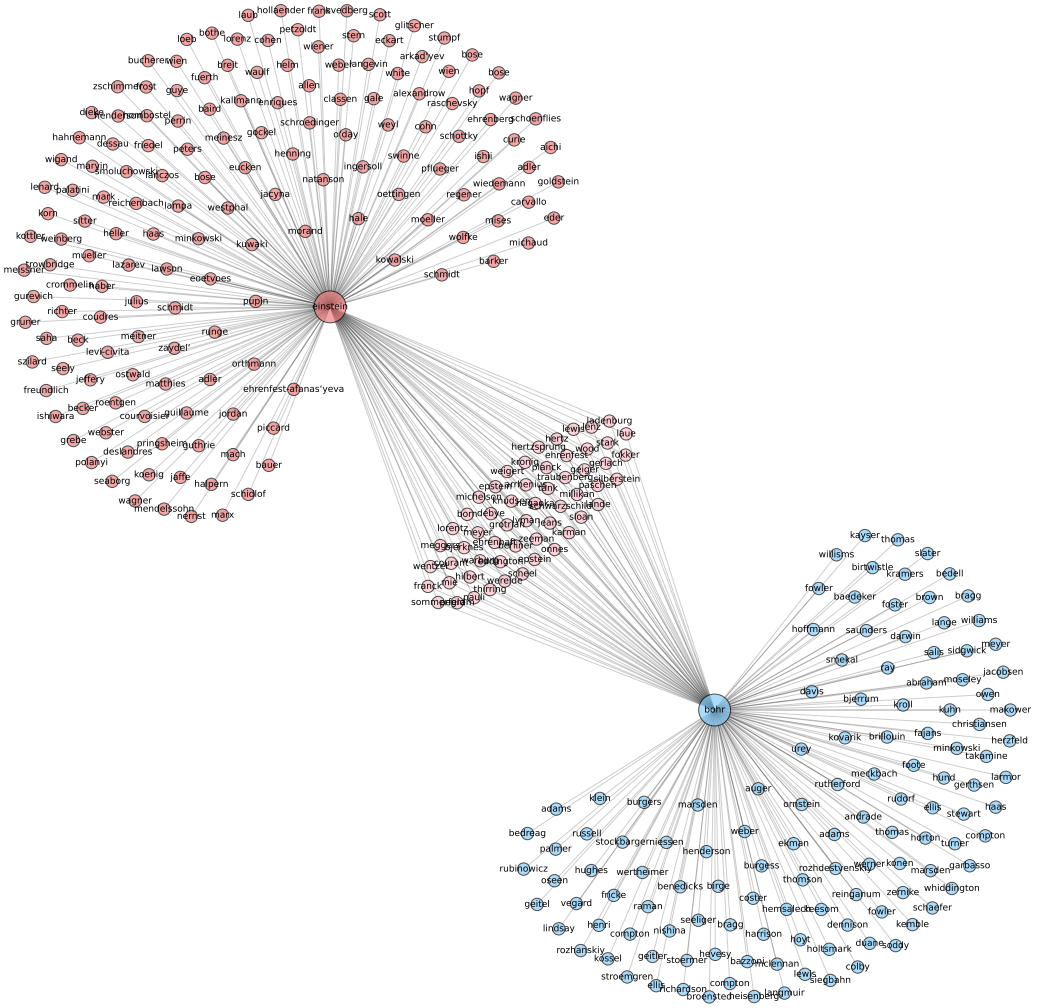


FIGURE 5. Social graph of core and their first neighbors.

Note: Undirected network. Years 1900-06, 1907-13, 1914-21, and 1922-25 are bunched together. Spring layout: k=0.35, iterations=200. Graph built using Python networkx library. Connections between the first neighbors (the second neighbors) are deliberately excluded to make accent on the overlap between the correspondents of Bohr and Einstein.

TABLE 2. Number of unique physicists by correspondence-network group, 1900-1925

Network group	**Number of unique physicists**
First neighbors	344
Second neighbors	1,085
Third neighbors	165
Fourth neighbors	8
Fifth neighbors	1
Core physicists	9
Total physicists	1,612

Notes: The core physicists are Bohr, Born, de Broglie, Dirac, Einstein, Heisenberg, Jordan, Pauli, and Schrödinger. Neighbor groups are defined according to the shortest-path distance from the core physicists in the historical correspondence network. The neighbor groups are non-overlapping.

The Figures **??** shows the cumulative years evolution of the correspondence networks of physicists (e.g., for years 1914-21, the figure depicts cumulatively edges from years 1900-06, 1907-13, and 1914-21).

TABLE 3. Top 20 physicists by alternative network-centrality measures, 1900–1925

Rank	Total degree	Betweenness centrality	Eigenvector centrality	PageRank centrality
1	Theodor L. Lyman [353] US; TT; 1874–1954	Theodor L. Lyman [0.100110] US; TT; 1874–1954	Johannes Stark [0.246392] GE; QG; 1874–1957	Johannes Stark [0.026434] GE; QG; 1874–1957
2	Johannes Stark [304] GE; QG; 1874–1957	Albert Einstein [0.074666] GE SZ US; QG; 1879–1955	Theodor L. Lyman [0.224546] US; TT; 1874–1954	Albert Einstein [0.020742] GE SZ US; QG; 1879–1955
3	Albert Einstein [293] GE SZ US; QG; 1879–1955	Niels Bohr [0.069173] DK; QG; 1885–1962	Niels Bohr [0.214892] DK; QG; 1885–1962	Theodor L. Lyman [0.018707] US; TT; 1874–1954
4	Niels Bohr [269] DK; QG; 1885–1962	Robert A. Millikan [0.063365] US; QG; 1868–1953	Robert A. Millikan [0.213846] US; QG; 1868–1953	Niels Bohr [0.016479] DK; QG; 1885–1962
5	Hendrik A. Lorentz [260] NE; QG; 1853–1928	Johannes Stark [0.054867] GE; QG; 1874–1957	Ernest Rutherford [0.209792] CA GB; NU; 1871–1937	Ernest Rutherford [0.015967] CA GB; NU; 1871–1937
6	Robert A. Millikan [250] US; QG; 1868–1953	Hendrik A. Lorentz [0.051542] NE; QG; 1853–1928	Hendrik A. Lorentz [0.203571] NE; QG; 1853–1928	Hendrik A. Lorentz [0.015071] NE; QG; 1853–1928
7	George E. Hale [243] US; AP; 1868–1938	George E. Hale [0.046475] US; AP; 1868–1938	Albert Einstein [0.197950] GE SZ US; QG; 1879–1955	Robert A. Millikan [0.013056] US; QG; 1868–1953
8	Arnold Sommerfeld [191] GE; QG; 1868–1951	Arnold Sommerfeld [0.033304] GE; QG; 1868–1951	Arnold Sommerfeld [0.184754] GE; QG; 1868–1951	Arnold Sommerfeld [0.012267] GE; QG; 1868–1951
9	Ernest Rutherford [172] CA GB; NU; 1871–1937	Otto H. Wiener [0.025947] GE; TT; 1862–1927	George E. Hale [0.176016] US; AP; 1868–1938	Joseph John Thomson [0.011175] GB; QG; 1856–1940
10	William F. Meggers [154] US; QG; 1888–1966	William F. Meggers [0.025866] US; QG; 1888–1966	Paul Ehrenfest [0.153075] AU UR NE; QG; 1880–1933	Thomas J. J. See [0.009602] US; TT; 1866–1962
11	Joseph John Thomson [148] GB; QG; 1856–1940	William Henry Bragg [0.025005] AL GB; QG; 1862–1942	Joseph John Thomson [0.139997] GB; QG; 1856–1940	Paul Ehrenfest [0.009595] AU UR NE; QG; 1880–1933
12	Paul Ehrenfest [142] AU UR NE; QG; 1880–1933	William Thomson [0.023476] GB; MA; 1824–1907	Henry Crew [0.139142] US; TT; 1859–1953	Owen W. Richardson [0.009511] GB US; QG; 1879–1959
13	Charles E. Mendenhall [137] US; QG; 1872–1935	Joseph John Thomson [0.023430] GB; QG; 1856–1940	Owen W. Richardson [0.125628] GB US; QG; 1879–1959	Joseph Larmor [0.009374] GB; QG; 1857–1942
14	William Henry Bragg [136] AL GB; QG; 1862–1942	Paul Ehrenfest [0.021285] AU UR NE; QG; 1880–1933	Heike Kamerlingh Onnes [0.122163] NE; SS; 1853–1926	George E. Hale [0.009322] US; AP; 1868–1938
15	Otto H. Wiener [135] GE; TT; 1862–1927	Owen W. Richardson [0.020893] GB US; QG; 1879–1959	Joseph Larmor [0.120232] GB; QG; 1857–1942	Svante Arrhenius [0.007997] SD; PC; 1859–1927
16	Joseph Larmor [134] GB; QG; 1857–1942	Karl T. Compton [0.020114] US; QG SS; 1887–1954	Charles E. Mendenhall [0.115711] US; QG; 1872–1935	Heike Kamerlingh Onnes [0.006715] NE; SS; 1853–1926
17	Owen W. Richardson [132] GB US; QG; 1879–1959	Joseph Larmor [0.018313] GB; QG; 1857–1942	Raymond T. Birge [0.101858] US; QG; 1887–1980	William F. Meggers [0.006638] US; QG; 1888–1966
18	Karl T. Compton [127] US; QG SS; 1887–1954	Heike Kamerlingh Onnes [0.017499] NE; SS; 1853–1926	William Henry Bragg [0.091549] AL GB; QG; 1862–1942	Raymond T. Birge [0.006617] US; QG; 1887–1980
19	Heike Kamerlingh Onnes [122] NE; SS; 1853–1926	Svante Arrhenius [0.016733] SD; PC; 1859–1927	Thomas J. J. See [0.090716] US; TT; 1866–1962	Carl Barus [0.006552] US; PC; 1856–1935
20	William Thomson [120] GB; MA; 1824–1907	Raymond T. Birge [0.015741] US; QG; 1887–1980	Carl D. T. Runge [0.090494] GE; QG; 1856–1927	Marie Sklodowska Curie [0.006423] FR; NU; 1867–1934

Note: Scientists are ranked in descending order within each centrality measure; the corresponding centrality value appears in square brackets. The second line of each entry reports country code, specialty code, and birth and death years. All measures were calculated on the directed and unweighted correspondence network for 1900–1925. Each directed correspondence tie therefore receives equal weight, and the number of letters exchanged between a pair of scientists is ignored. Total degree is the sum of in-degree and out-degree. Betweenness measures the extent to which a scientist lies on directed shortest paths; eigenvector centrality assigns higher values to scientists connected to other central scientists; and PageRank reflects the importance of incoming ties from other important scientists.

TABLE 4. Network coverage from Bohr and Einstein across correspondence periods

Period	Core physicist	All nodes	Reached nodes	Share reached	Unreached nodes
1900–1906	Bohr	609	583	0.957	26
1900–1906	Einstein	609	588	0.966	21
1907–1913	Bohr	836	821	0.982	15
1907–1913	Einstein	836	820	0.981	16
1914–1921	Bohr	989	980	0.991	9
1914–1921	Einstein	989	980	0.991	9
1922–1925	Bohr	954	934	0.979	20
1922–1925	Einstein	954	934	0.979	20
1914–1925	Bohr	1,325	1,312	0.990	13
1914–1925	Einstein	1,325	1,312	0.990	13
1900–1925	Bohr	1,626	1,612	0.991	14
1900–1925	Einstein	1,626	1,612	0.991	14

Notes: Reached nodes are physicists connected to the indicated core physicist through the correspondence network within the respective period, including successive network neighbors up to the fifth order. Share reached is the number of reached nodes divided by the total number of nodes present in the corresponding period. The network is treated as undirected.

TABLE 5. Direct-neighborhood comparison of Bohr and Einstein, 1900–1925

Network measure	Value
Bohr direct neighbors	181
Einstein direct neighbors	214
Shared direct neighbors	56
Bohr-only direct neighbors	125
Einstein-only direct neighbors	158
Jaccard similarity	0.165

Notes: Direct neighbors are physicists connected by at least one correspondence link to the respective scientist. Shared neighbors are directly connected to both Bohr and Einstein. Bohr-only and Einstein-only neighbors are directly connected to only one of the two scientists. Jaccard similarity is defined as the number of shared neighbors divided by the number of physicists in the union of the two direct neighborhoods; it ranges from 0 to 1. The network is treated as undirected.

TABLE 6. Bohr's strongest direct correspondence connections, 1900–1925

Physicist	ID	Specialty	Country	Letters
Paul Sigismondovich Ehrenfest	1289	QG	AU UR NE	89
James Franck	1536	QG	GE US	88
Wolfgang Pauli Jr.	3726	QG	AU SZ US	78
Gyorg von Hevesy	2133	PC	HU DK GE	77
Oskar Benjamin Klein	2629	QG	SD	67
Owen Willans Richardson	4053	QG	GB US	62
Ernest Rutherford	4226	NU	CA GB	60
Karl Friedrich F. C. Scheel	4316	DS	GE	53
Adalbert Wojciech Rubinowicz	4189	QG	PO	49
Alfred Lande	2839	QG	GE US	43
Hendrik Anthony Kramers	2736	QG	DK NE	41
Carl Wilhelm Oseen	3657	QG	SD	38
John Cunningham McLennan	3259	NU	CA GB	30
Hendrik Antoon Lorentz	3040	QG	NE	29
Alfred Fowler	1516	AP	GB	29
Max Karl Ernst Ludwig Planck	3830	QG	GE	26
Sophus Th. H. Weber	5263	TT	NE	22
Theodor L. Lyman	3082	TT	US	22
Ralph Howard Fowler	1521	QG	GB	19
Werner Heisenberg	2075	QG	GE	16

Notes: The table reports Niels Bohr's twenty direct neighbors with the largest total number of letters exchanged over 1900–1925. Letters are aggregated across all correspondence periods and across both directions of communication. Specialty and country codes follow the classifications used in the physicist-level data.

TABLE 7. Einstein's strongest direct correspondence connections, 1900–1925

Physicist	ID	Specialty	Country	Letters
Paul Sigismondovich Ehrenfest	1289	QG	AU UR NE	267
Hendrik Antoon Lorentz	3040	QG	NE	173
Arnold Johannes Wilhelm Sommerfeld	4670	QG	GE	83
Erwin Finlay Freundlich	1567	AP	GE	79
Hermann Weyl	5330	MA	SZ US	76
Jakob Johann Laub	2880	QG	AR GE	67
Max Karl Ernst Ludwig Planck	3830	QG	GE	51
Wilhelm Carl Werner Otto F. F. Wien	5366	QG	GE	42
Willem de Sitter	4572	AP	NE	41
Fritz Haber	1939	PC	GE	40
David Hilbert	2151	MA	GE	38
Max Theodor Felix von Laue	2882	QG	GE	38
Paul Langevin	2852	QG	FR	37
George Ellery Hale	1957	AP	US	34
Gustav Mie	3343	QG	GE	30
Max Born	548	QG	GE	27
Robert William Lawson	2893	NU	GB	27
Arnold Berliner	408	DS	GE	25
Tullio Levi-Civita	2944	MA	IT	23
Robert Andrews Millikan	3365	QG	US	22

Notes: The table reports Albert Einstein's twenty direct neighbors with the largest total number of letters exchanged over 1900–1925. Letters are aggregated across all correspondence periods and across both directions of communication. Specialty and country codes follow the classifications used in the physicist-level data.

TABLE 8. Physicists in the Bohr and Einstein correspondence neighborhoods, 1900–1925

	Mutually exclusive categories			Group totals		Unique
Network order	Bohr only	Einstein only	Shared	All Bohr	All Einstein	total
First order	125	158	56	181	214	339
Second order	344	269	751	1,095	1,020	1,364
Third order	152	193	168	320	361	513
Fourth order	6	7	8	14	15	21
Fifth order	0	0	1	1	1	1

Notes: Network order is measured separately from Bohr and Einstein in the undirected historical correspondence network. "Shared" denotes physicists appearing in both leaders' neighborhood sets at the given order. "Bohr only" and "Einstein only" exclude these shared physicists. All Bohr equals Bohr only plus shared; all Einstein equals Einstein only plus shared. The unique total is the union of the Bohr and Einstein sets within each row. Because distance is measured separately from each leader, a physicist may appear at different orders from Bohr and Einstein; consequently, the unique totals should not be summed across rows.

TABLE 9. Jaccard similarity between physicists' first-order correspondence neighborhoods, 1900–1925

Physicist pair	Jaccard similarity
Bohr–Einstein	0.1711
Bohr–Lyman	0.1222
Bohr–Stark	0.1636
Bohr–Millikan	0.1184
Bohr–Sommerfeld	0.2066
Bohr–Lorentz	0.1416
Bohr–Rutherford	0.1383
Bohr–Hale	0.1034
Bohr–Meggers	0.0992
Einstein–Lyman	0.0505
Einstein–Stark	0.1820
Einstein–Millikan	0.0812
Einstein–Sommerfeld	0.1765
Einstein–Lorentz	0.1955
Einstein–Rutherford	0.1121
Einstein–Hale	0.0969
Einstein–Meggers	0.0629
Lyman–Stark	0.0737
Lyman–Millikan	0.2063
Lyman–Sommerfeld	0.0731
Lyman–Lorentz	0.0821
Lyman–Rutherford	0.0914
Lyman–Hale	0.1563
Lyman–Meggers	0.1271
Stark–Millikan	0.0859
Stark–Sommerfeld	0.1778
Stark–Lorentz	0.2134
Stark–Rutherford	0.1776
Stark–Hale	0.1172
Stark–Meggers	0.0641
Millikan–Sommerfeld	0.1140
Millikan–Lorentz	0.1042
Millikan–Rutherford	0.1458
Millikan–Hale	0.2285
Millikan–Meggers	0.1379
Sommerfeld–Lorentz	0.2136
Sommerfeld–Rutherford	0.1241
Sommerfeld–Hale	0.1053
Sommerfeld–Meggers	0.0913
Lorentz–Rutherford	0.1774
Lorentz–Hale	0.1467
Lorentz–Meggers	0.0598
Rutherford–Hale	0.1213
Rutherford–Meggers	0.0734
Hale–Meggers	0.1548

Notes: Each neighborhood contains the unique first-order contacts of the respective physicist in the undirected and unweighted historical correspondence network. For physicists i and j, Jaccard similarity is calculated as $J_{ij} = |N_i \cap N_j|/|N_i \cup N_j|$, where N_i and N_j denote their respective sets of direct contacts. Higher values indicate greater overlap between the two correspondence neighborhoods.

5. Identification Strategy

The empirical analysis asks whether scientists with pre-existing access to the correspondence networks of the founders of quantum mechanics experienced a differential change in research output after the breakthrough of 1925 than otherwise comparable scientists without such access. To estimate the differential change, I use the Doubly Robust Difference-in-Difference estimator based on Sant'Anna and Zhao (2020) which allows a more flexible, non-parametric, way to account for covariates than a TWFE OLS model. For estimating the effect over years, I use an event study building on the machinery of Callaway and Sant'Anna (2021).

5.1. The Main Econometric Specification

The preferred estimator is Doubly Robust Inverse Propensity Weighting Difference-in-Differences estimator based on stabilized inverse probability weighting and ordinary least squares following Sant'Anna and Zhao (2020).[8] Below I provide a brief explanation of the estimator.

Let X_i collect the pre-treatment covariates and let $p(X_i) = \Pr(D_i = 1 \mid X_i)$ denote the propensity score with $D_i = 1$ being the indicator of the treatment group. In the inverse-probability-weighted estimator, treated scientists receive unit weight and comparison scientists are reweighted by the odds $p(X_i)/[1 - p(X_i)]$, with the weights normalized in estimation. This makes the weighted distribution of observed pre-treatment characteristics in the comparison group resemble that of the connected group, provided that the two groups have adequate common support. The doubly robust estimator combines this propensity-score model with an outcome-regression model; under the regularity conditions in Sant'Anna and Zhao (2020), it remains consistent when either nuisance model is correctly specified. Double robustness does not, however, remove the need for conditional parallel trends, overlap, and valid treatment timing.

I use a rich set of scientist-level covariates entering the propensity score model that could influence the researcher productivity in my setting. It is important to note that in the estimator, one must use observables measured *before* the treatment. Hence, I use the following pre-1925 scientist level observables[9]: specialty dummy(ies), country of

[8] The estimator is also known as the Augmented Inverse Propensity Weighting (AIPW) Difference-in-Differences.

[9] These are predominantly measured on the pre-1925 period; though specialty and country of work are the variables that I take from Wheaton (1993) which were coded by the historians based on all works of scientists. It is possible that some specialties and countries were selected on post-1925 works. See

work dummy(ies), first year of publication, scientist's experience defined as the 2-nd degree polynomial of the number of years passed since the first year of publication to 1925; an incumbent group indicator equal to 1 if scientist appears in the pre-1925 correspondence networks (exchanged any letters), and 0 otherwise; node characteristics from the network (logarithm of degree, betweenness centrality, eigenvector centrality, page-rank centrality)[10]. Also, I include a series of dummy variables (not mutually exclusive) that are equal to 1 if a direct correspondnece link was established with the following physicists pre-1925: Rutherford, Planck, Sommerfeld, and Einstein.[11]

The preferred dynamic analysis follows the conditional difference-in-differences framework of Callaway and Sant'Anna (2021) with important notice: my design is non-staggered (treatment – the Quantum Mechanics breakthrough – happens for everyone in the same year).[12] To display dynamics, I estimate event-time effects relative to 1924:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{e \in \mathcal{E},\, e \neq -1} \beta_e D_i \mathbf{1}\{t - 1925 = e\} + \varepsilon_{it}, \tag{3}$$

where $e = -1$ is the omitted reference year. In the weighted implementation, β_e is reported as the event-time average treatment effect for the connected group. The pre-period coefficients summarize differential changes relative to 1924; the post-period coefficients trace the evolution of the connected-group advantage after the breakthrough. The event-study figures report wild-bootstrap confidence intervals clustered at the scientist level.

5.2. Identifying Assumptions

Conditional parallel trends. The key identifying assumption is that conditional on observed pre-1925 characteristics (balanced via Doubly Robust weights), connected and unconnected physicists would have experienced the same trends in publications and citations after 1925, *had the Quantum Mechanics breakthrough not occurred in 1925*. There-

Appendix for discussion of this potential issue.

[10] These caracteristics are defined only for incumbent scientists, i.e. those who are observed in the pre-1925 correspondence networks.

[11] Rutherford and Planck were important pioneers of the older quantum physics; Sommerfeld was a leader of quantum physics center in Munich, while Einstein was an influential figure at the quantum physics center in Berlin.

[12] Callaway and Sant'Anna (2021) estimator is primarily designed for identification in staggered treatment design. Though, I use the machinery of CS 2021 as it is useful for producing event stuides. It is possible to choose a *never treated* option in the method to build the event study for the non-staggered design.

fore, the identification exploits the unexpected timing of the discovery and development of quantum mechanics in 1925–1927 as a common shock to the physics profession.

In the event study, the conditional parallel trends assumption states that, for every post-treatment year t,

$$\mathbb{E}\left[Y_{it}(0) - Y_{i,1924}(0) \mid D_i = 1, X_i\right] = \mathbb{E}\left[Y_{it}(0) - Y_{i,1924}(0) \mid D_i = 0, X_i\right]. \tag{4}$$

This states that, after conditioning on observed pre-1925 characteristics, connected and unconnected scientists would have experienced the same change in the outcome in the absence of the differential advantage supplied by direct founder access. Reweighting can improve balance in observed characteristics; it cannot "engineer" this counterfactual assumption or eliminate selection on unobserved time-varying determinants of productivity. The large unweighted differences in pre-1925 output documented below make overlap and weighted balance substantive diagnostics rather than mechanical preliminaries.

For identification of the causal effect of the interest, three additional conditions matter.

Common support. First, common support requires $0 < p(X_i) < 1$ for the treated population of interest; the distribution of propensity scores and the largest weights should therefore be reported. Second, no anticipation requires that scientists did not change their output before 1925 because they foresaw the breakthrough. The historical abruptness of matrix mechanics supports this assumption, but the timing of the correspondence measure must also be strictly pre-treatment. Third, the conventional estimator assumes no consequential interference across scientists. That condition is demanding in a correspondence network: an unconnected scientist may be indirectly exposed through a connected colleague. Methods for spillovers show that ignoring such interference can combine direct and indirect effects in nontransparent ways (Vazquez-Bare 2023). Accordingly, the estimates below should be read as the conditional differential change associated with a direct founder link, relative to scientists without such a link, and not as a pure individual-level treatment effect free of network diffusion.

5.2.1. Pre-Treatment Scientist Observables

Both in TWFE-OLS and Doubly Robust IPW DID models, I use the following pre-treatment time-invariant covariates.

Baseline covariates. *First year of publication* (inferred from articles published during

1900–1945 on a set of journals selected for the study); third degree polynomial of *experience* (number of years since the first year of publication until 1925); *physics specialty codes* (mutually non-exclusive indicators of physics specialties equal to 1 if a scientist has ever worked in the specialty, and 0 otherwise); *country codes* (mutually non-exclusive indicators of main countries of work of a physicist equal to 1 if the scientist has ever worked in the country, and 0 otherwise).

Pre-1925 research productivity. *Average slope of publications and citaitons* calculated on 1920–1924; *publication and citation levels* as of 1920[13]; and other scientist's research productivity measures aggregated over 1900–1924: mean of publications, mean of citations, and the 5th, 25th, 50th, 75th, and 95th percentiles of publications' and citations' respective distributions.

Pre-1925 network node characteristics. *Network centrality measures* (log of total degree, betweenness, PageRank, eigenvector, and Katz–Bonacich centralities, all calculated using the pre-1925 correspondence network); *indicators of pre-existing connections to Sommerfeld* and *Einstein* (equal to 1 if connected to the respective scientist); and two *measures of indirect exposure to Bohr and Einstein*:

$$J_i = \frac{\text{Number of } i\text{'s neighbors directly connected to Bohr}}{\text{Total number of } i\text{'s neighbors}}$$

and

$$K_i = \frac{\text{Number of } i\text{'s neighbors directly connected to Einstein}}{\text{Total number of } i\text{'s neighbors}}.$$

These two measures capture the shares of scientist i's neighbors who are directly connected to Bohr or Einstein, respectively, in the pre-1925 correspondence network. Both range from 0 to 1 for scientists with at least one neighbor.

All covariates are measured on the individual scientist level and on the pre-treatment period. The covariates are added to the TWFE-OLS model by interacting them with the post or year dummies (depending on specificaiton); and to the propensity score model to predict the probability of treatment (connection with the founders) when using the Doubly Robust IPW Difference-in-Differences estimator.

[13] For robustness, I also use other publication year levels in alternative specifications.

6. Main Results

This section presents the main estimation results of the effect of pre-existing connection (via correspondence) with the founders of quantum mechanics on publication and citation outcomes of connected scientists. Baseline treatment is defined as follows: scientist is treated if he has any amount of correspondence exchanged with any of the founders in years roughly from 1900 to 1925[14]. Observations for the founders, Sommerfeld and Einstein are removed from the estimation panel. For identification of the causal effect of connectedness with the founders, I prefer to use the Doubly Robust Inverse Propensity Weighting Difference-in-Differences estimator from SZ 2020, and I equip the machinery of CS 2021 to estimate the event study coefficients (see section 5 for details). I present main results in the form of event study figures, though I also report tables with estimated average effects.

6.1. The Effect of Preexisting Connection to Founders on Research Productivity

Below, I provide event study results and average treatment effect estimates. I start with a simple difference-in-differences estimator – the two-way fixed effects model, and then move to the preferred estimation method, the Doubly Robust IPW Difference-in-Differences estimator from SZ 2020 and CS 2021.

6.1.1. TWFE OLS Estimates

The purpose of this section is to show that the main results hold under a less robust model where conditional parallel trends assumption is more likely to be violated. Covariates used in the TWFE models are time-invariant, scientist-level and measured on pre-1925 period. Therefore, to include them in the regression, I interact each covariate with either the post indicator or year dummies, depending on specification. All TWFE OLS event study results are reported in the Appendix A.

I estimate three TWFE specifications which are provided in the equations 5, 6 and 7 below.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K},\, k \neq -1} \beta_k \left(\text{Connected}_i \times \mathbb{1}\{t - 1925 = k\}\right) + \left(\mathbf{X}_i' \boldsymbol{\gamma}\right) \times \text{Post}_t + u_{it}. \tag{5}$$

[14] Hence, the treatment captures the extensive margin of correspondence connection.

$$(6)\quad Y_{it} = \mu_i + \tau_t + \sum_{k \in \mathcal{K},\, k \neq -1} \theta_k \left(\text{Connected}_i \times \mathbb{1}\{t - 1925 = k\}\right) + \mathbf{Z}_i' \boldsymbol{\delta}_t + \left(\mathbf{W}_i' \boldsymbol{\pi}\right) \times \text{Post}_t + \nu_{it}.$$

$$(7)\quad Y_{it} = \eta_i + \xi_t + \sum_{k \in \mathcal{K},\, k \neq -1} \kappa_k \left(\text{Connected}_i \times \mathbb{1}\{t - 1925 = k\}\right) + \mathbf{V}_i' \boldsymbol{\rho}_t + \left(\mathbf{U}_i' \boldsymbol{\phi}\right) \times \text{Post}_t + \varepsilon_{it}.$$

Y_{it} is the citations outcome for scientist i in year t. The terms α_i, μ_i, and η_i denote scientist fixed effects, while λ_t, τ_t, and ξ_t denote year fixed effects in the respective models, and u_{it}, ν_{it}, and ε_{it} are the error terms. Standard errors are clustered at the scientist level in all three models. The indicator Connected_i identifies treated scientists (those with preexisting connection to the founders). The indicator $\text{Post}_t = \mathbb{1}\{t \geq 1925\}$ denotes the post-treatment period. The coefficients β_k, θ_k, and κ_k are the corresponding event-study coefficients that capture the treated–control difference at event time k, relative to the omitted year $k = -1$, which is 1924. The vectors $\mathbf{X}_i$, $\mathbf{Z}_i$, $\mathbf{W}_i$, $\mathbf{V}_i$, and $\mathbf{U}_i$ in respective models are the vectors of baseline characteristics which are explained in detail below. The coefficients $\boldsymbol{\gamma}$, $\boldsymbol{\pi}$ and $\boldsymbol{\phi}$ in respective models allow the pre-to-post change in the outcome to differ with scientists' baseline characteristics. These coefficients impose a common adjustment across all post-treatment years. Coefficients $\boldsymbol{\delta}_t$ and $\boldsymbol{\rho}_t$ in respective models vary by calendar year.

In the first model (equation (5)), $\mathbf{X}_i$ contains first publication year, specialty and country indicators, experience and network variables, slope of publications and slope of citations measured over 1920–1924, and the extended science covariates measured over 1915–1924: mean and the 5th, 25th, 50th, and 75th quantiles of publications and citations. All controls are interacted with Post_t.

Figure A1 presents the event study estimates for the first model. There is an obvious problem with the pre-trend, as pre-treatment coefficients are not statistically different from zero. The rising trend indicates that the connected group was getting more citations over years before the treatment. The results cannot be interpreted as causal since the conditional parallel trends assumption is violated.

Therefore, it might be helpful to account for the pre-treatment treated-control differences in publication and citation patterns in a more flexible way. In the second model (equation (6)), $\mathbf{Z}_i$ contains first publication year and specialty and country indicators, with coefficients that vary by calendar year. The vector $\mathbf{W}_i$ contains experience and

network variables, the slope of publications and the slope of citaitons, and the extended science covariates, each interacted with Post_t.

Figure A2 presents the event study for the second model. The estimates become better, with no-prends for relative time -2 to -4 (confidence intervals overlap with zero) though the rest of the pre-treatment coefficients remain statistically different from zero. Hence, allowing the pre-treatment treated-control differences in first pulication year and specialty and country indicators to vary by year does improve the estimation though not enough to establish the causality as pre-trends are not fully eliminated. For parallel trends assumption to plausibly hold, one needs to add in the regression science covariates *interacted with year dummies*. This should give the model enough flexibility, although the variation in the obsevations will decrease.

In the third and preferred TWFE model (equation (7)), $\mathbf{V}_i$ contains the slope of publications and the slope of citaitons, and the extended science covariates, with coefficients that vary by calendar year. The vector $\mathbf{U}_i$ contains first publication year, specialty and country indicators, and experience and network variables, each interacted with Post_t.

The event study estimates from the third model are reported in Figure 6 below. Once allowing treated-control differences in science covariates to vary by years, all the pre-treatment coefficients become statistically non-different from zero. The conditional parallel trends assumption seem to hold, giving credibility to the causal nature of the difference-in-differences estimates.

Overall, the TWFE-OLS event study estimates show that once flexibly controlling for baseline covariates, one can satisfy the no pre-trends assumtion and estimate the causal effect of connectedness with the founders on citaions. The effect is positive, quite large in magnitude and statistically significant. In the further sebsection, I move to the more sophisticaed estimator – the Doubly Robust IPW Difference-in-Differences estimator – to show that when pre-treatment baseline covariates are accounted for in a non-parametric way, the main results still hold and are similar to the TWFE OLS estimates.

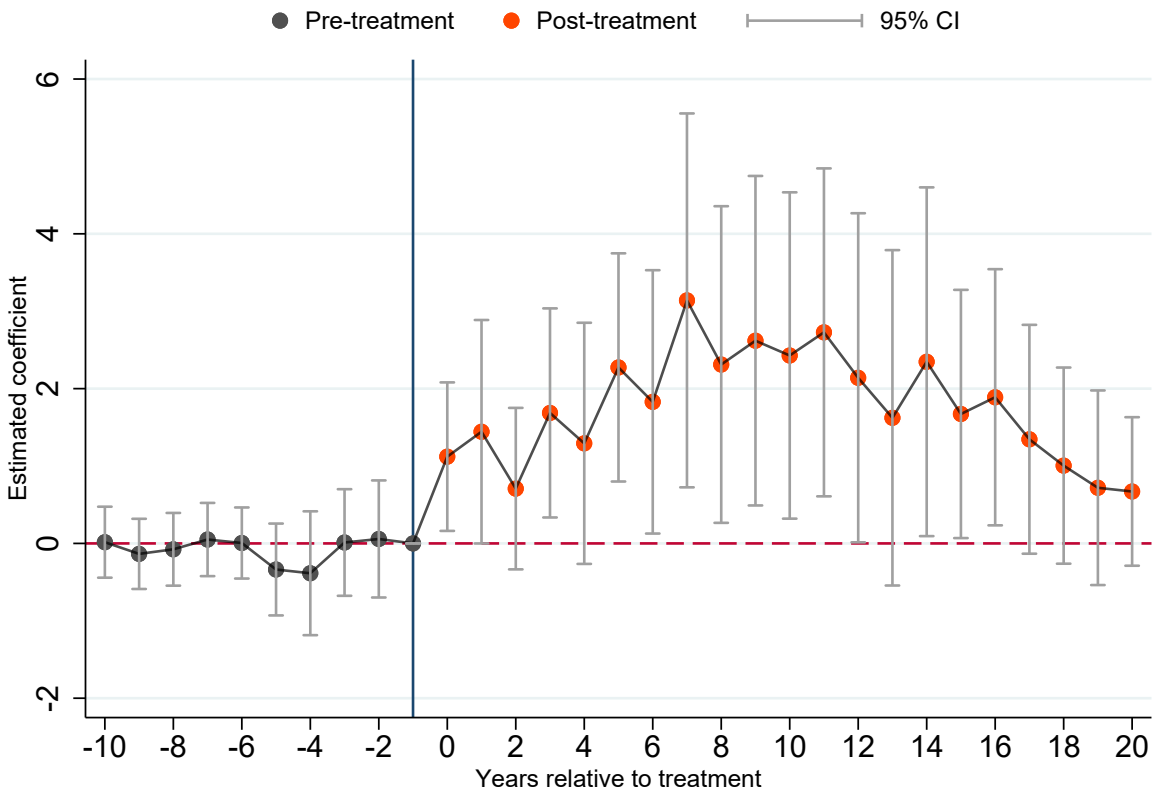


FIGURE 6. Baseline TWFE event-study estimates for citations.

Note: The figure reports event-time average treatment effects from a baseline TWFE model. The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

6.1.2. Doubly Robust IPW Difference-in-Differences Estimates

Figure 7 shows the estimated event study coefficients for the specification including the baseline scientist covariates, their pre-1925 research productivity covariates, and pre-1925 correspondence network node characteristics. The dependent variable as annual citations. For each calendar year, the outcome reports the amount of citations coming from all sources covered in Web of Science to all articles of a scientist published by the calendar year.[15]

From the Figure 7 it can be seen that there is a positive and large in magnitude (more than 50% change) statistically significant effect on citations of the connected group. Although there is especially large drop in the relative year 2 (corresponds to 1927), the effect is overall significant and persistent for roughly twelve years since 1925, and then gradually decreasing and becoming insignificant.

In the Appendix A, I report DRIPW Difference-in-Differences event studies for two other outcomes: number of publications (adjusted by the number of co-authors per publication) and probability of publishing (indicator equal to 1 if scientist i publishes at least one article in year t). It seems that connected scientists did not publish differentially more articles than the unconnected scienitsts after the breakthrough. Though, the estimation results are imprecise due to the pre-treatment coefficients not being close to zero: there seems to be issues with the pre-trend.

[15] Hence, after 1925, the outcome reports citations both to older and novel works. Furhter, I analyse citations to the older and novel works separately.

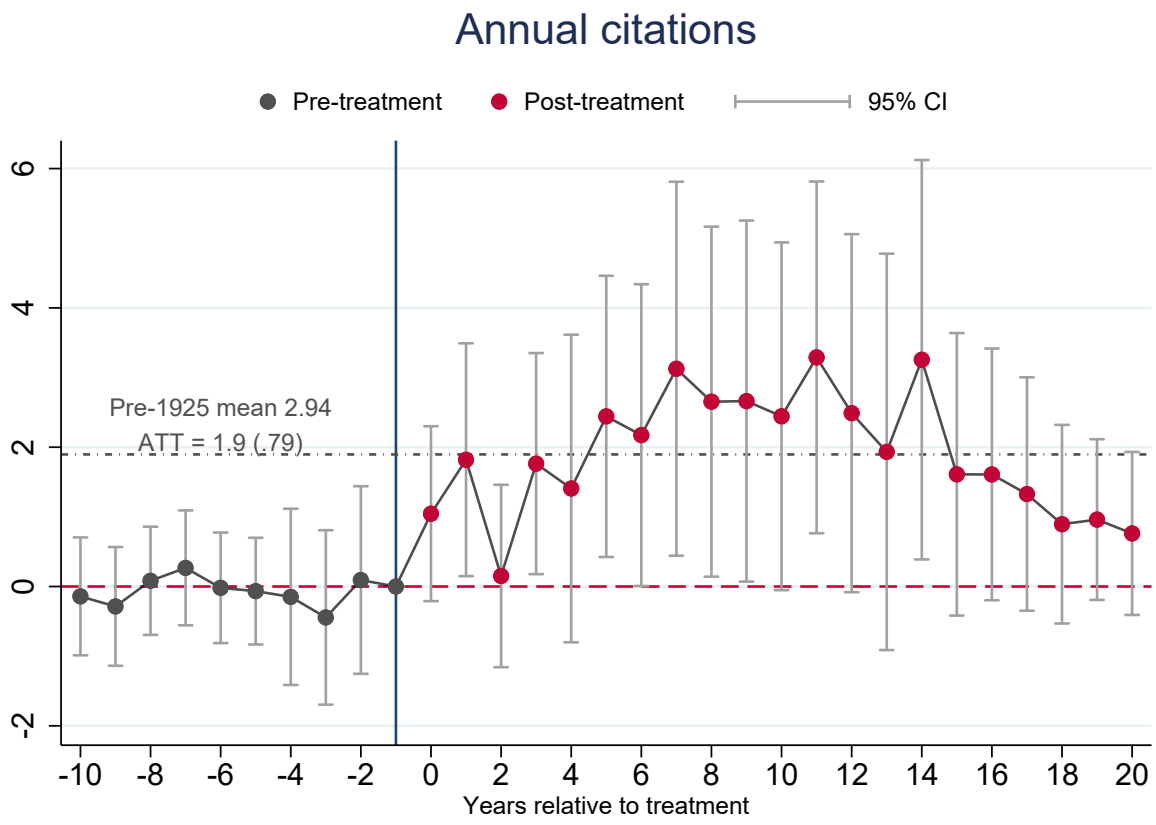


FIGURE 7. Baseline doubly robust inverse propensity weighting event-study estimates for citations.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

6.2. ATT of the Preexisting Connection to Founders: Various Specifications

Table ... presents results of the estimation of ATT – average treatment effect of the connection to founders – across three different specifications. The estimator is still the Augmented IPW Diff-in-Diff estimator, though the set of covarites used varies across specifications.

The first specification is using the set of only baseline covarites for the propenstiy score model: first year of publication, experience polynomial, physics specialty variables and country of work variables. There is one more indicator variable added to the first specification: the "incumbent" status. The indicator of the "incumbent" status of a scientist which is equal to 1 if the scientist is observed in the pre-1925 correspondence networks, and 0 otherwise[16].

The reason to include the last control variable, the incumbent status, is that in my panel, I observe physicists who already published before 1925 but whom I do *not* observe in the correspondence networks. Hence, it might be important to distinguish scientists that were involved in the pre-existing correspondence networks from scientists that were active in the profession but *did not* participate in the network.[17]

The second specification adds to the baseline covariates the extended research productivity covariates: average slope of publications and citaitons calculated on 1920–1924; publication and citation levels as of 1920; and the mean of publications (adjusted by the number of co-authors), mean of citations, and their 5th, 25th, 50th, 75th, and 95th percentiles. Including this covariates should be importnat for accounting for the pre-trend in the outcome because the treated group was not only producing more publications and getting more citations pre-1925, they were on a rising trend compared to the control group.

Finally, the third, most complete specification adds to the already used covariates the node characteristics from the pre-1925 correspondence networks: mainly, centrality measures, indeces of a direct exposure of one's neighbors to Bohr and Einstein, and

[16] More precisely, it is an indicator of belonging to the giant component of Bohr and Einstein in the network. The giant component of Bohr and Einstein is a sub-network starting from the two nodes – Bohr and Einstein – and expanding up to the fifth order of the neighborhood. When expanded to the the fifth order neighborhood, the pre-1925 correspondence network reaches almost all nodes in the network, which essentially makes the first-to-fifth order neighborhood sub-network almost a whole original network.

[17] Though, it is possbile that due to the selection bias the scientists are simply omitted in the correspondence catalog. It is likely that for some scientists not all letters were recorded in the catalog, though I assume that it is unlikely that there are many physicists for whom even a single letter was not recorded in the catalogue for the period of roughly twenty five years (1900–1925).

indicators of direct preexisting link to Sommerfeld and Einstein. It is important to note that these covariates are defined only for scientists in the giant component of the network, i.e. those who appear in the correspondence data. Hence, the regression specification will run only on a subsample of physicists who appeared in the pre-1925 correspondence networks, dropping scientist who were active in the profession but were not observed in the networks.

7. Robustness of the Main Results

7.1. Alternative Estimation Periods

I estimate the event study coefficients using preferred Doubly Robust IPW Difference-in-Differences estimator for alternative publishing year windows. Figure A10 and Figure A11 in the Appendix B present the results. Figure A10 shows the estimation results for the main outcome annual citations (to all published works, i.e. to older and novel works together) for the time window from 1919 to 1939 (starting after the War World I and before the War World II); and Figure A11 shows the estimation results for the time window from 1919 to 1932 (after the War World I and before the arrival of the Nazi to the power in Germany).[18]

Essentially, the direction and significance of the estimated effect on citations remains unchanged: it is large and significant. However, the magnitued now becomes even larger for the narrower estimation time windows. The ATT for estimation window 1919–1939 is 2.385331, which is larger in absolute terms than ATT for the baseline window 1915–1945 (1.896351). However, when measured as the share of the pre-ptreatment mean, it is essentially the same: 64.579098% for 1915–1945, and 64.191981% for 1919–1939. For 1919–1932 estimation window, the effect is lower in absolute terms (2.004156) and slightly lower in realtive terms (53.934126% of the pre-treatment mean) compared to the baseline estimation period results (1915–1945).

7.2. Dropping Correspondence with Founders in Years 1922–1925

Dropping correspondence in the years right before the arrival of matrix mechanics in 1925 (1922–1925) may be an important robustness check to possibly eliminate

[18]Waldinger studied the negative effects on publication activity of scientists staying in Germany after the expulsion of Jewish origin and "politically unreliable" scientists from German universities. Importantly, for each time window, an additional restriction is applied to scientist-year observations: the first year of publication must be always before or equal to the end of the window interval.

those scientist who connected to founders right on the eve of the discovery of matrix mechanics, possibly anticipating the event. This robustness check partly addresses possible violation of the no-anticipation assumption of the Doubly Robust Difference-in-Differences estimator. Although scientists could anticipate the general course of events towards the new mechanics, I argue that they were not anticipating the *pace* of the events related to the arrival of matrix mechanics. Hence, I eliminate those scientists who established correspondence connections with the founders only in 1922–1925. Table ... shows the list of such physicists.

7.3. Weighted event-study evidence

The inverse-probability-weighted event study for log(1 + twenty-five-year citations) yields pre-1925 estimates centered near zero, while every displayed post-1925 estimate is positive. The post-treatment profile rises in the first years after the breakthrough, is largest around event years three and four, and remains positive through 1932. The doubly robust specification produces a similar pattern. This stability across IPW and doubly robust estimators indicates that the dynamic result is not driven solely by one choice of nuisance model.

Two qualifications are important. First, the confidence intervals are wide at several horizons, so the year-to-year movements should not be overinterpreted. The event study is most informative about a sustained positive post-1925 pattern rather than the exact timing of a peak. Second, because the twenty-five-year citation window extends beyond the publication year, its pre-period coefficients cannot establish the absence of anticipatory behavior. They can partly capture the post-1925 recognition of papers published before the breakthrough. The cleaner evidence on pre-treatment trends must come from publication incidence, publication counts, and fractional authorship.

7.4. Event Study Diff-in-Diffs with IPW Results

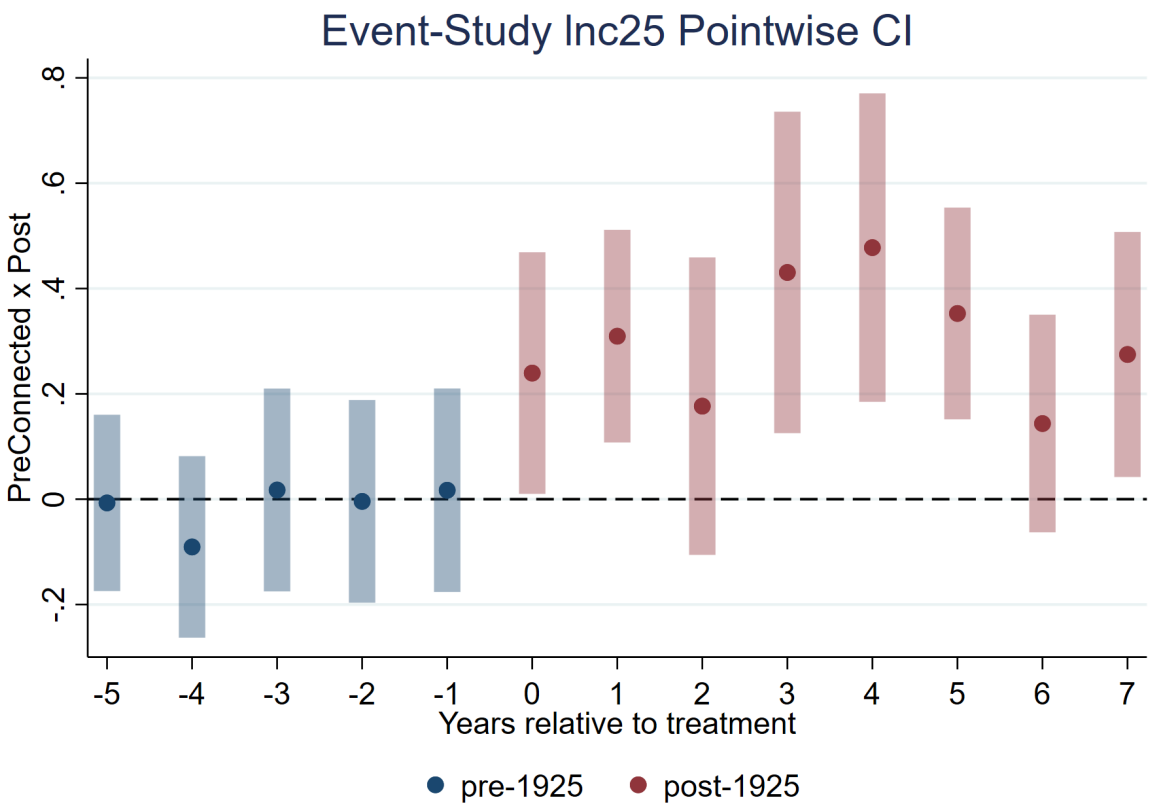


FIGURE 8. Inverse-probability-weighted event-study estimates for 25-year citations

Note: The figure reports event-time average treatment effects from an inverse-probability-weighted difference-in-differences estimator. The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the logarithm of one plus citation-weighted publications, with forward citations counted over the first 25 years after publication. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. Blue markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors; the dashed horizontal line denotes zero. The control group is reweighted using propensity scores based on pre-1925 mean publication share, probability of publishing, three- and five-year forward citations, scientific specialty, principal country of work, birth year, and first publication year. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

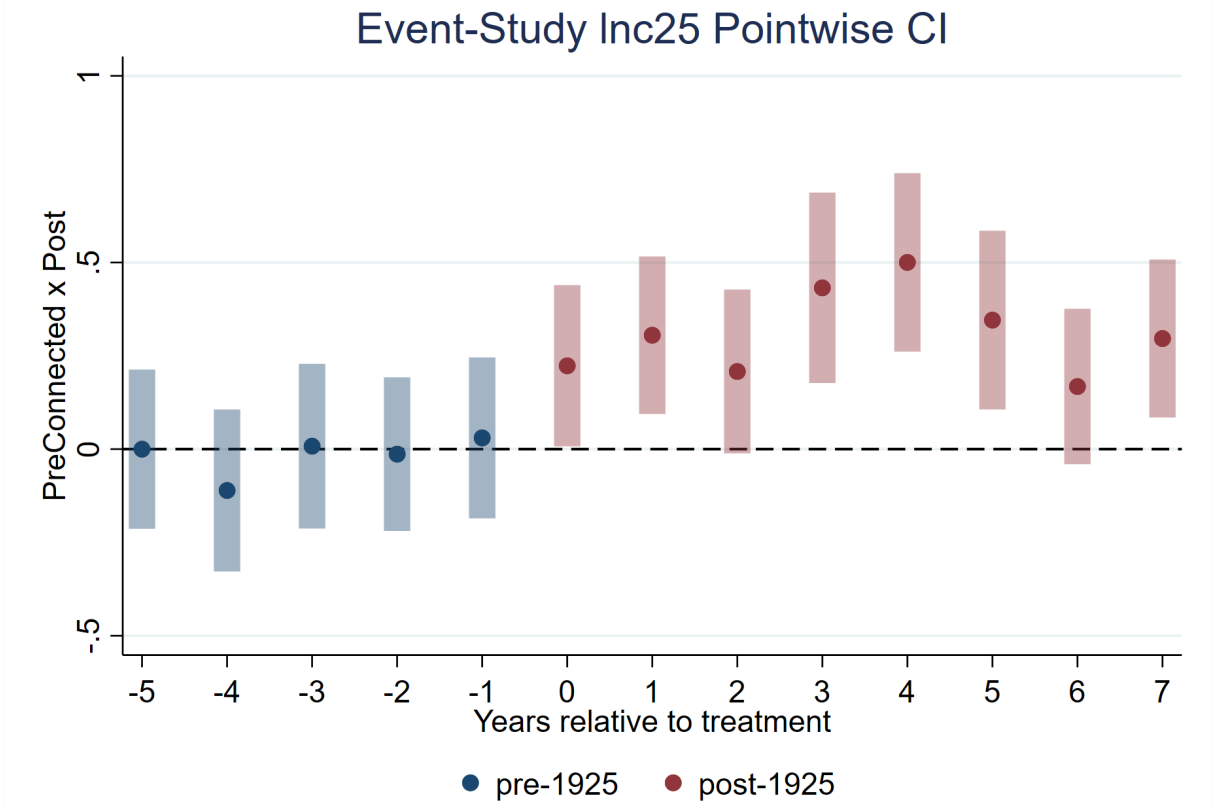


FIGURE 9. Doubly robust event-study estimates for 25-year citations

Note: The figure reports event-time average treatment effects from a doubly robust inverse-probability-weighted difference-in-differences estimator. Scientists preconnected through correspondence with at least one founder before 1925 form the treated group; scientists without such a connection form the control group. The outcome is the logarithm of one plus citation-weighted publications, with forward citations accumulated over 25 years after publication. Event time zero corresponds to 1925. Blue markers denote pre-treatment estimates, red markers denote post-treatment estimates, and the dashed horizontal line marks zero. Shaded bands are 95% pointwise confidence intervals constructed using wild-bootstrap standard errors. The estimator combines inverse-probability weights with an outcome-regression adjustment and uses pre-1925 publication share, publishing probability, three- and five-year forward citations, specialty, principal country of work, birth year, and first publication year as covariates. This doubly robust specification provides a robustness check on the baseline weighted estimates.

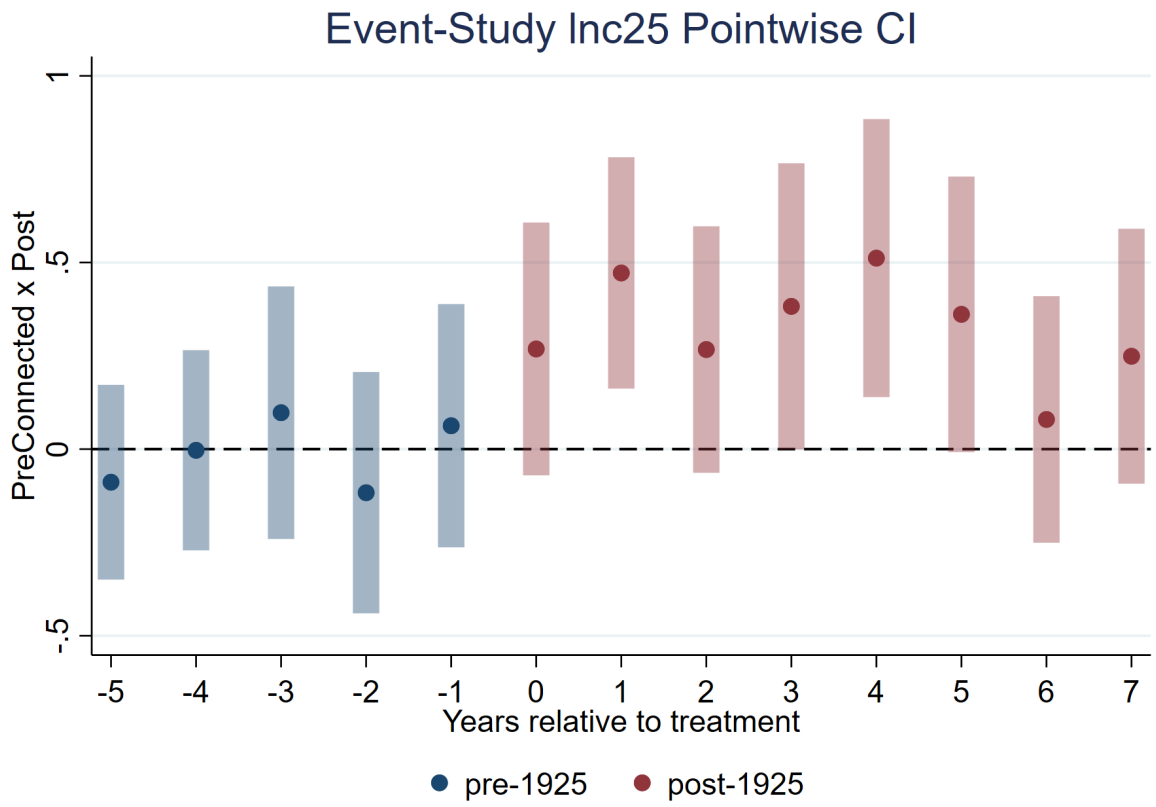


FIGURE 10. Doubly robust event-study estimates for theoretical quantum studies

Note: The figure repeats the doubly robust inverse-probability-weighted event study after restricting the sample to theoretical quantum studies. The restricted sample contains 8,008 scientist-year observations for 572 scientists, of whom 180 are treated and 392 are controls. Treatment is defined by a pre-1925 correspondence connection to at least one founder of quantum mechanics. The outcome is the logarithm of one plus citation-weighted publications, with citations measured over the first 25 years after publication, and event time zero denotes 1925. Blue markers show pre-treatment estimates and red markers show post-treatment estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors; the dashed horizontal line denotes zero. The estimator conditions on the same pre-discovery productivity and scientist characteristics as the full-sample specification. This restriction tests whether the estimated post-1925 citation response remains visible among scientists working most directly in the field affected by the breakthrough.

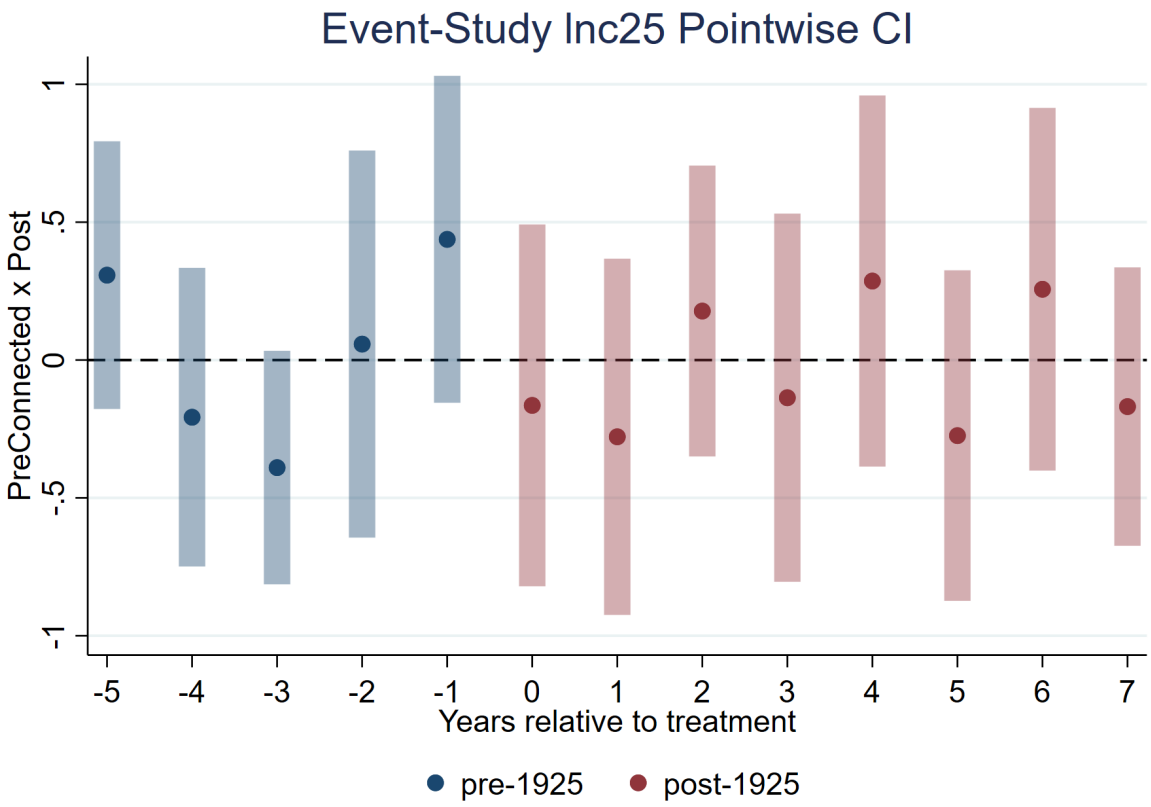


FIGURE 11. Doubly robust event-study estimates for astrophysics and physical chemistry

Note: The figure repeats the doubly robust inverse-probability-weighted event study for scientists classified in astrophysics or physical chemistry and excludes scientists classified in theoretical quantum studies. Treatment is defined by a pre-1925 correspondence connection to at least one founder of quantum mechanics. The outcome is the logarithm of one plus citation-weighted publications, using forward citations accumulated over 25 years after publication, and event time zero corresponds to 1925. Blue markers report pre-treatment estimates, red markers report post-treatment estimates, and the dashed horizontal line marks zero. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors. The estimator uses the same pre-1925 productivity measures and scientist characteristics as the baseline specification. The estimates are comparatively imprecise and fluctuate around zero before and after 1925, so this figure serves as a field-based comparison for the more concentrated response in theoretical quantum studies.

7.5. Citations to Previous Works versus to Novel Works

To distinguish the effect of preexisting connection between the preivous, existing works and the novel works (especially those publsihed after 1925), I look at the other outcome – forward citations. I build this measure following Waldinger (2012): for each cohort of publications, sum up citations to the publications coming from all Web of Science sources during the time window of 2, 3, 5 and 25 years since the year of publication + 1. In this subsection, I focus on short-term outcomes such as forawrd citations in the next 3 years, and I report results for forward citations in the next 2 and 5 years in the Appendix.

7.6. Field-specific evidence

7.7. Placebo comparisons

Restricting the sample to theoretical quantum studies leaves 8,008 scientist-year observations and 572 scientists, of whom 180 are connected and 392 are unconnected. In this more substantively proximate field, the pooled estimate for log(1+twenty-five-year citations) is approximately 0.13 log points, close to the full-sample benchmark. The doubly robust event-study coefficients are positive throughout the post-1925 period and are largest in the middle post-treatment years. This pattern is consistent with the proposed mechanism: pre-existing access to the founders should matter most for scientists working close to the new theory.

As a negative-control comparison, the corresponding event study for astrophysics and physical chemistry does not show a sustained positive post-1925 path. Its coefficients fluctuate around zero and are imprecisely estimated. The contrast with theoretical quantum studies is suggestive, although it is not a formal test of equality across fields and should not be interpreted as eliminating all field-specific confounding.

7.8. What the estimates establish

Across the fixed-effects benchmark and the weighted dynamic specifications, the available results show a positive post-1925 differential for scientists with direct correspondence links to the founders. The strongest evidence concerns the probability of publishing and the subsequent citation reception of their work. The persistence of the citation pattern in the theoretical-quantum subsample, together with the absence of a comparable pattern in the astrophysics and physical-chemistry comparison, is

consistent with a role for pre-existing social access in the diffusion and productive use of the new ideas.

The evidence remains conditional rather than design-based. Founder links were not randomly assigned; connected scientists were more productive before 1925; the treatment measure may include an interval that overlaps 1925; and ideas could diffuse to nominal controls through indirect network paths. IPW and doubly robust estimation address observed compositional differences but not selection on unobserved time-varying factors. For these reasons, the most defensible conclusion is that pre-existing direct correspondence with the founders predicts a larger post-1925 increase in scientific output and recognition among observably comparable physicists. Establishing a causal effect of personal communication more sharply requires strict pre-1925 dating of links, transparent common-support diagnostics, a consistently defined incumbent risk set, and an explicit treatment of network spillovers.

7.9. Diff-in-Diffs results accounting for spillovers in the network

This section will present the results with accounting for the interference in a simple and tractable way by introducing a measure of the Bohr-group exposure.

8. Heterogeneity and Mechanisms

This section presents the results of investigation of the potential mechanisms and heterogeneity channels that might drive the main results. The investigation is built around the following main mechanisms/channels: age of scientists (outcomes of younger *versus* older scientists), subset of journals (all journals in the sample *versus* only top-7 journals in physics), direction and intensity of mail correspondence with the founders; evidence on spillover effects across first, second and third neighbors of Niels Bohr; and last but not least – comparison of competing "camps" of quantum mechanics: physicists in the correspondence circle of Bohr and Heisenberg *versus* physicists in the correspondence circle of Einstein and Schrödinger.

Part of the archival correspondence is grouped into a 1922–1925 interval that cannot always distinguish contact formed before 1925 from contact initiated during the breakthrough year. Therefore, ...

9. Conclusion

This paper examines whether pre-existing personal communication via mail correspondence helped scientists respond to the quantum-mechanics revolution. I combine a newly digitized inventory of correspondence among physicists with publication and citation records, comparing scientists who corresponded with at least one quantum mechanics founder before 1925 with scientists who lacked a direct link. The setting captures a central feature of scientific production: personal access to the creators of new knowledge may affect how productively others use it.

The results show a large in magnitude and highly significant positive post-1925 differential for connected scientists. In the fixed-effects benchmark, their annual probability of publishing increased by 5.4 percentage points relative to the change among unconnected scientists. Their work received approximately 0.12–0.13 log points more citations over five- and twenty-five-year windows, while the publication-count estimate is positive but imprecise. Weighted and doubly robust event studies show a sustained positive citation differential. The pattern remains present in theoretical quantum studies, whereas astrophysics and physical chemistry show no comparable sustained response. These findings are consistent with prior access to the founders facilitating the production and recognition of research related to the new mechanics.

The evidence does not establish an unqualified causal effect. Scientists selected their correspondents, and connected scientists were more productive before the breakthrough. Reweighting improves comparability on observables but cannot remove selection on unobserved, changing factors. The timing of some links overlaps 1925, the letters' content is unknown, and indirect diffusion may expose scientists in the comparison group. Long citation windows also complicate pre-treatment coefficients. I therefore interpret the estimates as conditional differential changes associated with direct founder access, not as the total effect of the revolution or a pure effect of correspondence.

Further work can sharpen the design by dating links strictly before 1925, distinguishing sustained exchange from occasional contact, and documenting common support after weighting. Modelling indirect exposure would separate direct access from network spillovers, while cited-reference data could test whether connected scientists adopted the founders' ideas earlier. The central implication is that the productive use of a scientific breakthrough may depend not only on publication, but also on the social structure through which scientists communicate.

References

Bacciagaluppi, Guido, and Antony Valentini. 2009. *Quantum Theory at the Crossroads: Reconsidering the 1927 Solvay Conference*. Cambridge: Cambridge University Press.

Banerjee, Abhijit, Arun G Chandrasekhar, Esther Duflo, and Matthew O Jackson. 2013. "The diffusion of microfinance." *Science* 341 (6144): 1236498.

Borowiecki, Karol Jan. 2022. "Good reverberations? Teacher influence in music composition since 1450." *Journal of Political Economy* 130 (4): 991–1090.

Callaway, Brantly, and Pedro H.C. Sant'Anna. 2021. "Difference-in-Differences with multiple time periods." *Journal of Econometrics* 225 (2): 200–230. Themed Issue: Treatment Effect 1.

Forman, Paul. 1971. "Weimar Culture, Causality, and Quantum Theory, 1918–1927: Adaptation by German Physicists and Mathematicians to a Hostile Intellectual Environment." *Historical Studies in the Physical Sciences* 3: 1–115.

Heilbron, John L., and Thomas S. Kuhn. 1969. "The Genesis of the Bohr Atom." *Historical Studies in the Physical Sciences* 1: 211–290.

Heisenberg, Werner. 1925. "Zur Quantentheorie der Multiplettstruktur und der anomalen Zeemaneffekte." *Zeitschrift für Physik* 32 (1): 841–860.

Kojevnikov, Alexei. 2020. *The Copenhagen Network: The Birth of Quantum Mechanics from a Postdoctoral Perspective*. Berlin: Springer.

Kojevnikov, Alexei. 2022. "Quantum Historiography and Cultural History: Revisiting the Forman Thesis." In *The Oxford Handbook of the History of Quantum Interpretations*, edited by Olival Freire Jr. 887–908. Oxford: Oxford University Press.

Kuhn, Thomas S. 1962. "Historical Structure of Scientific Discovery: To the historian discovery is seldom a unit event attributable to some particular man, time, and place.." *Science* 136 (3518): 760–764.

Kuhn, Thomas S. 1970. *The Structure of Scientific Revolutions*. 2nd ed. Chicago: University of Chicago Press.

Kuhn, Thomas S, John L Heilbron, Paul L Forman, and Lini Allen. 1967. "Sources for History of Quantum Physics: An Inventory and Report (Philadelphia, 1967)." *Wolfgang Pauli, Wissenschaftlicher Briefwechsel mit Bohr, Einstein, Heisenberg, ua The first volume covering* 1929.

Mehra, Jagdish, and Helmut Rechenberg. 1982. *The Historical Development of Quantum Theory. Volume 2: The Discovery of Quantum Mechanics, 1925*. New York: Springer-Verlag.

Mehra, Jagdish, and Helmut Rechenberg. 2000. *The historical development of quantum theory*. vol. 1: Springer Science & Business Media.

Mokyr, Joel. 2016. "A culture of growth: The origins of the modern economy."

Sant'Anna, Pedro H. C., and Jun Zhao. 2020. "Doubly Robust Difference-in-Differences Estimators." *Journal of Econometrics* 219 (1): 101–122.

Vazquez-Bare, Gonzalo. 2023. "Identification and Estimation of Spillover Effects in Randomized Experiments." *Journal of Econometrics* 237 (1): 105237.

van der Waerden, B. L. edited 1967. *Sources of Quantum Mechanics*. Amsterdam: North-Holland Publishing Company.

Waldinger, Fabian. 2012. "Peer effects in science: Evidence from the dismissal of scientists in

Nazi Germany." *The review of economic studies* 79 (2): 838–861.
Wheaton, Bruce R. 1993. *Inventory of sources for history of 20th century physics (ISHTCP)*. vol. 2: Verlag für Geschichte der Naturwissenschaften und der Technik.

Appendix A. Additional results appendix

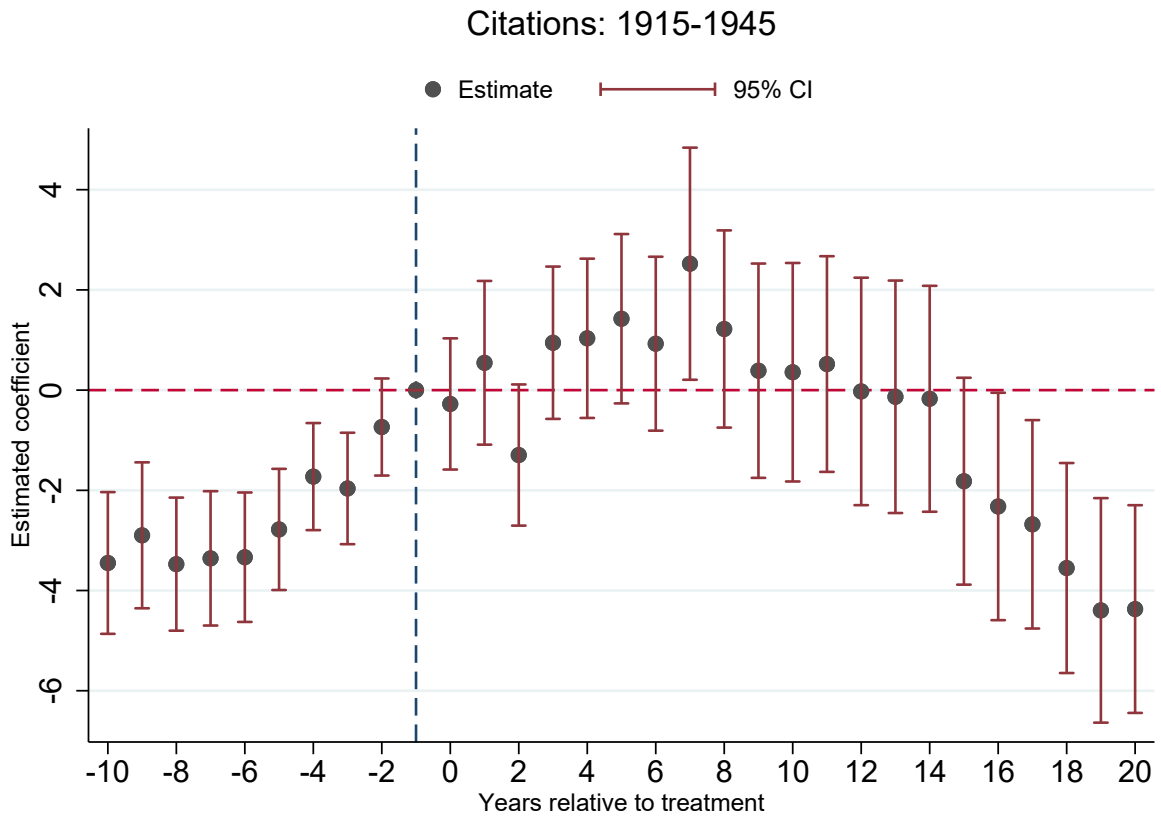


FIGURE A1. TWFE-OLS event-study for citations.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

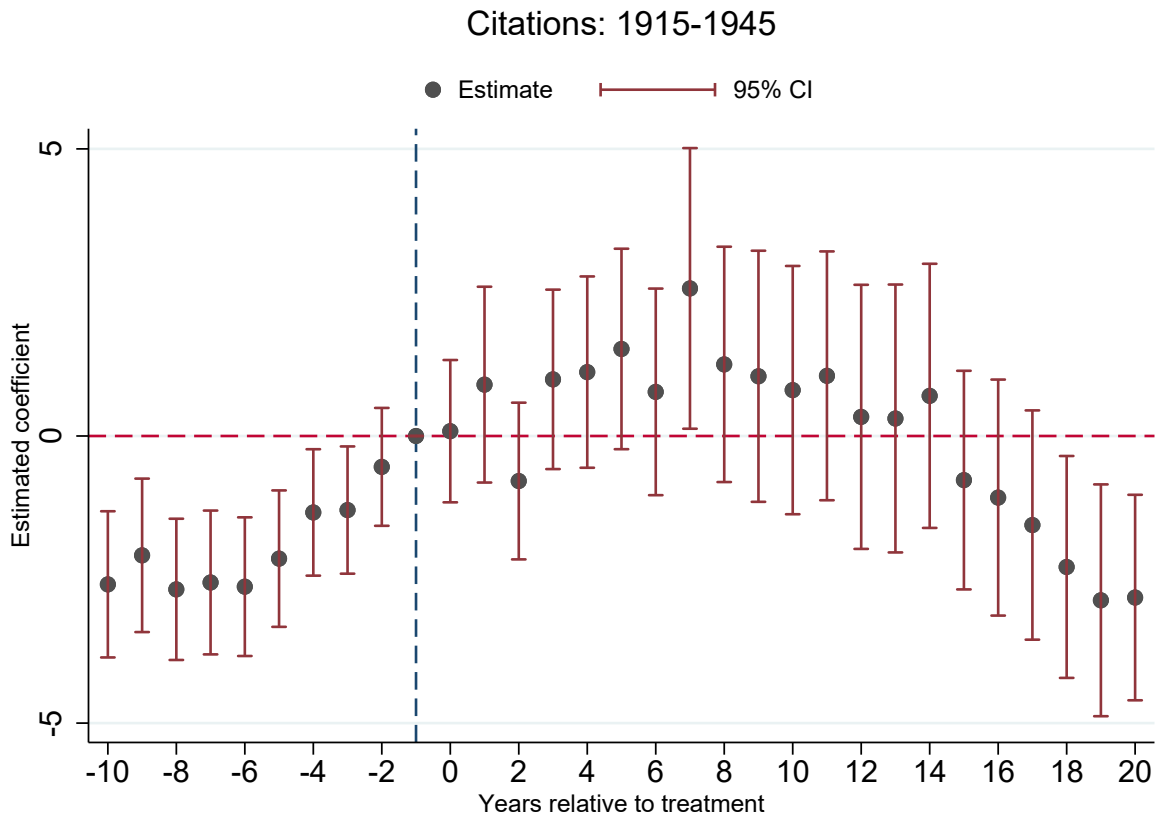


FIGURE A2. TWFE-OLS event-study for citations.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

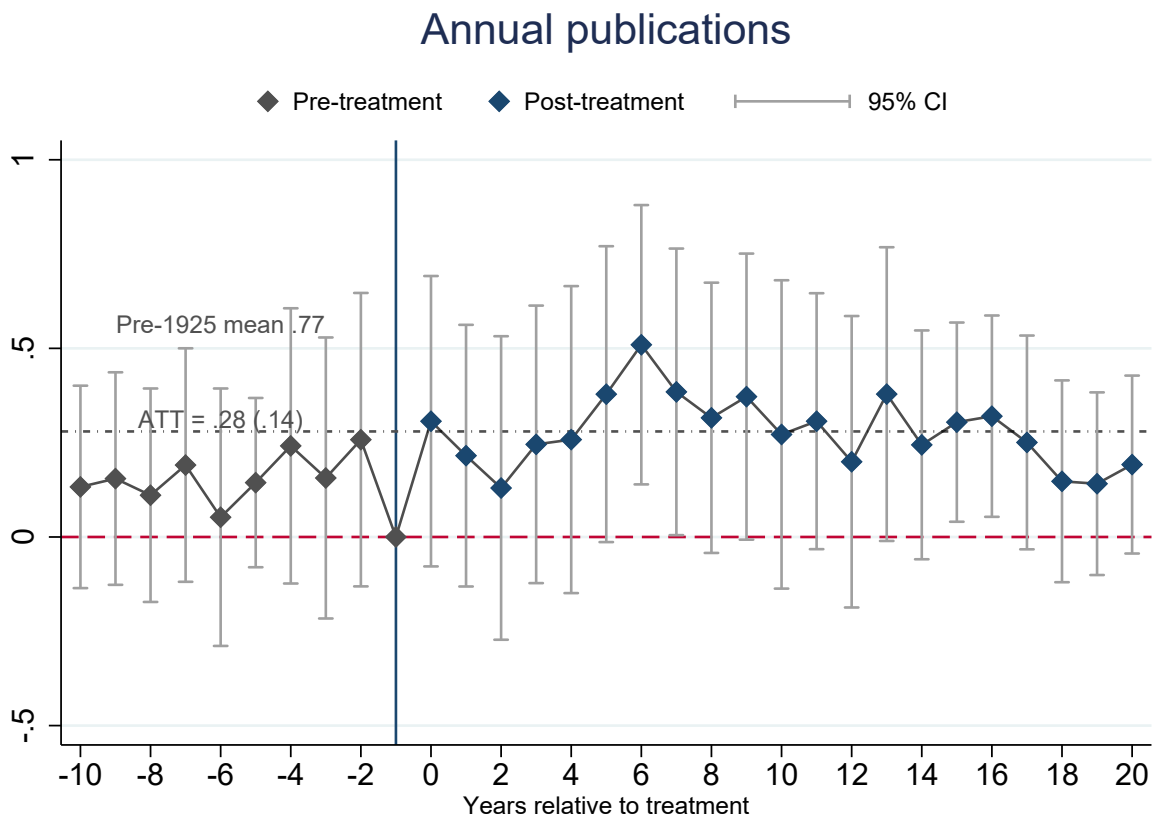


FIGURE A3. Doubly robust inverse-probability-weighting event-study for publications.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual publications (adjusted by the number of co-authors per article). Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey diamond markers report pre-1925 estimates and navy diamond markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

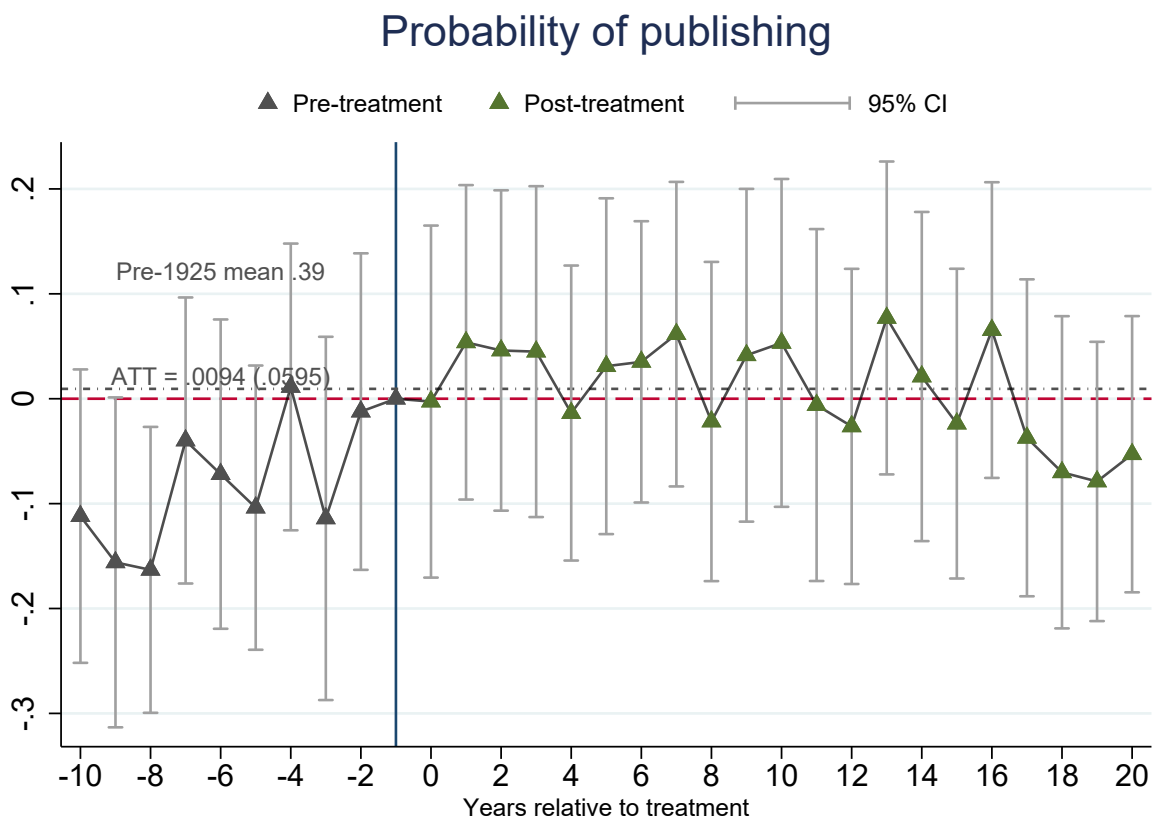


FIGURE A4. Doubly robust inverse-probability-weighting event-study for probability of publishing.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the indicator equal to 1 if scientist i publishes at least one article in year t. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey triangles report pre-1925 estimates and forest-green triangles report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

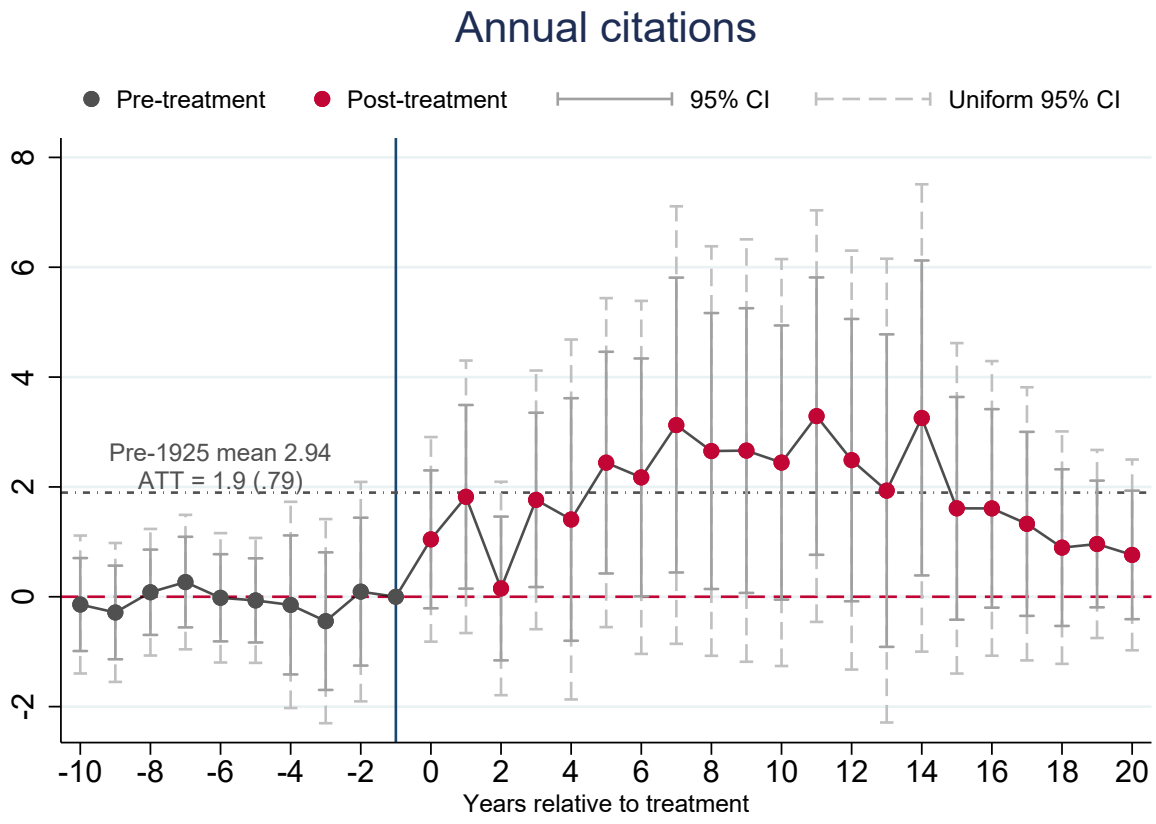


FIGURE A5. Doubly robust inverse-probability-weighting event-study for citations with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed shaded bands are 95% uniform confidence intervals based on clustered standard errors (by scientist). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

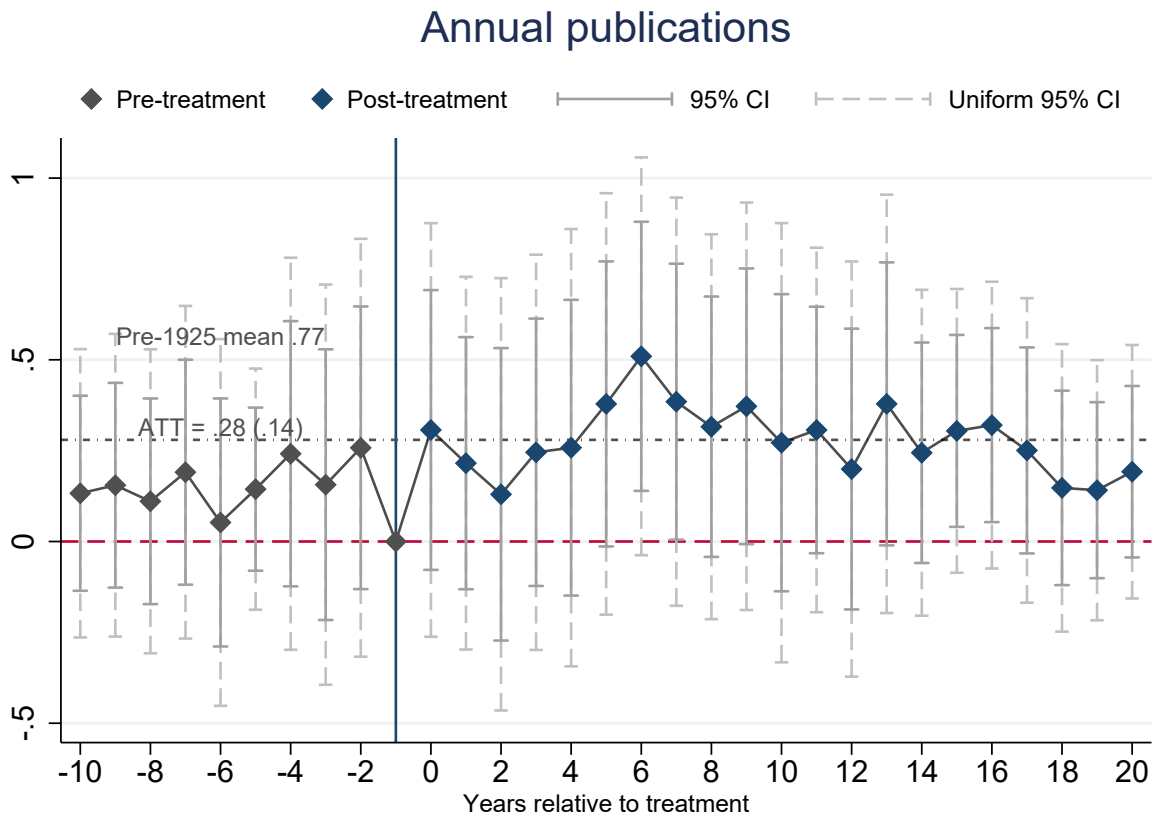


FIGURE A6. Doubly robust inverse-probability-weighting event-study for publications: with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual publications (adjusted by the number of co-authors per article). Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey diamond markers report pre-1925 estimates and navy diamond markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed shaded bands are 95% uniform confidence intervals based on clustered standard errors (by scientist). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

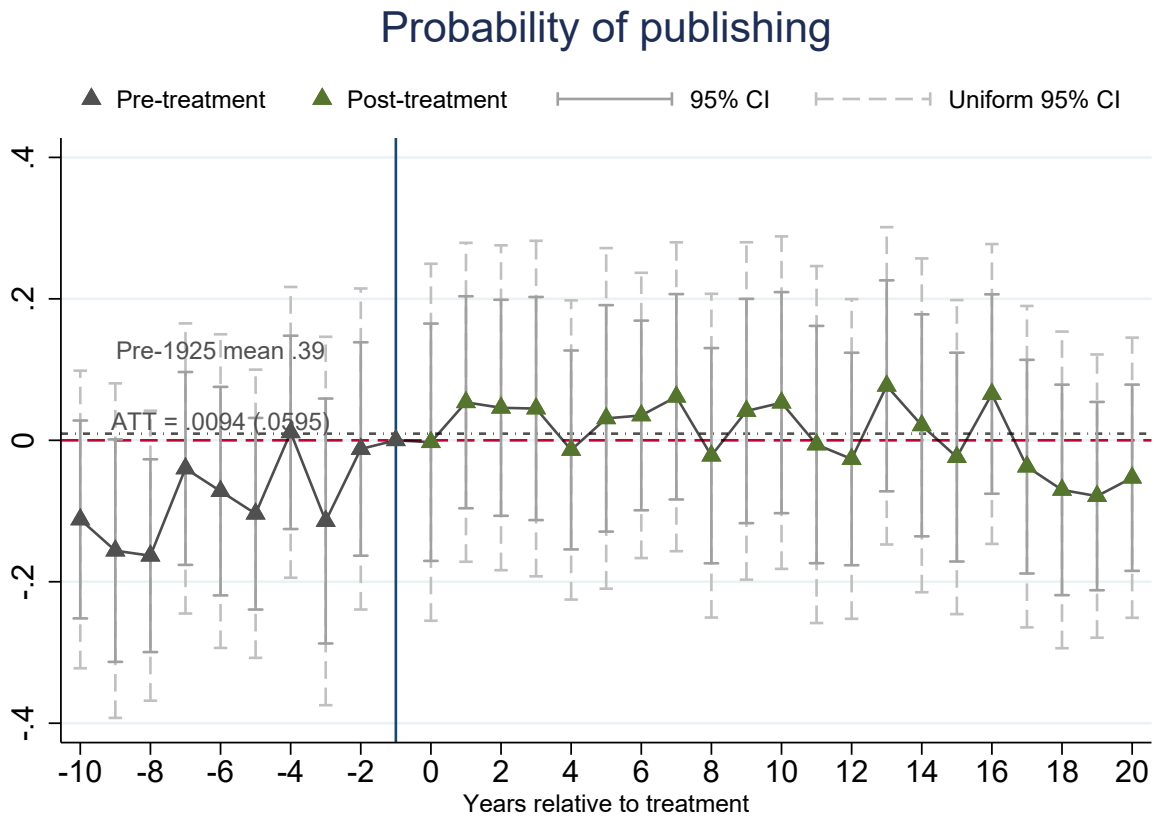


FIGURE A7. Doubly robust inverse-probability-weighting event-study for probability of publishing: with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the indicator equal to 1 if scientist *i* publishes at least one article in year *t*. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey triangles report pre-1925 estimates and forest-green triangles report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed shaded bands are 95% uniform confidence intervals based on clustered standard errors (by scientist). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

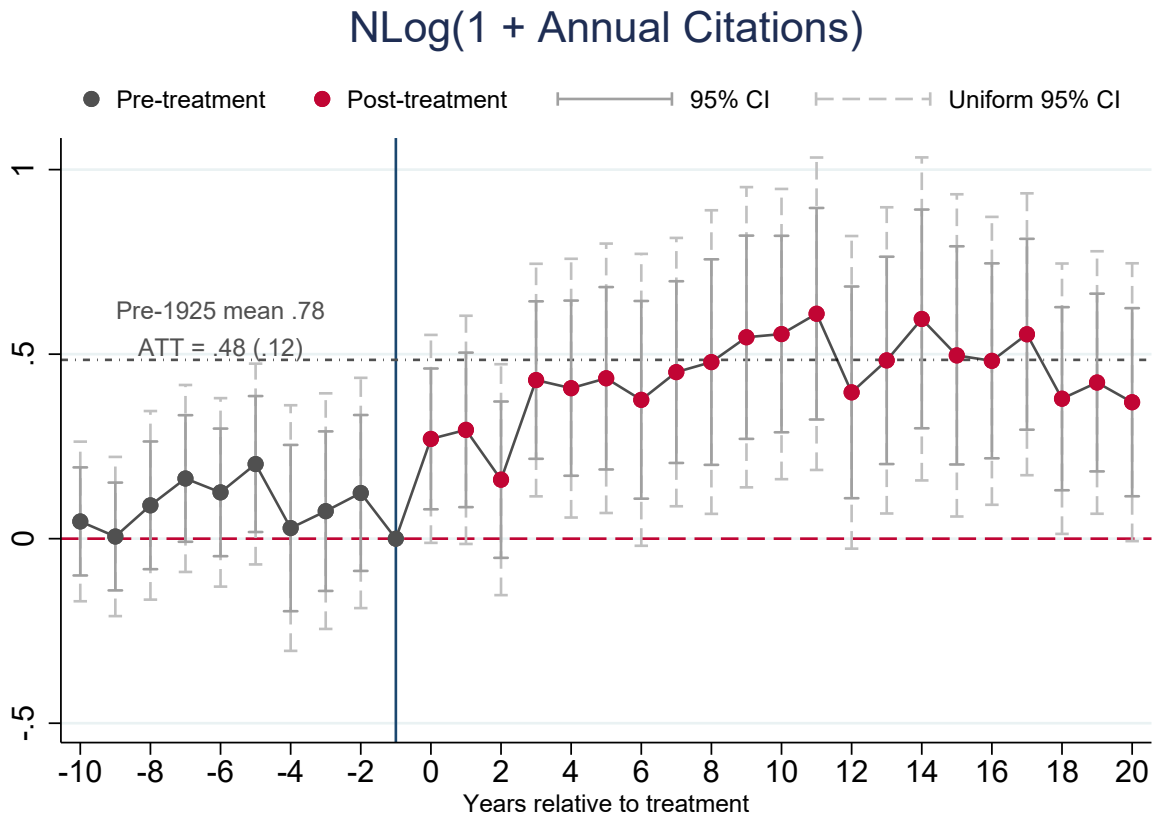


FIGURE A8. Doubly robust inverse-probability-weighting event-study for nlog(1 + citaitons) with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed shaded bands are 95% uniform confidence intervals based on clustered standard errors (by scientist). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

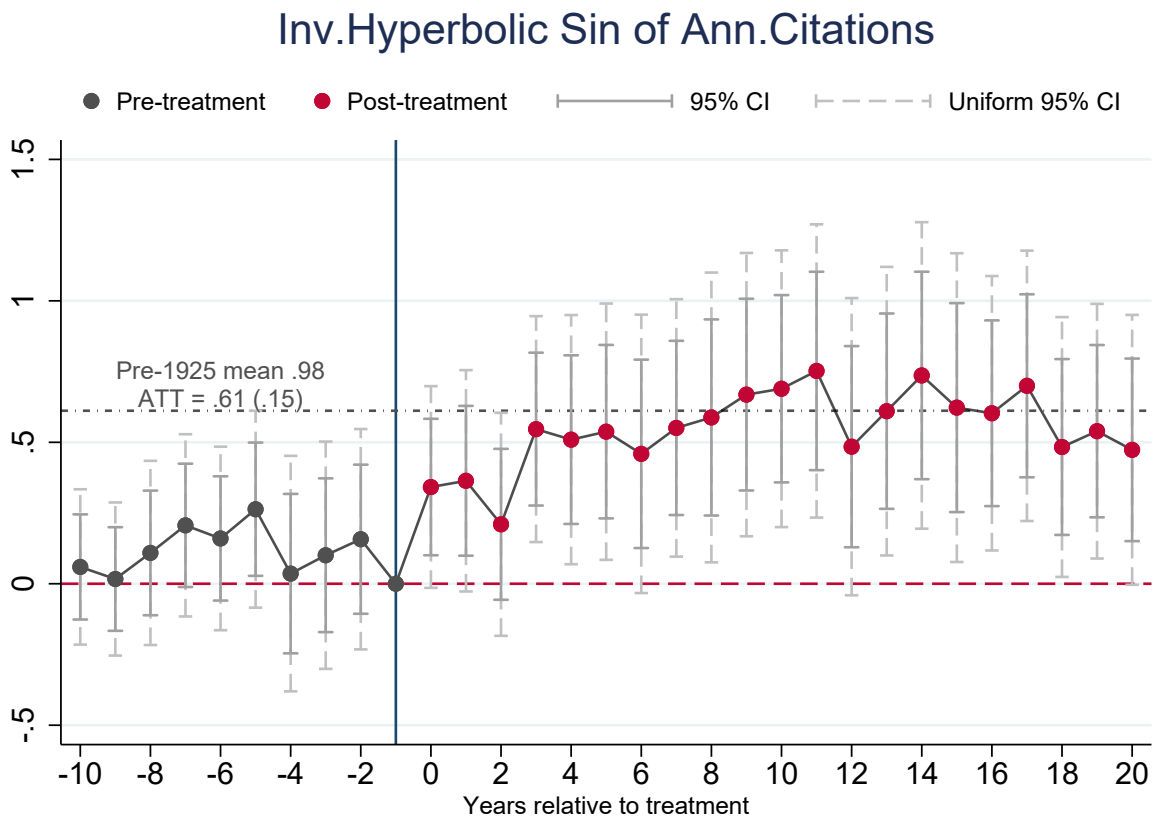


FIGURE A9. Doubly robust inverse-probability-weighting event-study for inv.hyperbolic sin transformation of citations with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed shaded bands are 95% uniform confidence intervals based on clustered standard errors (by scientist). The dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

Appendix B. Robustness appendix

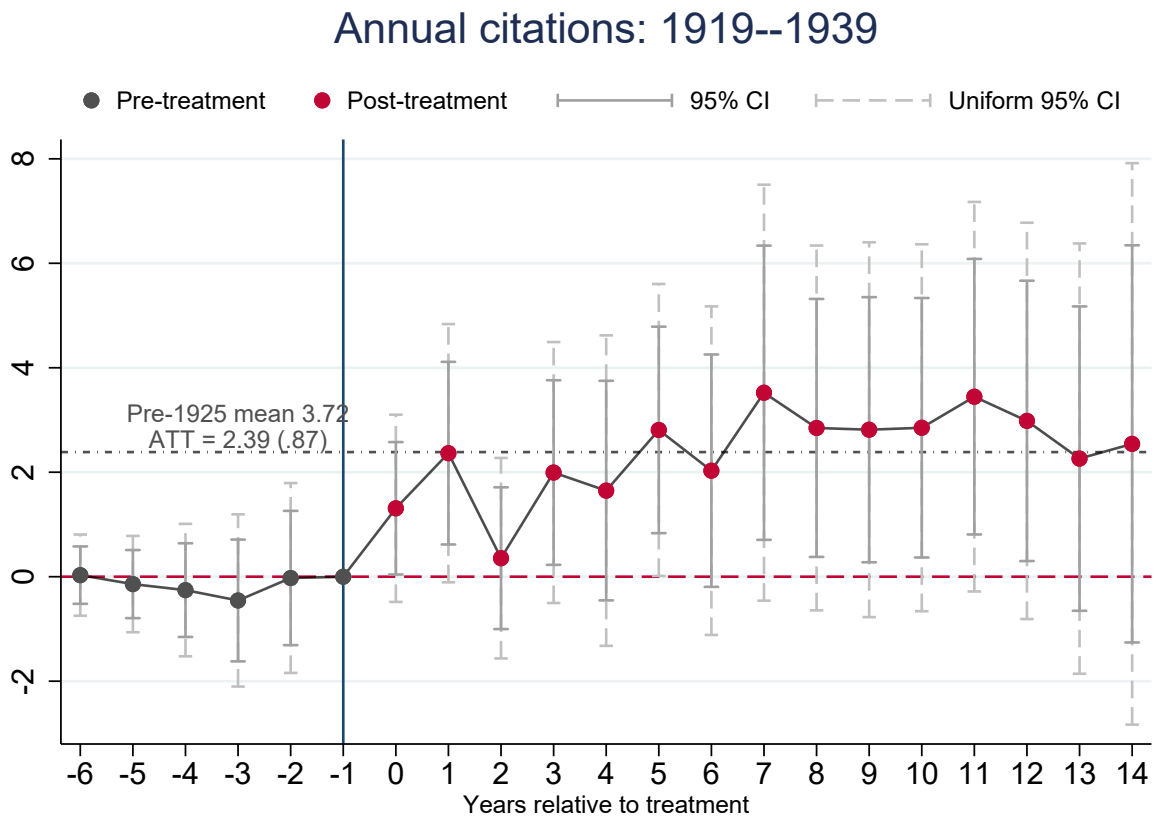


FIGURE A10. Doubly robust inverse-probability-weighting event-study for citations: 1919–1939 with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

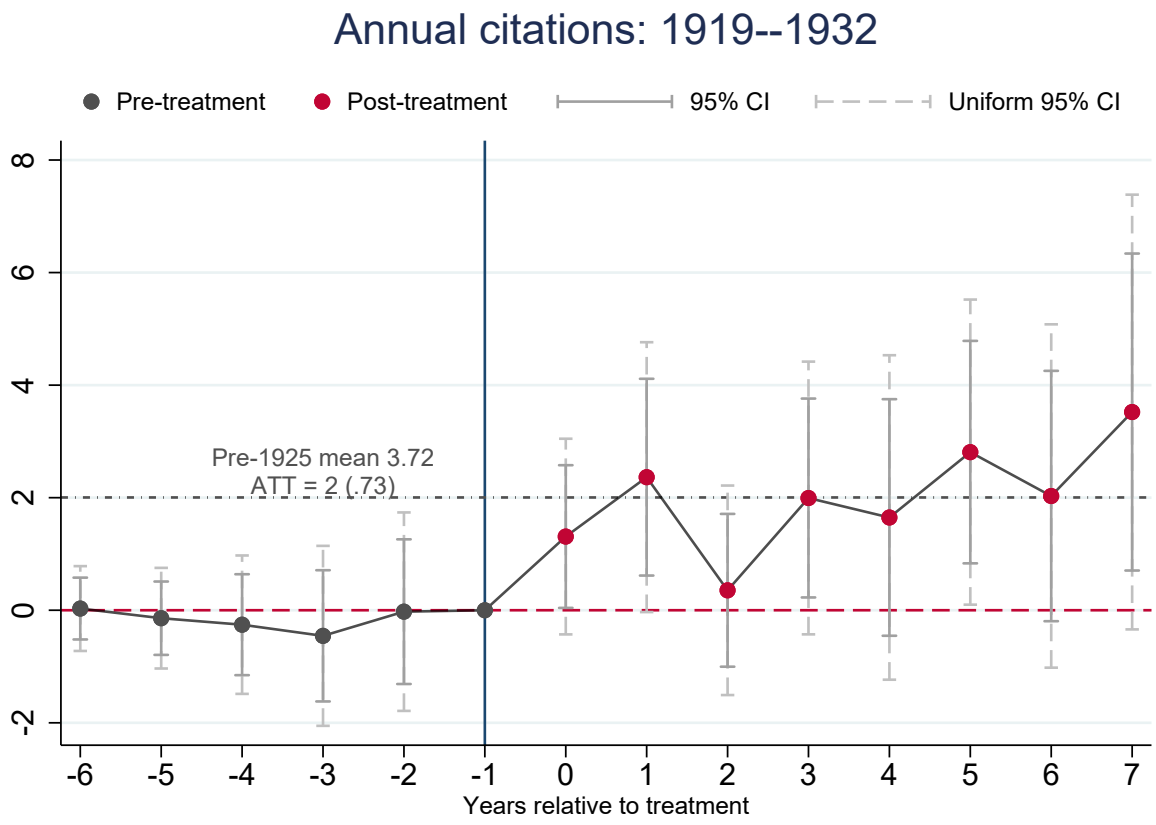


FIGURE A11. Doubly robust inverse-probability-weighting event-study for citations: 1919–1932 with uniform CIs.

Note: The figure reports event-time average treatment effects from an augmented inverse-probability-weighted difference-in-differences estimator (SZ 2020, CS 2021). The treated group consists of scientists classified as preconnected through correspondence with at least one founder of quantum mechanics before 1925; the control group consists of scientists without such a pre-1925 connection. The outcome is the annual citations to all existing works in levels. Event time zero corresponds to 1925, the year of the quantum-mechanics breakthrough used as the common treatment date. The vertical blue line set on x = -1 corresponds to 1924, a year before the treatment starts. Dark grey markers report pre-1925 estimates and red markers report post-1925 estimates. Shaded bands are 95% pointwise confidence intervals based on wild-bootstrap standard errors (9999 replications); the dashed horizontal line denotes zero; the shortdashed-dotted grey line with the text right above it is drawn at the level of the average estimated effect (ATT) and reports the value of the ATT and its standard error in brackets. The control group is reweighted using propensity scores based on a full set of pre-1925 scientist-level covariates outlined above. The pre-treatment estimates provide a diagnostic for conditional parallel trends, while the post-treatment estimates trace the dynamic effect of preexisting connections.

Appendix C. Data appendix

C.1. Inventory of Sources for History of the Twentieth Century Physics

Scope of the inventory. To the best of my knowledge, the ISHTCP is the largest source of collected information on historical mail correspondence among scientists in the twentieth century. Wheaton (1993) highlites the unprecedented scale of the inventory. The original goal of the inventory was to facilitate studying of the history of physics, with a particular interest to the history of quantum physics. Overall, the inventory presents information on nearly 750,000 letters written to or by physicists active in the period from 1896 to 1952. The main criterion for selection of a physicist into the inventory was that they actively published in one of the seven leading journals of pure physics in the period.[19]

The selection criterion may look rather strict since it likely biases the sample to include only top scientists in physicis over the period. However, the inventory consists of just over 5,600 physicists from all major fields in physics and from all over the world. Therefore, it is quite a universe of physicists, and there must be a substantial heterogeneity in the quality and quantity research output of the selected physicists. Moreover, it seems to be the case that often the most important innovations are created by a handsome number of scientists (super-stars), and most cited and influential works are published in leading journals – not only in physics but in science in general. The quantum physics revolution is a proper example of this. Therefore, it might be appropriate studying the production of knowledge only among upper-tail parts of scientist human capital.

Although this paper is in the setting of the quantum mechanics revolution, the inventory is not biased towards quantum physicists but instead covers all main areas in physics, including specialists in nuclear, solid-state, plasma, and low-temperature physics as well as the principal early contributors to acoustics, astrophysics, biophysics, and geophysics ((Wheaton 1993)). A more serious concern with respect to the data is that the inventory is limited to cover correspondence of mainly *theoretical* physicists, keeping the data on scientific correspondence of most technical and applied physicists excluded.

Wheaton (1993) underlines that the ISHTCP was designed to report all letters, regardless of collection, that a physicist exchanged with a given correspondent. Surely, due to

[19] *Annalen der Physik*; *Annales de Physique*; *Philosophical Magazine*; *Proceedings of the Royal Society of London A*; *Physical Review*; *Uspekhi Fizicheskikh Nauk*; and *Zeitschrift für Physik*.

survival bias (more famous physicists' correspondence is more likely to survive through years), the measurement error due to incomplete coverage of some correspondences (non-included but existing letters), etc.

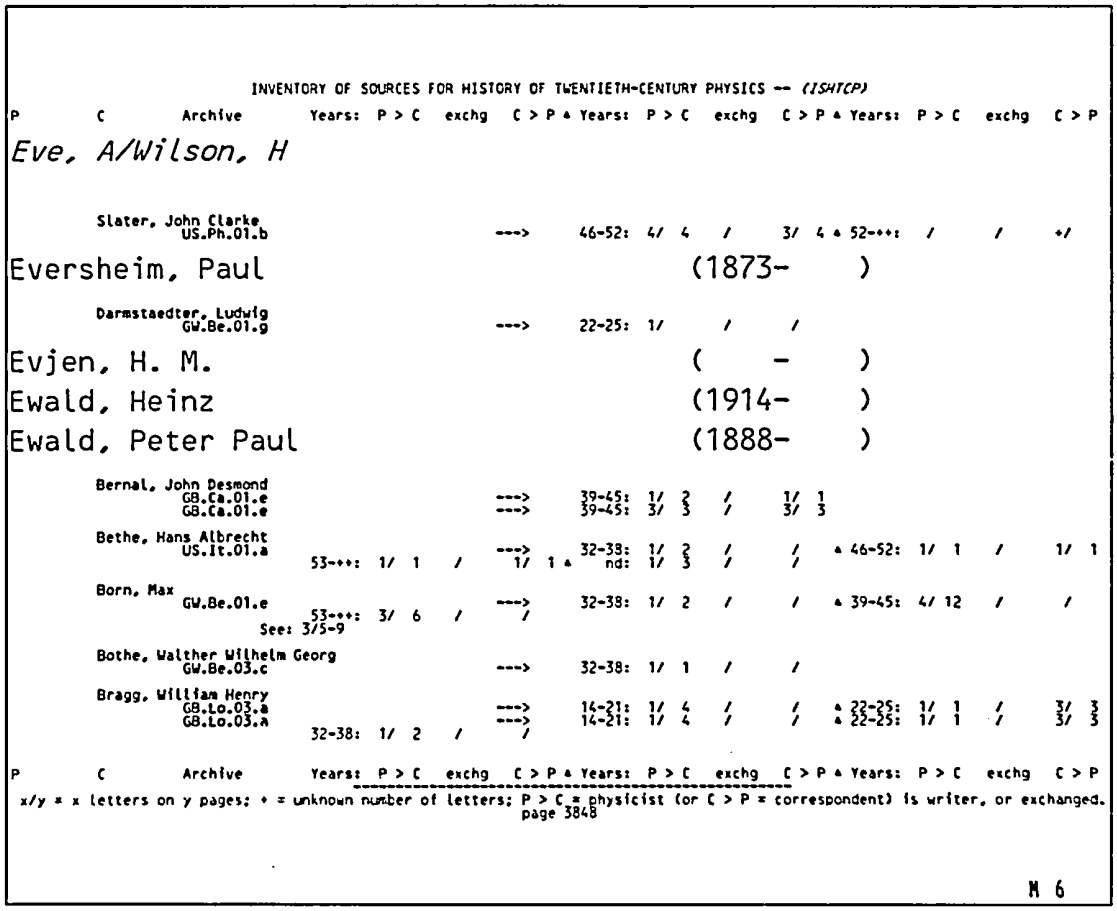
Structure of the inventory. The inventory consists of 12,698 pages on the microfilm reels. Each separate page presents a list of correspondents of the particular physicist (whose name is in bold font in the top of the page) and the number of *letters* / *pages* to and from the correspondents. The typical microfilm page also gives information on the repositories where the physical letters can be found: archive, institute, museum, private collection, etc.

Importantly, the counts of letters and pages are grouped by year intervals, typically of half-dozen year length. The largest part of the data is grouped by few standard year intervals (occur most often), though there are many more non-standard ones. The standard year intervals are: pre-1900 (very rarely), 1901-1906, 1907-1913, 1914-1921, 1922-1925, 1926-1932, 1933-1938, 1939-1945, 1945-1952, post-1952 (very rarely). In a typical page on a microfilm reel, one can see the number of letters and pages (if any) sent by physicist X to other physicists in each given year interval. For example, under a physicist Lev Davidovich Landau, one finds listed all the letters written either to or by Landau. Additionally, the inventory gives information about letters to and from other correspondents of the physicist X even if they are non-scientists. For example, under Albert Einstein one can find the information on letters exchanged with Charles Chaplin and F. D. Roosevelt. Figure A12 gives an example of a typical microfilm page from the inventory.[20]

Collecting and digitizing the inventory. The original microfilm reels of the inventory and an accompanying report were accessed at the The Saxon State and University Library Dresden (SLUB Dresden). I used a special machine for reading the microfilm reels, scanning the pages on the reels and exporting the scans to the PDF format. Once collection was completed, I proceeded with various approaches to digitize the data on pages into sturctured data sets of nodes and edges. These data sets were necessary to reconstruct the historical correspondence networks. My final choice to digitize the data was to use a Qwen's visual large language model (VLLM), which I accessed via the Ollama API subscription[21]. I used a detailed prompt written specifically to account for

[20] A typical page lists correspondence entries for a physicist whose name is of a larger bold font and usually appears in the top of the page, e.g., *Einstein, Albert*. However, since most pages are continuations from previous ones, the physicist's name for whom the correspondence information is listed in such pages is given in the form *Einstein, A/Bohr, N* carrying the last correspondent's name from the previous page after the character "/".

[21] The model was the Qwen VLLM, accessed during Fall 2025



INVENTORY OF SOURCES FOR HISTORY OF TWENTIETH-CENTURY PHYSICS -- *(ISHTCP)*

P	C	Archive	Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P
Eve, A/Wilson, H														
	Slater, John Clarke	US.Ph.01.b				--->	46-52:	4/ 4	/	3/ 4	▲ 52-++:	/	/	+/
Eversheim, Paul								(1873-)						
	Darmstaedter, Ludwig	GW.Be.01.g				--->	22-25:	1/	/	/				
Evjen, H. M.								(-)						
Ewald, Heinz								(1914-)						
Ewald, Peter Paul								(1888-)						
	Bernal, John Desmond	GB.Ca.01.e				--->	39-45:	1/ 2	/	1/ 1				
		GB.Ca.01.e				--->	39-45:	3/ 3	/	3/ 3				
	Bethe, Hans Albrecht	US.It.01.a				--->	32-33:	1/ 2	/	/	▲ 46-52:	1/ 1	/	1/ 1
			53-++:	1/ 1	/	1/ 1 ▲	nd:	1/ 3	/	/				
	Born, Max	GW.Be.01.e				--->	32-38:	1/ 2	/	/	▲ 39-45:	4/ 12	/	/
			53-++:	3/ 6	/	/								
		See: 3/5-9												
	Bothe, Walther Wilhelm Georg	GW.Be.03.c				--->	32-38:	1/ 1	/	/				
	Bragg, William Henry	GB.Lo.03.a				--->	14-21:	1/ 4	/	/	▲ 22-25:	1/ 1	/	3/ 3
		GB.Lo.03.a				--->	14-21:	1/ 4	/	/	▲ 22-25:	1/ 1	/	3/ 3
			32-38:	1/ 2	/	/								
P	C	Archive	Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P

x/y = x letters on y pages; + = unknown number of letters; P > C = physicist (or C > P = correspondent) is writer, or exchanged.

page 3848

M 6

FIGURE A12. A page from the microfilm reel in the ISHTCP

Note: The figure presents a typical page from the microfilm reel inventory (page 3,848 out of 12,600+). The colors of the image made black-and-white, and any artifacts in the image removed with Python.

Source: Wheaton (1993)

the unique features of the pages in the inventory to extract and store the information from each page in the most correct and systematic way. Then, I manually verified all the digitized entires. See Appendix C for detail.

C.2. Other correspondence and biographical data on physicists

For robustness checks, I use supplementary data on selected physicists' daily correspondence and biographical data from another catalog of scientific correspondence – *Sources for the History of Quantum Physics (SHQP)* in the Kuhn et al. (1967)[22]. For details, see Appendix.

For additional data on physicists – biographical data and daily correspondence and info on unpublished manuscripts – I rely on another catalog of scientific correspondence and manuscripts – *Sources for the History of Quantum Physics (SHQP)* in the Kuhn et al. (1967). The full content of the book can be accessed in the digital form on the website of the American Philosophical Society.[23] The microfilm catalog contains a collection of materials – a selected list of manuscripts and – primarily – letters of 280 physicists who contributed at least marginally to quantum studies in physics. The catalog is incomplete and misses information on some letters. However, the advantage of the data is that the data is presented by dates of sent/received letters, and that for each physicist, there is a schematic biography with all career movements information available at the end date of the SHQP project.

The catalog was collected by T. Kuhn, P. Forman, J. Heilbron and L. Allen in the project under the original name *Sources for history of quantum physics: An inventory and report (Philadelphia, 1967)* published in the form of a book. This was a pioneering project of the scale which goal was to preserve the information about the storage of unpublished materials related to the history of quantum physics across libraries, universities, archives and private collections all over the world. The resulting catalog is an index and a report to the inventory of sources for studying the history of quantum physics. The largest and primary part of the catalog is scientific correspondence (letters).

The AHQP project focused on preserving information on those manuscripts and letters which had not been published by the time of the project, and the authors of the documents had to be physicists whose work at least occasionally concerned quantum physics. As a result, the project covered 280 names, of which all most important contributors to quantum physics, such as M. Planck, W. Heisenberg, E. Schrödinger, L. de

[22] The alternative title is *Archive for the History of Quantum Physics (AHQP)*.
[23] https://www.amphilsoc.org/guides/ahqp/.

Broglie, P. Jordan, P. Dirac, W. Pauli, N. Bohr, A. Einstein and others are necessarily included. The list of names was created based on the condition that a physicist included must have contributed at least occasionally to the topics on quantum studies.

The most covered period of correspondence is approximately in 1900-1930s. Most of time, the catalog gives information on scientific correspondence of physicists in the following format:

- The number of letters sent from physicist X to physicist Y, together with the exact calendar date of each letter.
- The archival source of the correspondence, i.e., the collection in which the original letters are physically preserved.

A typical entry in the Authors section is a name of a physicist under whose name there are surnames of other physicists also presented in the catalog, and the dates of letters sent by the author to the physicists. An individual entry (date) is an individual letter. See Figure A.4 in the Appendix for a typical author entry on the website. The AHQP data sets of scientific correspondence of 280 quantum physicists has few important advantages as well as some drawbacks. The advantages are that the dataset gives detailed information on letters exchanged between physicists on a daily basis. The level of detail allows us to build networks in dynamics.

The main drawback of the AHQP catalog data is that the data is only a part of full scientific correspondence presented in depositories all over the world. To remind, the goal of the project was to gain information on manuscripts which have not been cataloged properly by the start of the project in 1962. Hence, although the social graphs built on this data can provide interesting descriptive statistics, there are likely a significant number of missing connections as well as the total number of letters exchanged between physicists can be missing. The drawback of the AHQP data can be down-scaled to a large degree by exploring only a part of the data, more precisely the data in which **continue**

It is important to make a note on the relation of the ISHTCP inventory and the AHQP: the latter is entirely subsumed in the ISHTCP inventory (Wheaton 1993).

I have web-scrapped the full data from the AHQP website with the help of Python, and combined it into properly structured dataframes. I extract schematic biographies of the 280 physicists provided in the AHQP.

C.3. Publication and citation data

Scientist productivity measures. One of the research questions in this paper is what is the effect of being in a social proximity to the superstar, key founders of a break-through new theory in science – the quantum mechanics – on subsequent general productivity of scientists? To answer this question, I introduce three measures of scientist-level productivity outcomes.

I realize that the focusing on only leading pure journals in physics is limiting my analysis of scientific productivity.

Construction of the final panel data set. I merge the physicists list from the inventory report with the Web of Science authors' full names using a combination of fuzzy matching techniques in Python. As a result, I obtain a balanced panel of ... scientists observed over 1915–1945 with their individual charateristics, inclduing treatment status and other correspondence network variables, and scientific outcomes.

Cited references data. To study questions about the diffusion of knowledge, I need data not only on forward citations (total citations that an article received in a given year from all source) but on the cited references – the list of sources (other articles) cited in a given article. The cited references data will allow to track cross-citations between the papers and authors - who is citing whom, and which year. Obviously, the mutual citations are not a perfect measure of the diffusion of knowledge, and alternative measured based on the titles and abstracts/texts of papers could be better ones for studying the diffusion of particular concepts and ideas. However, WoS provides full abstract, not even mentioning text of paper, for a very few articles in the period of my study; and even the OpenAlex database, although giving abstracts for more papers than WoS, also has lots of missing data. Therefore, for simplicity and for preliminary analysis, I rely on the data on cited references.

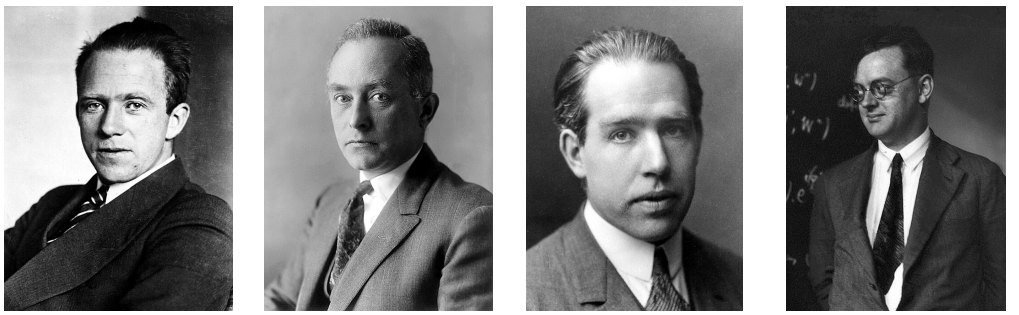
Extracting data on cited references is not a straightforward task to do in the WoS web UI. The use of API provided to the subscribed universities also does not help.[24] Instead, I rely on free open-source database of scientific works and citations – OpenAlex.[25] I access the database and download all data on works of type "article" and their cited references, as well as authorship, affiliations of authors and topics and concepts (when available) for the works published in top journals in physics in the period 1900–1952. The journals are: **place list of journals**.

[24] To the best of my knowledge, the WoS API allows to extract references for maximum one document ID at once, which makes it impractical to collect cited references for thousands of articles.

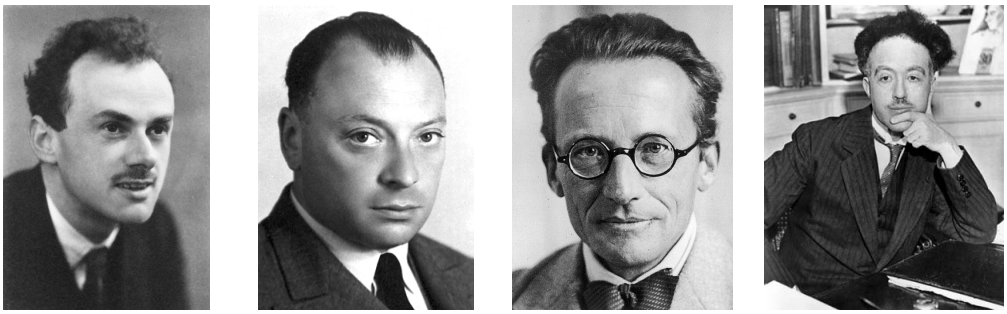
[25] I accessed the SQL slice of the OpenAlex database on the heavyhorse stations at CERGE-EI.

The data on cited references is invaluable for studying the diffusion of knowledge in science because it provides information on who cites whose papers, which can be considered as a measure of reception of/ access to the codified knowledge carried by the cited paper.

Appendix D. Additional Figures Appendix



Heisenberg **Born** **Bohr** **Jordan**



Dirac **Pauli** **Schrödinger** **de Broglie**

FIGURE A13. Founders of quantum mechanics.

Source: Portraits obtained from Wikipedia.

TABLE A1. Effect of Prussian education reforms on academic births (1730–1855)

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth) *Patriots sample: Academics whose birth state is the same as their death state.*			
Panel A – Baseline			
Connected × Post-1925	1.203***	0.902***	0.469*
	(0.249)	(0.219)	(0.244)
Specialty × Post		✓	✓
Country × Post		✓	✓
Career controls × Post		✓	✓
Network controls × Post			✓
Scientist FE	✓	✓	✓
Year FE	✓	✓	✓
# Obs.	17,074	12,908	9,238
R^2	0.784	0.824	0.850
Panel B – Cities produced an elite academic ever			
Connected × Post-1925	3.643***	3.200***	2.660***
	(0.765)	(0.825)	(0.839)
Specialty × Post		✓	✓
Country × Post		✓	✓
Career controls × Post		✓	✓
Network controls × Post			✓
Scientist FE	✓	✓	✓
Year FE	✓	✓	✓
# Obs.	18,102	13,510	9,520
R^2	0.568	0.597	0.619
Panel C – Cities with baseline pop (1790) above 5,000 residents			
Connected × Post-1925	2.622***	2.391***	1.661***
	(0.517)	(0.551)	(0.547)
Specialty × Post		✓	✓
Country × Post		✓	✓
Career controls × Post		✓	✓
Network controls × Post			✓
Scientist FE	✓	✓	✓
Year FE	✓	✓	✓
# Obs.	18,102	13,510	9,520
R^2	0.943	0.952	0.961

Notes: DiD regression absorbing city and birth-cohort fixed effects. Outcome: raw count of births of promi

TABLE A2. Effect of Pre-1925 Connections on Scientists' Productivity

	(1)	(2)	(3)
Panel A – Cumulative citations			
Connected × Post-1925	32.115***	30.028***	20.110***
	(5.069)	(5.515)	(5.420)
Specialty × Post		✓	✓
Country × Post		✓	✓
Career controls × Post		✓	✓
Network controls × Post			✓
Scientist FE	✓	✓	✓
Year FE	✓	✓	✓
# Obs.	18,102	13,510	9,520
R^2	0.798	0.816	0.838
Panel B – Annual citations			
Connected × Post-1925	3.643***	3.200***	2.660***
	(0.765)	(0.825)	(0.839)
Specialty × Post		✓	✓
Country × Post		✓	✓
Career controls × Post		✓	✓
Network controls × Post			✓
Scientist FE	✓	✓	✓
Year FE	✓	✓	✓
# Obs.	18,102	13,510	9,520
R^2	0.568	0.597	0.619
Panel C – CPA share			
Connected × Post-1925	1.203***	0.902***	0.469*
	(0.249)	(0.219)	(0.244)
Specialty × Post		✓	✓
Country × Post		✓	✓
Career controls × Post		✓	✓
Network controls × Post			✓
Scientist FE	✓	✓	✓
Year FE	✓	✓	✓
# Obs.	17,074	12,908	9,238
R^2	0.784	0.824	0.850

*Note*The table reports two-way fixed-effects DiD regressions. All specifications include scientist and year fixed effects. Column (1) reports the baseline specification. Column (2) additionally controls for specialty × post-treatment, country × post-treatment, first publication year × post-treatment, and birth year × post-treatment. Column (3) additionally includes network characteristics interacted with the post-treatment indicator. Standard errors are clustered at the scientist level. The treatment variable equals one for scientists connected to the relevant incumbent network before 1925, and Post-1925 equals one from 1925 onward. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.1. Evolution of correspondence connections between physicists over time

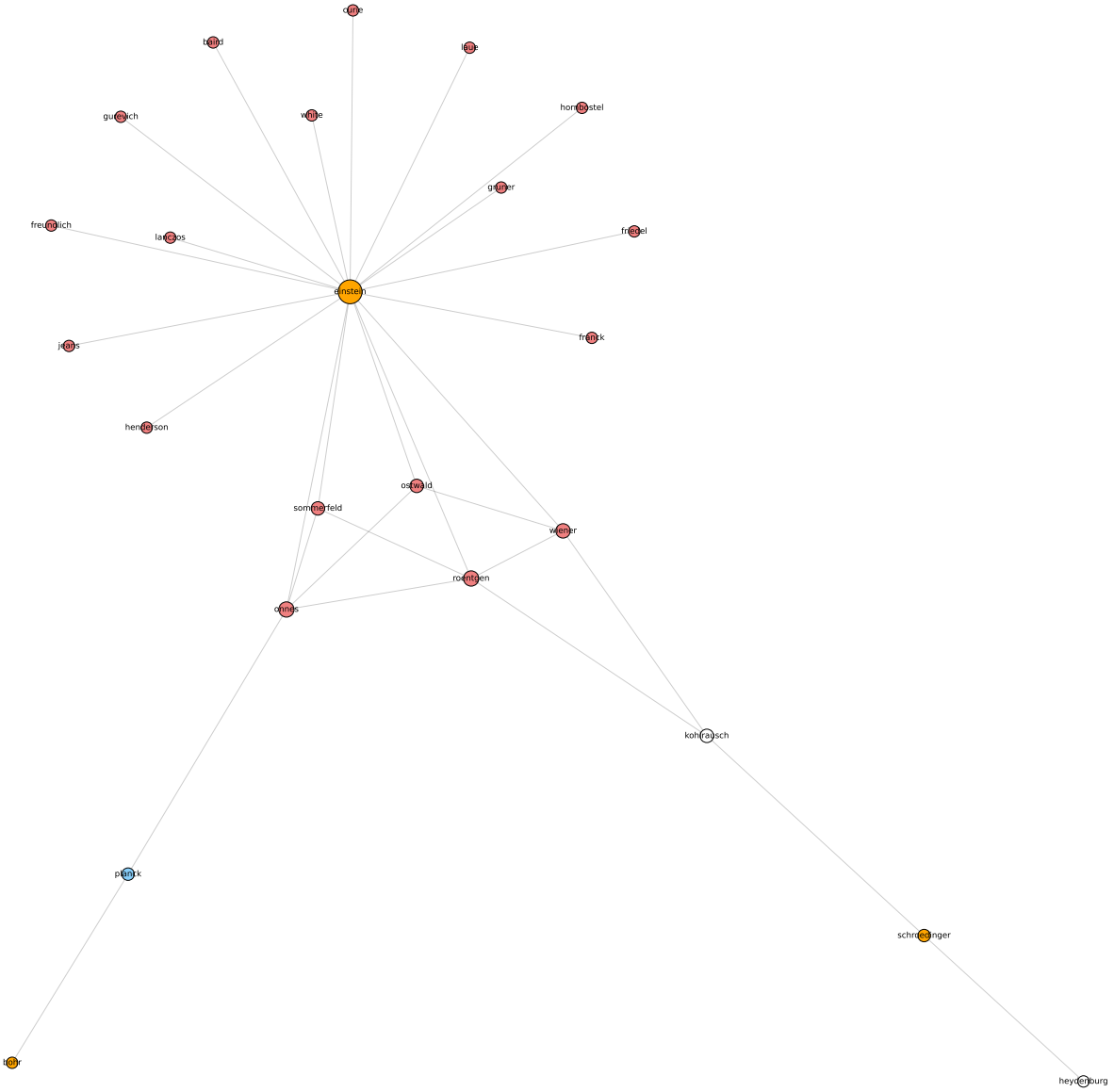


FIGURE A14. Correspondence network of core physicists and their first neighbors, 1900–1906.

Note: Undirected network.

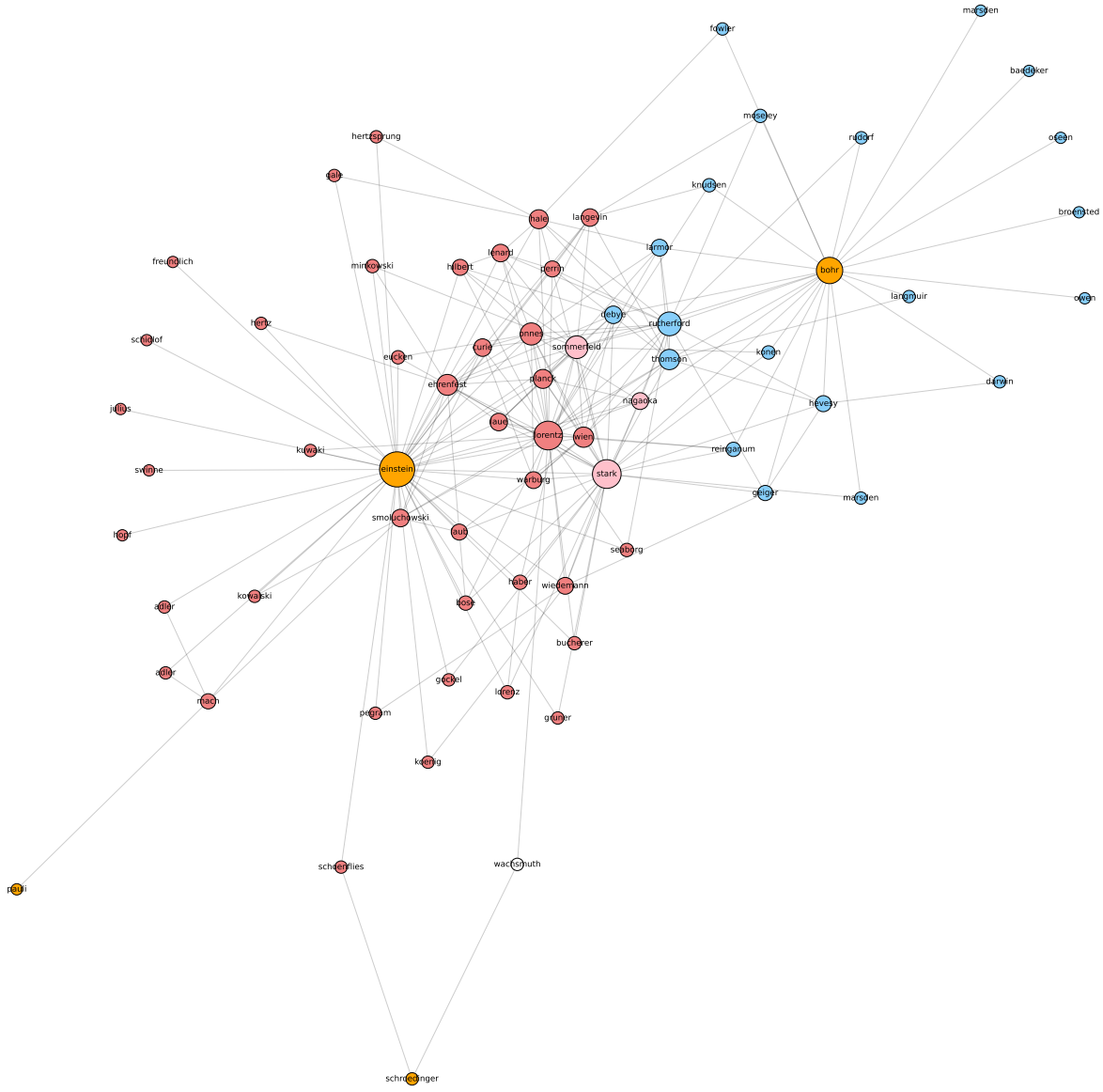


FIGURE A15. Correspondence network of core physicists and their first neighbors, 1907–1913.

Note: Undirected network.

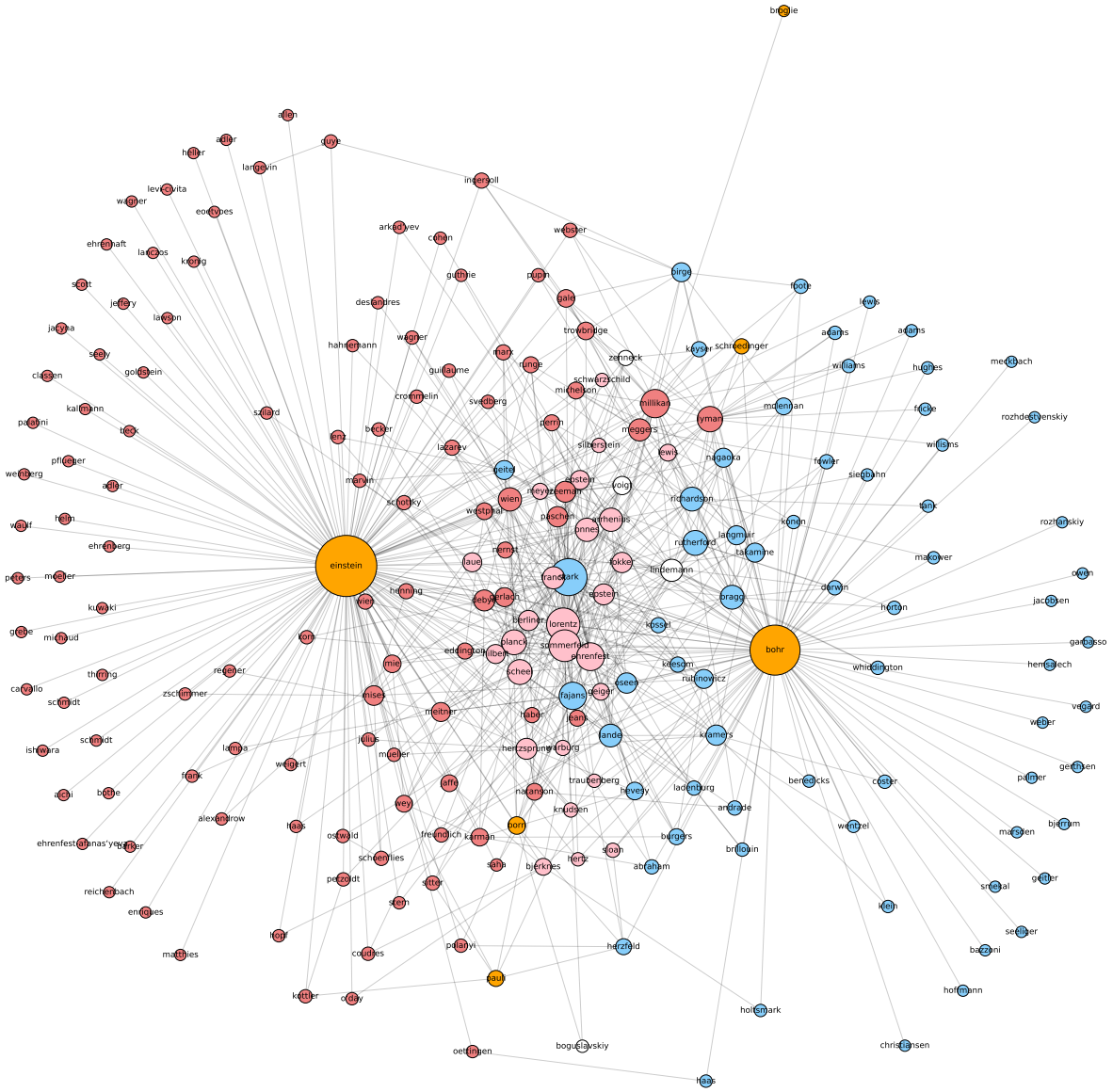


FIGURE A16. Correspondence network of core physicists and their first neighbors, 1914–1921.

Note: Undirected network.

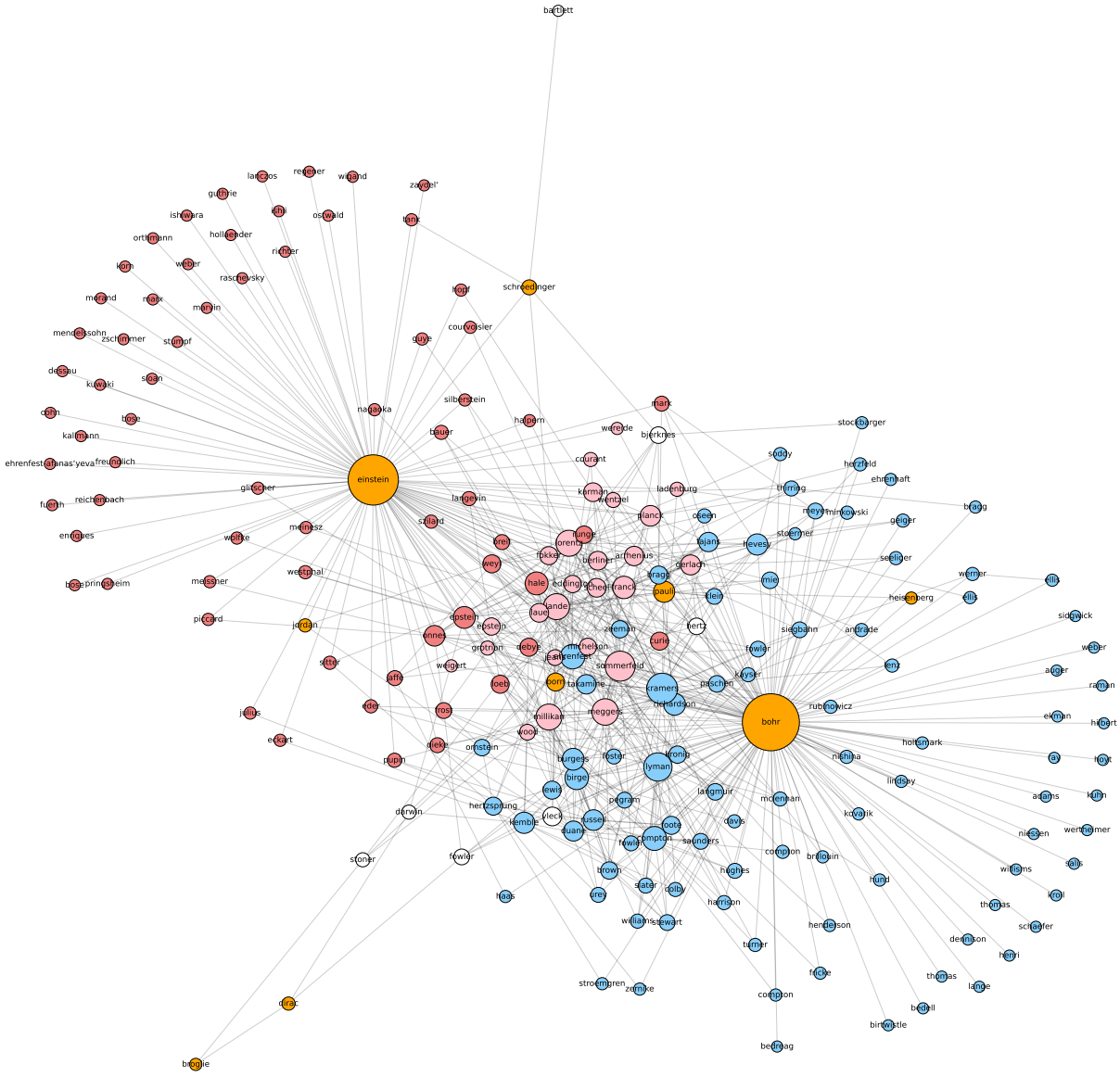


FIGURE A17. Correspondence network of core physicists and their first neighbors, 1922–1925.

Note: Undirected network.



A. 1900–1906

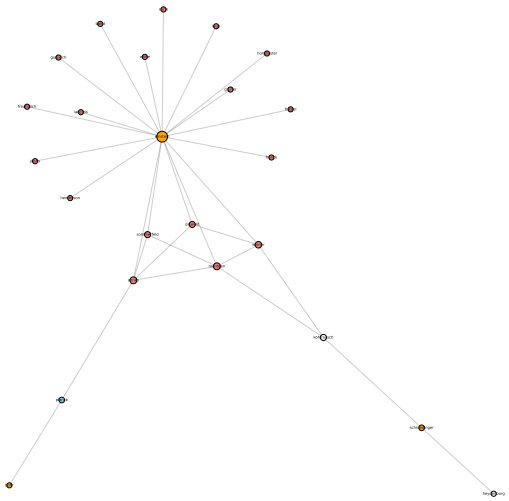
B. 1907–1913

C. 1914–1921

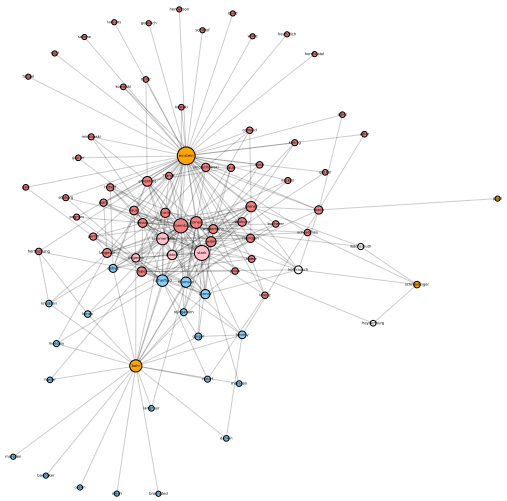
D. 1922–1925

FIGURE A18. Evolution of the correspondence network of core physicists and their first neighbors, 1900–1925.

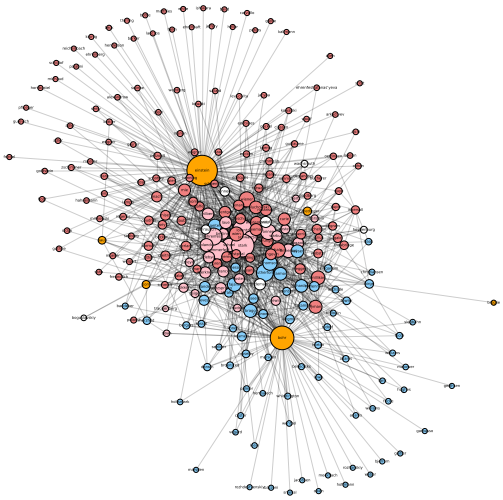
Note: Undirected networks.



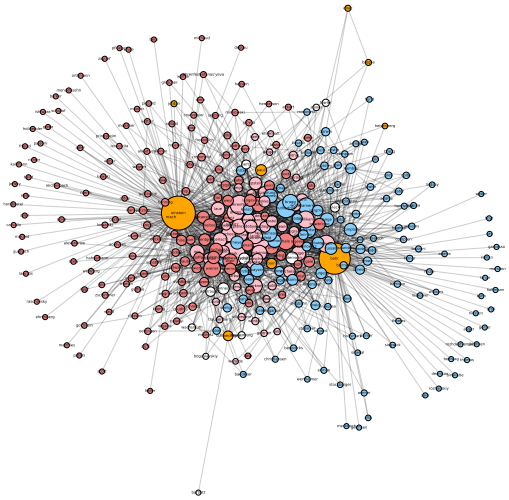
A. 1900–1906



B. 1900–1906, 1907–1913



C. 1900–1906, 1907–1913, 1914–1921



D. 1900–1906, 1907–1913, 1914–1921, 1922–1925

FIGURE A19. Cumulative-by-years evolution of the correspondence network of core physicists and their first neighbors, 1900–1925.

Note: Undirected networks. Cumulative means that for each given year period, the figure depicts edges from all the previous periods together with the given period.

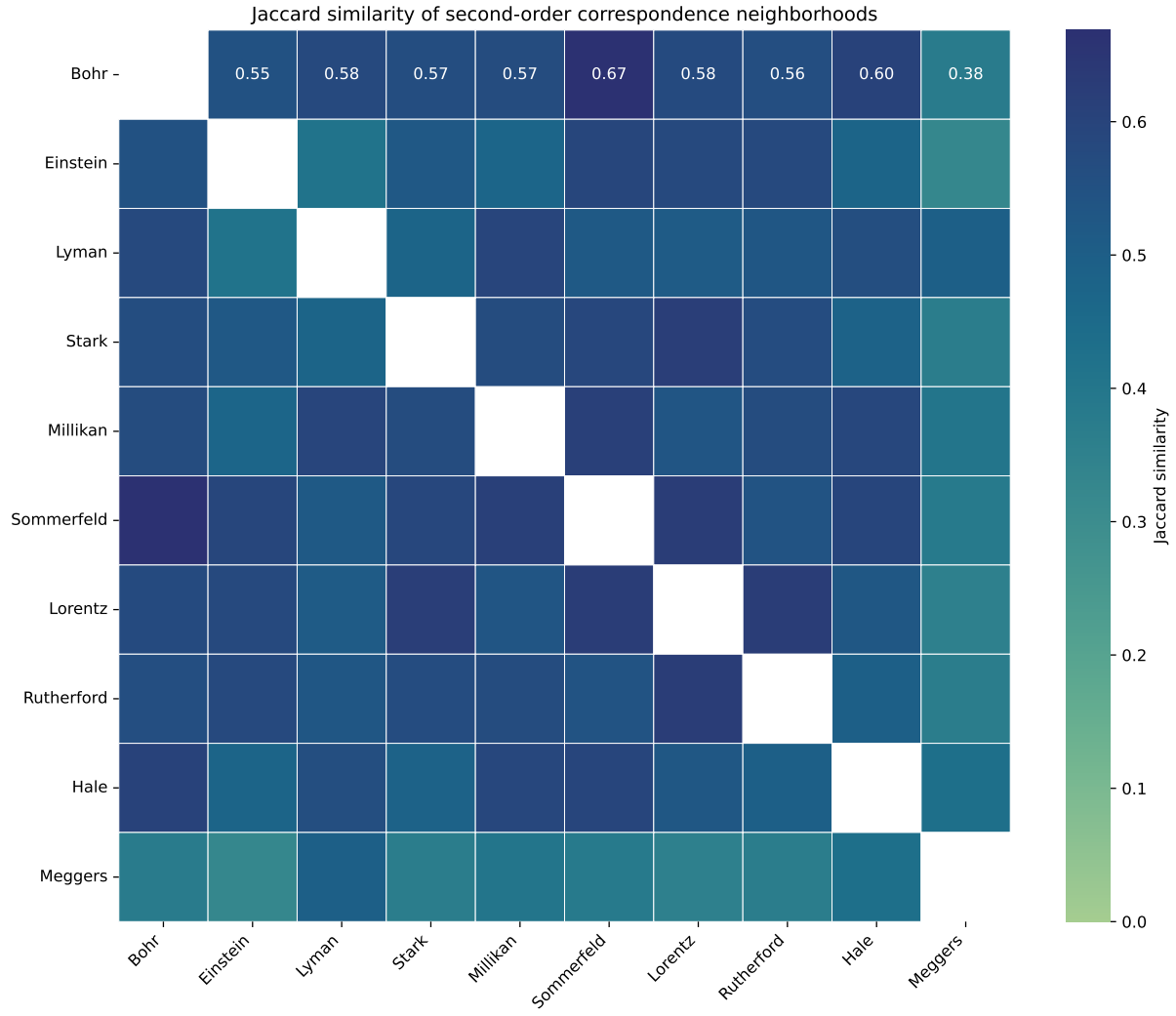


FIGURE A20. Heatmap of second neighbours overlap (Jaccard similairty).

Note: The figure reports pairwise Jaccard similarities between physicists' sets of second-order contacts in the undirected and unweighted correspondence network for 1900–1925. For each focal physicist, the second-order neighborhood contains contacts of their direct contacts, after removing the focal physicist and all first-order contacts. It therefore represents scientists reached at exactly two steps from the focal physicist. Jaccard similarity is the number of second-order contacts shared by two physicists divided by the total number of unique second-order contacts belonging to either physicist's network. Darker shading indicates greater neighborhood overlap. The matrix is symmetric, and diagonal values are omitted because each neighborhood is perfectly similar to itself.

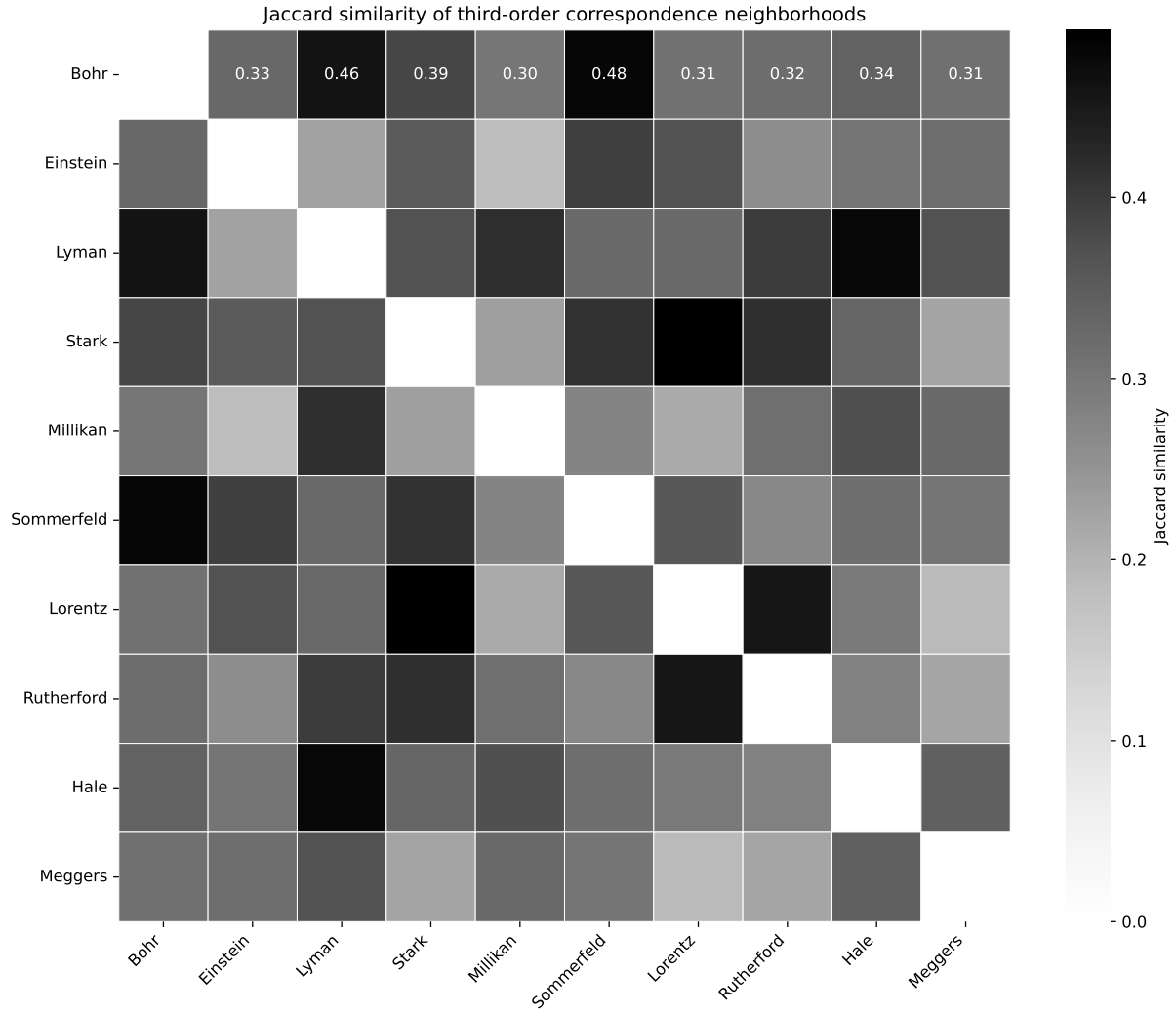


FIGURE A21. Heatmap of third neighbours overlap (Jaccard similairty).

Note: The figure reports pairwise Jaccard similarities between physicists' sets of third-order contacts in the undirected and unweighted correspondence network for 1900–1925. For each focal physicist, the third-order neighborhood contains contacts of their second-order contacts, after removing the focal physicist and all first- and second-order contacts. It therefore represents scientists reached at exactly three steps from the focal physicist. Jaccard similarity is the number of third-order contacts shared by two physicists divided by the total number of unique third-order contacts belonging to either physicist's network. Darker shading indicates greater neighborhood overlap. The matrix is symmetric, and diagonal values are omitted because each neighborhood is perfectly similar to itself.

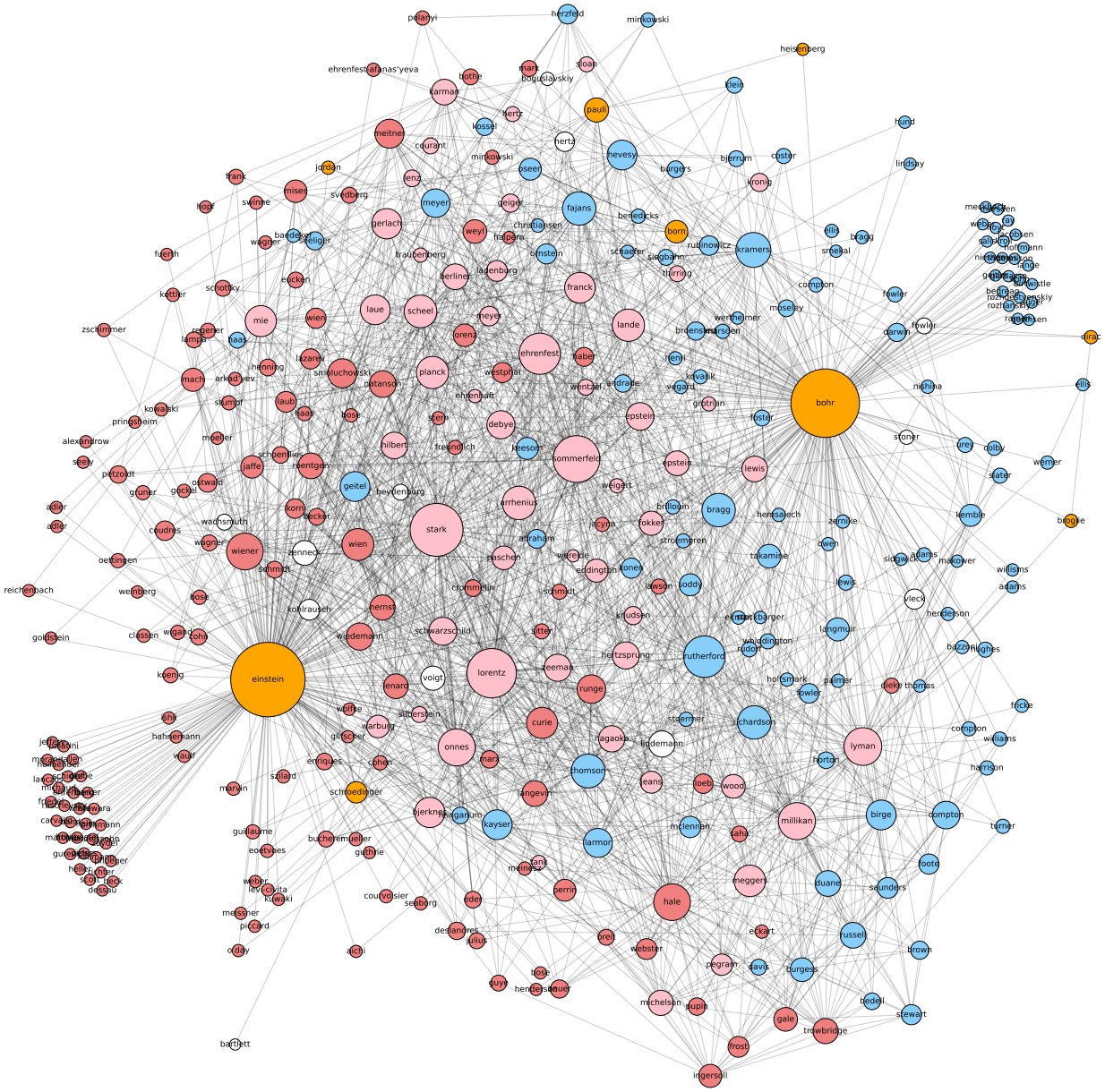


FIGURE A22. Heatmap of second neighbours overlap (Jaccard similairty).

Note: Node positions are generated using the ForceAtlas2 layout algorithm with 1,000 iterations and a fixed random seed for reproducibility. The network is undirected and unweighted, so every correspondence tie contributes equally. ForceAtlas2 is designed to produce readable visualizations of relatively large networks and to reveal densely connected communities. Node positions represent network structure rather than geographical location.